# The Efficacy of Exercise Therapy in Reducing Neck Pain and Fatigue in CH-146 CF Aircrew

A Thesis

Submitted to the Faculty of Graduate Studies and Research

in Partial Fulfillment of the Requirements

for the Degree of

Master of Science

in

Kinesiology and Health Studies

University of Regina

by

Danielle M. Salmon

Regina, Saskatchewan

July, 2009

Copyright 2009: D.M. Salmon

Library and Archives Canada

Published Heritage Branch

395 Wellington Street
Ottawa ON K1A 0N4
Canada

Bibliothèque et Archives Canada

Direction du Patrimoine de l'édition

395, rue Wellington
Ottawa ON K1A 0N4
Canada

*Your file Votre référence*
*ISBN:* 978-0-494-55094-6
*Our file Notre référence*
*ISBN:* 978-0-494-55094-6

NOTICE:

The author has granted a non-exclusive license allowing Library and Archives Canada to reproduce, publish, archive, preserve, conserve, communicate to the public by telecommunication or on the Internet, loan, distribute and sell theses worldwide, for commercial or non-commercial purposes, in microform, paper, electronic and/or any other formats.

The author retains copyright ownership and moral rights in this thesis. Neither the thesis nor substantial extracts from it may be printed or otherwise reproduced without the author's permission.

AVIS:

L'auteur a accordé une licence non exclusive permettant à la Bibliothèque et Archives Canada de reproduire, publier, archiver, sauvegarder, conserver, transmettre au public par télécommunication ou par l'Internet, prêter, distribuer et vendre des thèses partout dans le monde, à des fins commerciales ou autres, sur support microforme, papier, électronique et/ou autres formats.

L'auteur conserve la propriété du droit d'auteur et des droits moraux qui protège cette thèse. Ni la thèse ni des extraits substantiels de celle-ci ne doivent être imprimés ou autrement reproduits sans son autorisation.

In compliance with the Canadian Privacy Act some supporting forms may have been removed from this thesis.

While these forms may be included in the document page count, their removal does not represent any loss of content from the thesis.

Conformément à la loi canadienne sur la protection de la vie privée, quelques formulaires secondaires ont été enlevés de cette thèse.

Bien que ces formulaires aient inclus dans la pagination, il n'y aura aucun contenu manquant.

Canada

**UNIVERSITY OF REGINA**

**FACULTY OF GRADUATE STUDIES AND RESEARCH**

**SUPERVISORY AND EXAMINING COMMITTEE**

Danielle Margaret Salmon, candidate for the degree of Master of Science in Kinesiology and Health Studies, has presented a thesis titled, ***The Efficacy of Exercise Therapy in Reducing Neck Pain and Fatigue in CH-146 CF Aircrew,*** in an oral examination held on April 17, 2009. The following committee members have found the thesis acceptable in form and content, and that the candidate demonstrated satisfactory knowledge of the subject material.

| | |
|---|---|
| External Examiner: | Dr. Mark Brigham, Department of Biology |
| Supervisor: | Dr. Patrick Neary, Faculty of Kinesiology and Health Studies |
| Committee Member: | Dr. Darren Candow,<br>Faculty of Kinesiology and Health Studies |
| Committee Member: | Dr. Donald Sharpe, Department of Psychology |
| Chair of Defense: | Dr. Hilary Horan, Faculty of Business Administration |

## ABSTRACT

The prevalence of neck pain related to the occupational environment of flight in CH-146 Griffon aircrew has been identified in the range of 81-84% (Adam, 2004). In an attempt to mitigate this problem, we quantified the adaptations of cervical muscle isometric strength using a 12-week training program. Subjects were recruited on a volunteer basis from Canadian Forces (CF) CH-146 Griffon aircrew in Gagetown, NB, and randomized into either a general neck strength and muscle coordination training program (CTP; $n$=10), or a neck muscle resistance training program (ETP; $n$=11). Volunteers for a non-treatment control (NTC; $n$=8) were recruited from CH-146 Griffon aircrew base in Valcartier, PQ. Baseline assessments were performed using isometric contractions to determine maximal voluntary contraction (MVC) strength of the cervical musculature (flexion, extension, left lateral flexion and right lateral flexion). Endurance capacity was measured using isometric submaximal contractions to fatigue at a resistance level of 70% of their MVC values. The ETP subjects performed dynamic contraction at 30% of their MVC in the four testing directions using a head harness and Thera-band tubing. The CTP consisted of exercises that focused on strengthening the deep cervical musculature using weight of the head as resistance, progressing to exercises that incorporated the superficial cervical muscles using a resistance load of 30% MVC. After completion of the 12-week intervention, the CTP produced the greatest improvement in maximal force reaching statistically significant for flexion (13.76%) and right lateral flexion (15.92%). However, the ETP achieved the only significant increase when compared to the NTC in right lateral flexion (14.35%). Improved times to fatigue were achieved by the CTP for flexion, $t(9) = 3.04$, $p = 0.01$, left lateral flexion, $t(9) = 2.1$, $p = 0.03$ and right lateral flexion, $t(9) = 3.51$ $p = 0.00$. When compared to the NTC only, left and right lateral flexion improved significantly. The ETP produced substantially smaller improvements, none of which were significant. For the area under the endurance force curve the CTP produced significant

right phase shifts for flexion, $t(9) = 2.67$, $p = 0.01$ and right lateral flexion, $t(9) = 2.78$, $p = 0.01$. When compared to the NTC, significant phase shifts were found for left and right lateral flexion in the CTP and in right lateral flexion for the ETP. Both the ETP and the CTP achieved clinically important reductions in self-reported neck pain on a visual analog scale (VAS), with the greatest decreases reported for 'pain at worst'. A significant reduction in self-reported neck pain was achieved by the CTP for 'pain in general' ($t(9) = 2.47$, $p = 0.02$) and 'pain at worst' ($t(9) = 2.06$, $p = 0.04$), compared to the ETP, where the reduction was limited to 'pain at worst' ($t(10) = 2.18$, $p = 0.03$). A relative pain reduction of 50% was also used to determine whether the treatment was successful in reducing the occurrence of neck pain. The ETP achieved a 50% reduction in VAS scores in 18% of the subjects for both 'pain at present' and 'pain in general' and in 45% of the subjects for 'pain at worst'. This decrease was achieved by 30% of the CTP cohort for 'pain at present', 40% for 'pain in general', and 60% for 'pain at worst'. The provision of an ETP and CTP three times a week for 12 weeks resulted in a positive trend towards improved maximal isometric force and muscular endurance in the cervical musculature. The greatest improvements were found for those subjects assigned to the CTP treatment. By performing ETP or CTP, participants were able to reduce self-reported neck pain, with the most profound effects occurring for 'pain at worst'. Future research should be directed toward quantifying the results of the training programs using neuromuscular and metabolic measures.

**Support: DND Canada** (Canadian Forces Quality of Life Grant #1725981 & Military Health Program #W3931-050513-001/SV)

**ACKNOWLEDGEMENT**

I wish to acknowledge and thank my supervisor Dr. J. Patrick Neary for his time, effort and unwavering support throughout this project. I would like to recognize my committee members Dr. Darren Candow and Dr. Donald Sharpe and thank them for their contributions and insight during the writing of my thesis. I would like to thank those who were not involved in an official capacity, but played an influential role in the success of this project: Dr. James C. Croll, Dr. Victoria L. Chester, Dr. Usha Kuruganti, Major Mario Coutu, Captain Dan O'Neil, Captain Cheryl A. Elvidge, Major Corporal Joseph L. Hussey and Kelsey Forde. I would particularly like to acknowledge Mr. Gregory T. Dickson and Dr. Michael F. Harrison for their outstanding help and technical support with NVG-Lap test and data acquisition. In the development of the general neck strength and muscle coordination training program, I would like to acknowledge the guidance of Carol Kennedy BScPT, DipManipPT, FCAMT, who played an immeasurable role in the development of the program. I am very grateful to the Kinesiology Faculty of the University of New Brunswick for including me as if I were a part of the graduate program and providing me with desk space. Thank you to the University of Regina for allowing me to conduct my research in Fredericton, NB. I would also like to thank all the volunteers from CFB Gagetown 403 Squadron and the Valcartier 430 Squadron without whom, this research would not have been possible. I wish to acknowledge the Canadian Forces further as a source of funding that enabled this project to occur (Canadian Forces Quality of Life Grant #1725981 & Military Health Program #W3931-050513-001/SV). Finally, I would like to thank my fellow graduate students at both the University of New Brunswick and the University of Regina whose knowledge and insights throughout this process have been greatly appreciated.

## DEDICATION

I would like to dedicate this paper to my mom and dad whose love and support throughout this process has been unconditional.

# TABLE OF CONTENTS


| | Page |
|---|---|
| ABSTRACT | i |
| ACKNOWLEDGEMENT | iii |
| DEDICATION | iv |
| LIST OF TABLES | vii |
| LIST OF FIGURES | viii |
| REVIEW OF THE LITERATURE | 1 |
| 1.1 Neck pain and its prevalence | 1 |
| 1.2 Critical load of the cervical spine | 2 |
| 1.3 The cervical spine | 3 |
| 1.4 The cervical musculoskeletal system | 3 |
| 1.5 Hypotheses for the origins of neck pain | 5 |
| 1.6 Differential mechanisms for development of neck pain in aircrew | 10 |
| 1.7 Factors contributing to flight related neck pain in helicopter aircrew | 11 |
| 1.8 Canadian forces CH-146 helicopter pilots and flight engineers | 20 |
| 1.9 Use of exercise therapy in pilot populations | 21 |
| 1.10 Chronic neck pain and presence of strength and weaknesses | 24 |
| 1.11 Neck strength in fighter pilots and helicopter pilots with neck pain | 25 |
| 1.12 Research relating neck pain and exercise training in general population | 26 |
| 1.13 Neck pain and exercise training regimes in the air force | 27 |
| 1.14 Changes in cervical muscles in response to resistance training | 31 |
| INTRODUCTION | 33 |
| 2.1 Training Programs | 33 |
| 2.2 Assessment of strength and endurance and neck pain | 36 |
| METHODS | 39 |
| 3.1 Subjects | 39 |
| 3.2 Laboratory assessment protocol (LAP)- procedures | 40 |
| 3.3 NVG-LAP test measurements instrumentation – apparatus | 43 |
| 3.4 Training program interventions | 44 |
| 3.5 Data analysis | 52 |
| RESULTS | 53 |
| 4.1 Characteristics of sample | 53 |
| 4.2 Maximal voluntary contraction | 53 |
| 4.3 70% endurance trial time to fatigue | 62 |
| 4.4 Area under the 70% endurance trial force curve | 67 |
| 4.5 Visual analog scale (VAS) self-reported neck pain scores | 73 |

DISCUSSION 78
5.1 Maximal voluntary contractions 78
5.2 70% endurance trial time to fatigue 84
5.3 Area under the 705 endurance trial force curve 86
5.4 Summary 87
5.5 Visual analog scale self-reported neck pain scores 90
5.6 Limitations 93
5.7 Future directions 95

CONCLUSION 96

REFERENCES 97

APPENDIX A 110

APPENDIX B 111

APPENDIX C 112

APPENDIX D 115

## List of Tables

Page

1. Table 1: Baseline anthropometric values, personal characteristics and flight experience for endurance, coordination training groups and control group volunteers. Values are in means (*M*) and standard deviations (*SD*). 54

2. Table 2: Summary statistics for isometric MVC (lbs) measured pre- and post-training intervention for the ETP, CTP and NTC. 56

3. Table 3: Time (seconds) to volitional fatigue for the 70% isometric endurance trial pre- and post-test for ETP, CTP and NTC 63

4. Table 4: Area under the force curve (lbs/sec) for the 70% endurance trial for the ETP, CTP and NTC Groups. Values are in means (*M*) and standard deviations (*SD*). Delta change (Δ) between the pre- and post-test scores is reported. 68

## List of Figures

Page

Figure 1: Front view of the Canadian Forces standard issue CH-146 flight helmet with night vision goggles. 13

Figure 2: Lateral view of the Canadian Forces standard issue CH-146 flight helmet with night vision goggles. 13

Figure 3: Pilot seated in CH-146 helicopter while wearing night vision goggles. 16

Figure 4: CH-146 pilot demonstrating posture adopted to check instrument panel while wearing night vision goggles. 16

Figure 5: Routine duties performed during the course of a flight for Ch-146 griffon flight engineer illustrating cervical postures. 22

Figure 6: The standard CH-146 cock pit seat used during laboratory assessment protocol procedures in the four cervical testing directions during pre- and post-test assessments. 42

Figure 7: Exercises comprising the endurance training program top left posture adopted for right and left lateral flexion, top right extension, bottom flexion contraction. 48

Figure 8: Used in the coordination training program: top left deep neck flexor head nod, top right deep neck flexor head nod with head lift, middle seated return to neutral, bottom left return to neutral in four point kneeling, bottom right external shoulder rotation. 51

Figure 9: Summary statistics for isometric MVC (lbs) measured pre- and post-training intervention for the ETP, CTP and NTC. 55

Figure 10: Mean force outputs (lbs) for a 5 second seated maximal voluntary isometric contraction in the four tested directions pre- and post-intervention for the ETP, CTP and NTC groups. 59

Figure 11: Mean isometric MVC (lbs) differences for the ETP, CTP and the NTC groups pre- and post-treatment intervention 61

Figure 12: Mean times (seconds) produced during a fatigue trial conducted for the four tested directions using a resistance load of 70% an individual's MVC pre- and post-treatment for the ETP, CTP and NTC. 66

Figure 13: Area under the 70% endurance force curve (lbs/sec) for the pre- and post-test for flexion (flx), extension (ext), left lateral flexion (lt flx) and right lateral flexion (rt flx) in the ETP, CTP, and NTC groups 69

Figure 14: The mean areas under the 70% endurance force curve (lbs/sec) for the four tested directions in the ETP, CTP, and NTC groups during pre- and post-test assessments 72

Figure 15: Mean pre- and post-test scores for self-reported neck pain on visual analog scale for 'pain at present', 'pain in general' and 'pain at worst' for the ETP, CTP and NTC 76

Figure 16: Mean self-reported VAS neck pain scores (mm) for 'pain in general', 'pain at present' and 'pain at worst' pre- and post-intervention for the ETP, CTP and NTC groups 77

# 1. Review of the Literature

## *1.1 Neck Pain and its Prevalence*

Non-specific neck pain has been defined as pain perceived as originating in the region bounded superiorly by the superior nuchal line and inferiorly by an invisible transverse line traveling through the spinous process of the first thoracic vertebra (Bogduk, 2003). In most instances, the origin and precise pathophysiological mechanism(s) of chronic neck pain remains obscure. Most research is indicative of multifactorial origins, including external psychosocial and physical loading factors; as well as the psychological and biological characteristics of the particular individual (Bogduck, 2003; Bonfort et al., 2001 Nikander et al., 2006; Oldervoll, Rø, Zwart, & Svebak, 2001; Ylinen et al., 2003; Ylinen et al., 2006). Some plausible causative factors have been identified, such as muscle degeneration and/or impaired neuromuscular function resulting from chronic overuse, which is frequently accompanied by symptoms of pain, muscular weakness and fatigue (Conley, Stone, Nimmons, & Dudley, 1997b). Degenerative changes in the cervical vertebrae and disks, and nerve impingement may also lead to the expression of neck pain (Ylinen et al., 2003).

At any given time, 10-20% of the general population are experiencing neck pain (Holmstrange of motion, Lindell, & Moritz, 1992). The prevalence of neck pain in society has been previously reported to be about 67% (Côté, Cassidy, & Carroll, 1998). Previous research conducted using CF CH-146 military helicopter pilots found that 81% of aircrew had reported experiencing neck pain (Adam, 2004). This inflated value occurs despite the higher level of physical fitness exhibited by this population ($n$ = 40 CH-146 helicopter pilots and flight engineer), who recorded predicted $VO_2$Max scores of 40.9 ± 5.1 $ml \cdot kg^{-1} \cdot min^{-1}$ (Harrison et al, *in press-b*). The mean recorded $VO_2$Max score for this sample falls within the normative standard range of 40-42 ml/kg/min which defines this

population's fitness level as *Good* (The Physical Fitness Specialist Certification Manual, The Cooper Institute for Aerobic Research, Dallas TX, revised 1997).

### *1.2 Critical Load of the Cervical Spine*

The cervical region of the human spine has evolved to perform three basic functions: carry large loads, allow the head to move in multiple directions, and protect the nerves located within the spinal canal while performing the other two functions. To accomplish these tasks, the cervical spine must be mechanically stable in not only static but also dynamic movements. This stability is comprised "of the inherent passive stability of spinal column and the highly developed active stability provided by the surrounding muscles" (Panjabi et al., 1998, p. 11). An example which clarifies the roles of the osteoligamentous system and cervical musculature in providing stability to the cervical spine is as follows. The head of an adult is approximately equivalent to 7% of their body mass (Panjabi et al., 1998). Previous work by Harrison et al (*in press*-b) with CH-146 aircrew reported a mean mass of 82.7 kg for this sample population, which is equivalent to 811 N. Given the mass of the head is 7% of the body weight, the cervical spine would be required to support a load of 56.8 N. In addition to the mass of the head, a helicopter pilot is also required to wear a helmet and during night flights night vision goggles (NVG) leading to a combined load of 3.68 kg (Wierstra, 2001), equivalent to 36.1 N. In a low visibility situation where NVG are required, the helicopter pilot's cervical spine must support a total load of 92.9N. In an *in vitro* experimental study including only the osteoligamentous system, Panjabi et al. (1998) determined the critical load of an adult human cervical spine, to be 10.5N; any increase in mass beyond this load and the curvature of cervical spine would collapse. Given the above example, the cervical musculature of an 82.7 kg helicopter pilot would be responsible for supporting 82.4 N or 88.7% of the load placed on the neck. This highlights the importance of the cervical

muscle in maintaining the stability of the cervical spine in these aircrew members, and the problems that will result if weakness or pain compromises the ability of the muscle system to develop contractions.

### *1.3 The Cervical Spine*

The cervical vertebrae, identified as C1 – C7, are the smallest and lightest vertebrae of the spine (Marieb & Hoehn, 2007). The first two cervical vertebrae, the atlas and axis, are unique in that they have no intervertebral disc separating them, and are highly modified, reflecting their specialized function. The remaining cervical vertebrae C3 through to C7 are known as *typical* cervical vertebrae and have: an oval body, a *bifid* spinous process that projects directly backwards with the exception of C7, a vertebral foramen that is large and transverse processes that contains the transverse foramen. Due to the unique characteristics of the cervical vertebrae, the osseoligamentous system, and the surrounding musculature, the cervical region has the greatest range of motion. This range consists of flexion, extension, rotation, and lateral flexion movements with range of motions of approximately 145° in flexion/extension and 180° in rotation (Marieb & Hoehn, 2007). Most of the rotation permitted by the cervical vertebrae occurs along the C1-C2 joint, while C3-C7 joints are primarily responsible for flexion and extension (Forde, 2008). The increased mobility of the cervical vertebrae results in decreased stability of this section of spine, leading to an increased potential for injury relative to the other regions of the spine (Marieb & Hoehn, 2007).

### *1.4 Cervical Musculoskeletal System.*

The musculoskeletal system of the cervical spine is one of the most complex and dynamic systems in the human body. The cervical muscles are required to demonstrate marked morphological diversity to enable and permit the control of the wide variety of

head movements. Panjabi et al., (1998) found the osseoligamentous system contributed to 20% of the overall stability while the remaining 80% was provided by the surrounding neck musculature. In the osseoligamentous system, the ligaments' role in stabilization is provided mainly at the end of the range of motion (Harms-Ringdahl, Ekholm, Schuldt, Nemeth & Arborelius, 1986) while the muscles function to "supply dynamic support in activities around the neutral and mid-range postures, which are commonly adopted during functional daily tasks" (Falla, 2004, p. 125). Through the use of non-invasive magnetic resonance imaging (MRI), Conley, Meyer, Bloomberg, Feeback and Dudley (1995) identified the muscles and muscle pairs of the cervical spine responsible for specific movements. They quantified shifts in signal relaxation times of T2-weighted magnetic resonance images to identify the muscles or muscle pairs extensively used to perform head movements. They were:

1. Extension- Semispinalis capitis, Semispinalis cervicis, Levator scapulae, Longissimus capitis, Longissimus cervicis, Scalenus medius and anterior, Splenius capitis, and Multifidus muscles
2. Flexion- Sternocleidomastoid, Longus capitis, Longus colli
3. Rotation- Splenius capitis, Levator scapulae, Scalenus, Semispinalis capitis ipsilateral to the rotation and Sternocleidomastoid contralateral to the rotation
4. Lateral Flexion- Sternocleidomastoid (SCM)

Based on their moment-generating capacity, the muscles of the cervical spine can be classified as either segmental stabilizers or prime movers. The musculature surrounding the cervical vertebrae of the spinal column are classified as segmental stabilizers, whose role is to provide the postural support for cervical lordosis and the cervical joints (Winters & Peles, 1990). The deep cervical flexors are the longus colli, longus capitis, rectus capitis anterior and rectus capitis lateralis (Falla, 2004). The deep cervical extensors are the semispinalis capitis, semispinalis cervicis, longissimus cervicis and the posterior

posterior suboccipitals (Kennedy, 2007). Current research has indentified pronounced weakness in these deep segmental muscles of patients with chronic neck pain, in particular the deep cervical flexors (Falla, Jull, & Hodges, 2004). The superficial and intermediate layers of the cervical muscles are known as prime movers, whose primary role is movement of the head and cervical spine (Kennedy, 2007). Fall, Rainoldi, Merletti, and Jull (2003-b) identified the SCM and anterior scalene muscles as the prime movers of cervical flexion. Based on MRI technology the splenius capitis, semispinalis capitis, semispinalis cervicis and multifidus muscles are the primary head extensors (Conley et al., 1997a). However, Marieb and Hoehn (2007) also identified the upper fibres of the trapezius as a prime mover responsible for head extension, when the scapulae were fixed.

### *1.5 Hypotheses for the Origins of Neck Pain*

The origins of neck pain are likely multifactorial (Falla, 2004; Sarig-Bahat, 2003; Visser & Dieën; Ylinen et al., 2006). Many hypotheses have been proposed. Visser and Dieën (2006) reviewed the proposed pathophysiological mechanism(s) for the development of neck pain; seven of which are highlighted:

*1.51 Cinderella hypothesis.* The *Cinderella Hypothesis* is viewed as the most influential in the development of muscle damage due to the performance of low-intensity tasks (Hägg, 2000). Within skeletal muscle there exists a heterogeneous population of muscle fibres with various histochemical and biological characteristics. The three major fibres types have been defined based on these individual differences are Type I, Type IIa and Type IIx fibres. Type I fibres also known as slow-twitch fibres contain a large number of oxidative enzymes, are surrounded by more capillaries than any other fibre, have a large capacity for aerobic metabolism and are highly resistant to fatigue. Type IIx or fast-glycolytic fibres have a small number of mitochondria, limited capacity for aerobic metabolism, are rich in glycolytic enzymes, and rely mainly on anaerobic

metabolism. Type IIa are also known as intermediate or fast-oxidative glycolytic fibres whose biochemical and fatigue properties range between that of Type IIx and Type I. Skeletal muscle fibres are capable of a wide range of contractile and biochemical properties, based on this fibre types I, IIa, IIx represent a continuum, rather than discrete and distinct types of fibres (Staron, 1997; Powers & Howley, 2004).

Sub-maximal muscular force engages only a fraction of the motor-units available for a contraction, the recruitment patterns of these motor-units is thought to be stereotypical. Thus, the continuous activation of the same motor-units during low-intensity sustained tasks is hypothesized to be the precursor for future damage to these motor-units (Henneman, Somjen, & Carpenter, 1965). Henneman et al. (1965) proposed a *size principle* which suggested that small Type I fibres were continuously activated during sustained low-intensity contractions. This provides a possible explanation for the predominance of damage to these small Type I fibres evident in muscle biopsy experiments of the trapezius (Forsman, Kadefors, Zhand, Birch, & Palmerud, 1999; Kadefors, Frosman, Zoega, & Herberts, 1999; Thorn, Frosman. Zhang, & Taoda, 2002; Westgaard & DeLuca, 1999; Zennaro, Laubli, Krebs, Klipstein, & Krueger, 2003). Although the *Cinderella hypothesis* is a plausible explanation for selective loading of Type I fibres, it does not explain how the actual development of muscle fibre damage occurs (Visser & Dieën, 2006).

*1.52 Muscle damage due to low-intensity loading.* Experimental animal research has established concrete evidence that low-intensity loading can result in muscle damage, provided that the load imposed is static and of long duration (Barbe, Barr, Gorzelany, Amin, Gaughan, & Safadi, 2003; Lexell, Jarivs, Downham, & Salmons, 1993).

*1.53 $Ca^{2+}$ accumulation.* Gissel (2000) suggested that $Ca^{2+}$ accumulation in response to sustained MU activity may play a causative role in the development of muscle disorders. Furthermore, $Ca^{2+}$ accumulation may lead to mitochondrial $Ca^{2+}$ reabsorption,

which is hypothesized to result in structural damage and energy depletion in the mitochondria (Visser & Dieën, 2006).

*1.54 Ischemia.* Decreased blood flow leading to the reduction of muscle tissue oxygenation during sustained repetitive work has been hypothesized to lead to the development of neck pain (Carayon, Smith, & Haims, 1999; Galen, Müller, Meulenbroek, & Gemmert, 2002). This hypothesis states that blood flow is compromised due to compression on the brachial artery in head and shoulder forward posture (Keller, Corbertt, & Nichols, 1998). A more widely supported variant of this hypothesis is that increased intramuscular pressure impedes microcirculation (Järvholm, Palmerud, Karlsson, Herberts, & Kadefors, 1991; Jensen et al., 1995). Intramuscular pressure is the product of the muscular force generated during a muscular contraction and depends on the shape and location of the contracting muscle (Sejersted, Hargens, Kardel, Blom, Jensen, & Hermansen, 1984). Low-intensity work often coincides with a relatively low level of intramuscular pressure (Järvholm et al., 1991). However, the local intramuscular pressure maybe much higher in parts of the muscle where motor-units are active relative to the overall intramuscular pressure of the muscle (Sjøgaard, Kiens, Jorgensen, & Saltin, 1986). This maybe the case when Type I motor-units are spatially clustered in muscle compartments (Zuylen, Gielen, & Van der Gon, 1988) as seen in the trapezius muscle (Hermans & Spaepen, 1965; Mathiassen & Winkel, 1990).

*1.55 Blood vessel-nociceptor interaction.* Knardahl (2002) proposed a theory on the origins of pain without muscle cell contraction being the primary mechanism. Rather this origin of pain was based on the locations of the free nerve endings and nociceptive afferent nerves found in the connective tissue. These nerve fibres are found in close proximity to the arteries and arterioles feeding the muscles of interest. Knardahl proposed three possible options for interactions:

> (a) Arterial vasodilatation stretches the blood vessel wall, producing mechanical activation, (b) arterial vasodilatation causes vascular production and release of pain producing substances, like bradykinin and nitric oxide, which can activate nociceptors, (c) arterial vasodilatation causes release of inflammatory mediators, such as histamine and algogenic factors from the plasma that may activate or sensitize nociceptors (Visser & Dieën, 2006, p. 7).

This hypothesis was not supported by Pritchard, Pugh, Wright and Brownlee (1999), who found that local micro-circulation decreased around muscle fibres in the trapezius in individuals suffering from chronic pain.

*1.56 Intramuscular shear forces.* Vøllestad and Roe (2003) purposed that low-intensity static contractions lead to the development of higher shear loading than high-intensity dynamic contractions. They hypothesized that only a small number of muscle fibres are needed to contract within a muscle when a low-intensity force production is required. The shortening and lengthening of these motor-units are not synchronized, contributing to the movement of some fibres relative other non-contracting fibres. The nociceptors located between these muscle fibres are undergoing repeated exposure to shear stresses during these contractions, leading to the sensation of pain. Currently there is no scientific evidence supporting this hypothesis (Visser & Dieën, 2006).

*1.57 Nitric oxide/oxygen ratio hypothesis.* The nitric oxide/oxygen ($NO/O_2$) ratio hypothesis incorporates many of the above hypotheses to explain the development of neck pain. Pain is hypothesized to occur when low-intensity muscle contractions in the trapezius are combined with a decrease in capillary blood flow due to high sympathetic nerve activity (Holte & Westgaard, 2002). Sympathetic vasoconstriction is maintained as long as muscle activity is below 20-30% of maximal voluntary contraction (MVC) (Hansen, 2002). Other evidence presented links increased sympathetic discharge with the occurrence of psychological distress, which occurs in situations of high quantitative work

demands, low social support and prolonged head down neck flexion (Ray, Hume, & Shortt, 1997; Shortt & Ray, 1997). These low level contractions and head down flexion are prominent risk factors for helicopter pilots.

The occurrence of low-intensity muscle contractions increases the intracellular concentration of $Ca^{2+}$ released from the sarcoplasmic reticulum. The release of $Ca^{2+}$ stimulates the production of nitric oxide (NO) from amino acid L-arginine by a family of enzymes termed NOS in the muscle cell (Alderton, Cooper, & Knowles, 2001). NO is a highly diffusible, gaseous messenger molecule that plays a vital role in many biological functions but mainly in local vasoregulation (NO is a vasodilator), neurotransmission, and immunological responses (Moncada & Erusalimsky, 2002). Reduced capillary blood flow due to sympathetic vasoconstriction inhibits destruction of NO by molecules such as myoglobin (Beckman & Koppenol, 1996). A fall in intracellular oxygen concentration coincides with decreased blood flow. These events combine to contribute to the rise in the ratio of $NO/O_2$ in the muscle cells, facilitating the reversible and competitive inhibition of cytochrome oxidase (CytOx), the terminal enzyme in the mitochondrial electron transport chain, by NO (Moncada & Erusalimsky, 2002).

When CytOx is inhibited, its capacity to generate adenosine triphosphate (ATP) is reduced. At the same time, the muscle is performing low intensity contractions which further deplete ATP stores. With a drop in the availability of ATP, the muscle relies on anaerobic glycolysis to generate ATP, which results in the production of lactic acid (Juel, 2001; Spriet, Howlett, & Heigenhauser, 2000). The efflux of lactic acid from the cell activates proton-sensitive nociceptive fibres, resulting in myalgic pain (Immke & McCleskey, 2001; Issberner, Reeh, & Steen, 1996; Sutherland, Cook, & McCleskey, 2000). With sustained inhibition of CytOx by NO for a periods lasting hours, peroxynitrile may be generated. Peroxynitrile is capable of irreversible inactivation of some enzymes in the electron transport chain. If this is repeated over an extended period

such as months or years, the production of ragged red fibres also known as *moth-eaten* fibres that lack the enzymatic capacity for cellular aerobic respiration can result (Brown & Borutaite, 2002; Moncada, Erusalimsky, 2002; Radi, Cassina, Hodara, Quijano, & Castro, 2002).

### *1.6 Differential Mechanisms for Development of Neck Pain in Aircrew*

Neck pain is a significant aeromedical problem facing all aircrews and it has the capacity to interfere with a pilot's flying abilities, concentration, safety, overall health status and leisure time activity (Alricsson, Harms-Ringdahl, Larsson, Linder, & Werner, 2004; Äng, Linder, Harms-Ringdahl, 2005; Hämäläinen, 1999). There are two distinct sub-groups of neck pain related to flight: acute and chronic neck pain. Acute neck pain occurs during or shortly after a flight in response to isolated events such as a high gravitational (+Gz) turn with an unsupported neck postures. Chronic neck pain is defined as occurring over prolonged periods and is associated with a loss of function or in some cases premature degenerative changes in the cervical spine (Green & Brown, 2004). Below, I discuss the differences between the occurrences of neck pain in helicopter aircrew versus fast jet aircrew.

There exists a large body of literature examining the role of neck pain in fighter pilots (Alricsson, et al., 2004; Burnett, Naumann, & Burton, 2004; Green, & Brown, 2004; Hämäläinen, Heinijoki, & Vanharanata, 1998; Hämäläinen, 1999; Jones et al., 2000; Naumann, Grant, & Dhaliwal, 2004; Sovelius et al., 2006). The stimulus for this research is in response to frequent reports of discomfort in the cervical region resulting from deviated head postures, unpreparedness for high +Gz manoeuvres, and/or repeated exposure to large +Gz (over +4Gz) forces (Burnett et al., 2004; Hämäläinen, Vanharanata, & Bloigu, 1993). For fighter pilots, there has been a failure to identify predictive factors relating to the development of flight related neck pain (Wickes &

Greeves, 2005). Butler (2000) found that neck pain correlated positively with the fighter pilots age, flight hours, +Gz level and onset rate, duration of exposure, muscle strength, number of repeated exposures, and duration of rest between exposures. This data supports the acute neck pain onset hypothesis, where predictability of neck pain occurrence when anthropometric, demographic or lifestyle factors is limited (Wickes & Greeves, 2005).

In contrast to acute neck pain development in fighter pilots, neck pain in rotary winged aircrew is typically defined as more chronic. The chronic nature of the neck pain experienced by helicopter pilots and flight engineers has been related to suboptimal postures adopted during flight, use of NVG, vibration, and maintenance of a low level contractions for extended periods (Adam, 2004, Äng et al., 2005, Forde, 2008; Harrison et al., 2007; Thuresson, Äng, Linder, & Harms-Ringdahl, 2003; Thuresson, Äng, Linder, & Harms-Ringdahl, 2005; Wickes & Greeves, 2005). While in flight, helicopter pilots experience smaller +Gz forces but are exposed to greater vibration frequencies and magnitudes compared to fighter pilots (Chen et al., 2007). These vibrations frequencies are among the highest of all occupations, and have been linked to the premature onset of some spinal abnormalities (Butler, 2000). To determine the existence of predictive factors relating to neck pain in CF rotary winged aircrew, research was conducted examining fitness, fight history, and cervical isometric strength and endurance of CH-146 helicopter pilots and flight engineers. The data were analyzed using logistic regression to derive a predictive equation. Height and longest single NVG mission recorded in the respondent's logbook were the only variables that predicted which individuals would suffer from neck pain (Harrison et al., *in press-c*).

### *1.7 Factors Contributing to Flight Related Neck Pain in Helicopter Aircrew*

Recent technological advances in airframes, composite materials, propulsion systems, flight controls and avionics allow helicopters and jets to travel faster, longer and higher (Seng, Lam & Lee, 2003). Innovations have also extended to other technologies

such as flight helmets, head display units (HDU) and NVG, which are critical for operational success (Adam, 2004). The flight helmet was designed initially as protective equipment but as technology improves its role has evolved to include a mounting platform for devices such as HDU and NVG (Sovelius et al., 2006).

*1.71 Night vision goggle usage.* Equipment affixed to the helmet is becoming more prevalent in helicopter pilots and flight engineers to allow flights in low visibility conditions (i.e. during cloudy conditions or under the cover of darkness; Adam, 2004). NVG use near-infrared light to enhance ambient light reflected off objects within the field of view (refer to *Figures 1* and *2*). One drawback to NVG use is a reduction in the user's peripheral vision. The AN/AVS-9 model used by CH-146 aircrew has a 40° field of view (Task & Filpovich, 1997). Forde (2008) measured postures adopted while wearing NVG during a simulated night flights. She found that subjects spent increased time in mild (74%) and severe (10%) cervical flexed postures when compared to day flights without NVG. Increased time spent in mild axial twist postures was also identified in CH-146 helicopter pilots when comparing simulated day (35%) to night (63%) flights. These findings indicate that pilots spend a higher percentage of their work cycle in high risk postures (severe cervical flexion and axial twist postures) resulting in higher peak and cumulative loads placed upon the cervical spine, especially during night flights (Forde, 2008).

Use of head worn equipment increases the mechanical load placed on the cervical vertebrae and the skeletal musculature supporting the neck. Increased load alters the center of mass of the head forward and upward with respect to the motion axis of the cervical spine (Harms-Ringdahl, Linder, Spångberg, & Burton, 1999). This additional

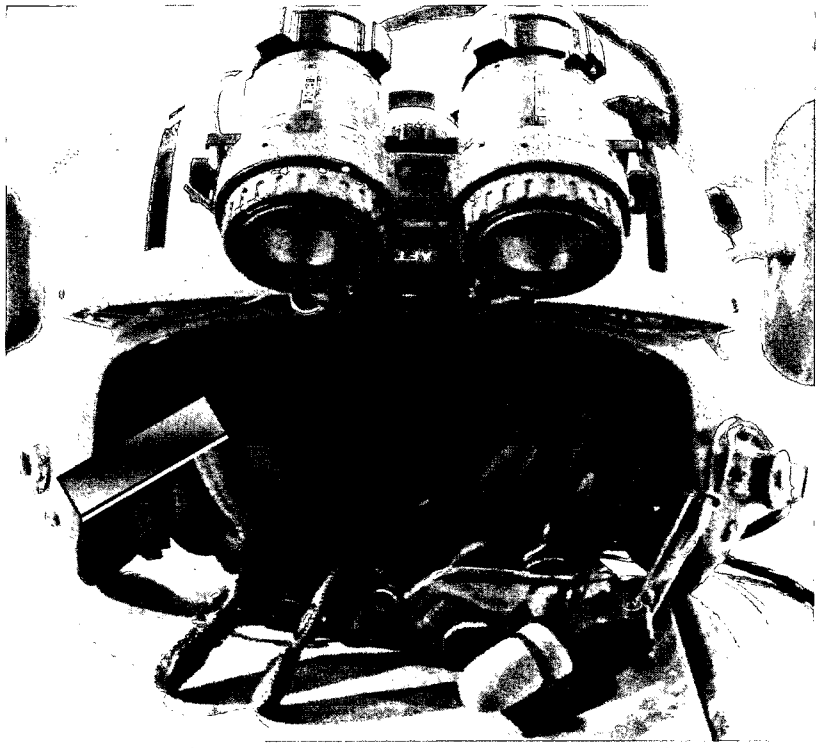


*Figure 1:* Front view of the Canadian Forces standard issue CH-146 flight helmet with night vision goggles

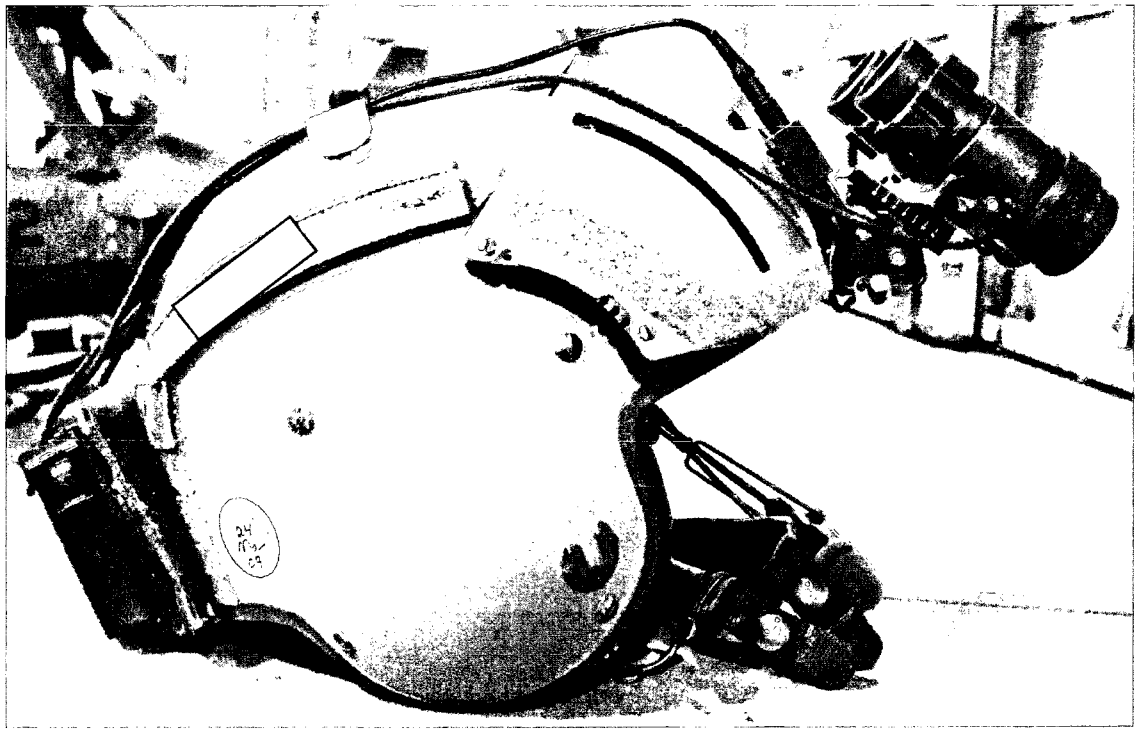


*Figure 2:* Lateral view of the Canadian Forces standard issue CH-146 flight helmet with night vision goggles

flexion moment on the neck increases the work required from the posterior cervical extensors to control movement and head posture (Sovelius et al., 2008). Any additional equipment mounted on the helmet, except on the normal center of mass for the head, places added load on the muscles and the cervical vertebrae (Forde, 2008). The impact of impulsive (0-4) +Gz forces (trampoline) on a neutral spine generated using a trampoline, increased muscle strain in SCM, trapezius and the cervical erector spinae in response to the helmet mass alone. The addition of NVG to the helmet significantly increased muscle strain in the SCM, and cervical erector spinae (Sovelius et al., 2008).

Using myoelectric manifestations as an index of fatigue found that NVG induced an increase in cervical muscle strain that was greatest for the muscles that were already specifically loaded (Sovelius et al., 2008). The increased frontal load induced by NVG was not evenly distributed across the cervical musculature. It appeared to affect those muscles that were already subjected to the highest loading, as seen in the SCM during back bouncing and the cervical extensors during hand and knee bouncing on a trampoline (Sovelius et al., 2008). In an attempt to quantify a dose-response relationship, Jonsson (1982) calculated a muscular strength utilization ratio (MUR). MUR is the ratio between the load moment produced by the mass of the neck, head, helmet, equipment worn on the helmet and the maximum cervical strength multiplied by 100. If the a load was to be maintained for a period longer than an hour an upper limit of 5% of the MUR was set. In a laboratory environment examining the EMG activity of upper and lower neck musculature induced by different head worn equipment and neck postures, Thuresson et al. (2005) found in a neutral posture that % MUR exceeded the 5% recommendations with any head born equipment, with the exception of the helmet only condition. In a cervical flexed position wearing a helmet and NVG, a MUR of 16% was recorded, which is well exceeds the recommended level.

Harrison et al. (2007) assessed the metabolic stress placed on the cervical spine as a result of NVG use with near infrared spectroscopy (NIRS). An increased blood volume and oxygenation response was recorded in the trapezius when NVG simulated flight was compared to a helmet only condition. The use of NVG induced sufficient metabolic stress on the traps that increased blood volume was required to meet the oxygen demands on the contracting muscles. This shows that the use of NVG during night flights places increased metabolic demands on the trapezius muscle.

*1.72 Posture.* Research has established that helmet mass increases muscle strain (Sovelius, Oksa, Rintala, Huhtala, & Sitonen, 2008) and metabolic responses (Harrison et al., 2007) experienced by cervical musculature; however, the role posture and head movements play in muscle strain development maybe even more significant. Laboratory based EMG analysis revealed that neck and body position had a substantially greater effect on neck muscular activity than the mechanical load of head worn equipment (Thuresson et al., 2003; Thuresson et al., 2005). The ergonomic design of the helicopter cockpit, forces pilots to adopt a forward flexed position of the trunk and cervical spine (refer to *Figures 3 and 4*). The major portion of this forward flexion occurs in the lumbar and thoracic spine and has been cited as a plausible cause of low back pain (Pelham, White, Holt, & Lee, 2005). Lopez-Lopez et al. (2001) described the posture adopted by helicopter pilots as being slightly forward and rotated to the left. Adoption of this kyphotic lumbar and thoracic posture (convexly curved) forces the lower cervical spine into flexion and the upper cervical spine into extension. In this position, the upper cervical spine musculature is continuously activated, which over extended periods can be fatiguing (Adam, 2004). This constant state of muscular contraction could potentially lead to damage and anterior compression of the intervertebral discs, which with repeated exposures would contribute to the development of neck pain (Pelham et al., 2005).



*Figure 3:* Pilot seated in CH-146 helicopter while wearing night vision goggles



*Figure 4:* CH-146 pilot demonstrating posture used to check instrument panel while wearing night vision goggles

The vertebral column is composed of four curvatures which generates an "S" or sinusoid shape. The cervical and lumbar curvatures are posteriorly concave while the thoracic and sacral curvatures are posteriorly convexed. These curvatures function to increase the resilience and flexibility of the spine, allowing it to act as a spring rather than a rigid rod (Marieb & Hoehn, 2007). Any deviations in the sinusoidal curvature resulting from improper posture as seen during head forward flexion, places additional strain upon the spine (Pelham et al., 2005). Pelham et al. (2005) hypothesized that microtrauma occurs after prolonged and repeated exposure of the spine to these mechanical stressors. This accumulation of microtrauma over time may eventually lead to the development of macrotrauma, which can lead to the occurrence of dysfunction and pain. Restraint of the torso and shoulders by the 4-point safety harness, standard issue for helicopter pilots, also functions to restrict movement of the thoracic and lumbar regions of the spine. This restriction forces a greater proportion of required movement be accommodated for by the upper thoracic and cervical vertebrae. This accommodation places considerable stress on the neck during excessive movements such as the *check 6 position* (i.e. rotation of the cervical vertebrae in looking towards the rear of the aircraft) and is exacerbated by the addition of NVG and other head mounted equipment (Wickes and Greeves, 2005). The results of posture matching based on video records of CH-146 helicopter pilots during simulated night and day flights concluded that "pilots may be at a increased risk of developing neck strain during NVG flights since the majority of their time is spent in flexed postures greater than 15 degrees" (Forde, 2008, p. 50).

To identify specific stresses placed on the spine due to cockpit design, research using EMG (to examine the neuromuscular component) and NIRS (to examine the metabolic component) has been conducted to quantify the loads placed on the bilateral musculature. Dieën (1996) investigated static posture with trunk rotation of 15°, 30°, and 45°, with the development of torque equivalent to 20%, 40%, 60% and 100% of MVC.

Dieën found more activity contralateral to the side of rotation starting at 20% of MVC. Based on this study one would expect to see higher levels of contraction of the right side of the spine. This was supported by the study conducted by Lopez-Lopez et al. (2001) who used surface EMG to measure the back muscle activity of helicopter pilots during flight. Lopez-Lopez et al. found a higher prevalence of muscular activity on the right side, concluding that the posture adopted during flight imposed a higher metabolic demand on the right than the left side. In contrast de Oliveira, Simpson, and Nadal (2001), found higher albeit not significant EMG activity on the left rather than the right during a short flight. Data from longer flights (approximately 2 hrs) found no differences between EMG signals between the left and right sides (de Oliveria & Nadal, 2004).

Harrison et al., (2007) found differences between the left and right trapezius for total haemoglobin (tHb), oxygenated haemoglobin (HbO2), and deoxygentated haemoglobin (HHb) in 33 helicopter pilots using NIRS during a simulated flight in a CH-146 Griffon helicopter. Higher values of tHb and HbO2 occurred on the right side suggesting an increased response to meet increased metabolic demands. This further supports the idea of a higher level of contraction on the right relative to the left side.

*1.73 Whole body vibration.* Whole-body vibration (WBV) is vibration transmitted to the entire body from a vibrating surface such as a seat in a vehicle such as a truck, bus or aircraft (Maikala & Bhambhani, 2004). While in the seated position "human exposure to vertical whole-body vibration has consistently shown the principal resonance frequency of the upper body to be in the vicinity of 5 Hz" (de Oliveira & Nadal, 2005, p. 1). Chen, Wickramasinghe, and Zimcik (2007) measured the vibration spectrum of the CH-146 Griffon helicopter. They found that the principle harmonic frequency was approximately 5Hz. During whole body vibration, maximum energy transfer to the human spine occurs within the range of 4.5-5Hz (Maikala & Bhambhani, 2004). Given the vibration frequency of the CH-146 aircraft, individuals who fly it are likely

experiencing maximum energy transfer in their spine. Chen et al. (2007) evaluated the displacement of the helmeted head of a CH-146 pilot exposed to a 5Hz vibration frequency. They found that the pilot's head underwent Z-axis displacements twice the magnitude of the Z-axis displacement occurring on the floor of the aircraft. It has also been demonstrated that vibration transmission from the buttocks to the head in a seated posture increased in an exponentially (Wilder Woodworth, Frymoyer, & Pope, 1982). Therefore, the vibration frequency experienced by the neck is likely higher than that experienced by the lower back. Also of primary significance during whole body vibrations is the posture assumed by pilots, described as flexed and rotated to the left; this awkward body position combined with seated vibrations poses the greatest risk for the development of back problems (Kitazaki & Griffin, 1995).

Bonger et al., (1990) found vibrations experienced by a helicopter pilot to be a prominent cause of low back pain. Development of localized fatigue due to a variety of postures and vibrations frequencies was consistent with EMG data to assess the effect of WBV on the paraspinal muscles in the low back (Hansson, Magnusson, & Brange of motionan, 1991; Pope, Magnusson, & Wilder, 1998; Wilder et al., 1982). Maikala and Bhambhani (2004) found WBV induced a *localized vasoconstriction stimulus* in the lumbar regions of the erector spinae muscles. A reduction in blood flow resulted in lowered blood volume based on greater deoxygenation in the area determined by NIRS. In contrast, Pope, Wilder and Donnermijer (1986) found that posture, not vibration, played the most significant role in the development of back muscle fatigue. This hypothesis was supported by research conducted by de Oliveria and Nadal (2004) who failed to isolate a change in muscle activity due to vibration. However, more data in both of these areas are needed to elucidate the contribution of vibration and posture.

Another factor that may promote the development of neck pain in helicopter aircrew is the application of additional mass on the helmet. This has become a standard

practice for the CF during night missions where all helicopter aircrew are required to use NVG. Wikström, Kjellberg and Landström (1994) identified an association between increased neck and shoulder muscle load and a tendency for increased problems in those areas after exposure to whole body vibration. An alternative hypothesis is that the effects of vibration lead to disturbances in the diffusion of nutrients to the vertebral disk, resulting in accelerated degeneration rather than the development of muscular fatigue (Dupuis & Zerlet, 1987).

### *1.8 Canadian Forces CH 146 Helicopter Pilots and Flight Engineers*

Adam (2004) administered a NVG-induced neck strain questionnaire to identify the percentage of flight related neck pain in this population. Flight related neck pain was reported in 81.2% of 138 CH-146 Griffon pilots. Of those who reported neck pain during flight, nearly 40% had experienced more than 10 episodes, meaning that the pain was more or less chronic. Of those reporting neck pains during flight, over 15% felt that their worst episode was severe enough during flight to be incapacitating. This was defined as "pain that rendered the respondent unable to perform normal duties" (Adam, 2004, p. 39). Approximately 45% of pilots reported the occurrence of neck pain more than 10 times after a flight had occurred. Close to 50% of these pilots reported their worst episode of neck pain lasted in excess of a full day and in many cases up to four days (Adam, 2004). Given these statistics, the impact that neck pain is having on this population of helicopter pilots is large enough to be having a major impact on the performance of their occupation duties as well as their quality of life.

To better understand the development of neck pain, Griffon CH-146 helicopter pilots were asked to list the factors they perceived as being causal. The most common response reported was forward flexion of the cervical spine to view or use the control display unit, center console or a map. The effect of this posture was reported to be

exacerbated by the use of NVG; however, not all responses were clear on the importance of this as an issue. The second most commonly reported factor was extended flying hours in a 24-hour period. The third risk factor was poor posture required during flying due to cockpit ergonomics and task requirements (Adam, 2004).

Within the CH-146 Griffon flight engineers, 84.5% reported neck pain related to flying. At least 50% reported experiencing more than 10 episodes, indicating their pain was chronic or on its way to becoming so. The most disturbing figure was that almost 100% reported neck pain after flight. Compared to the pilots, neck pain experienced after flight by flight engineers was severe enough to be considered incapacitating in 25% of cases, which is slightly higher than their pilot counterparts (Adam, 2004).

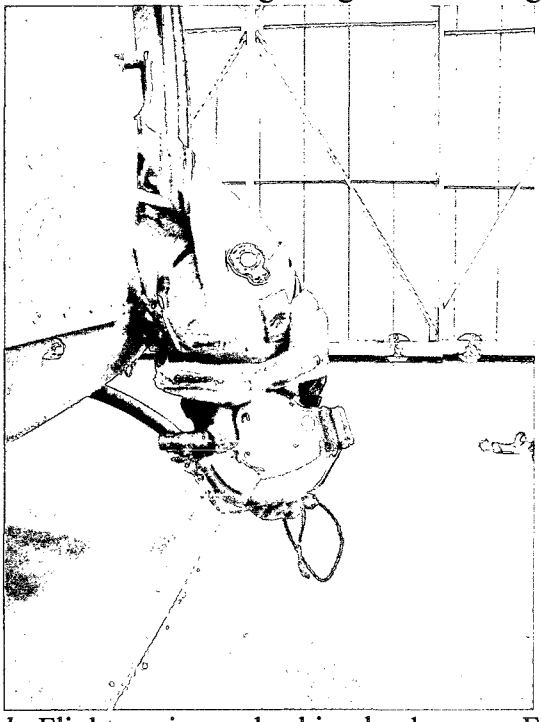
Griffon flight engineers reported that low and moderate +Gz flights with NVG were the leading factor in the development of neck pain. The most frequently self-reported comments for activities that aggravated neck pain for this aircrew were operations that involved positioning the head out of the side door and performing duties such as slopes, clearance, hoisting or slinging. These duties require flight engineers to flex, extend and rotate the cervical and thoracic spine in unsupported positions (Weirstra, 2001), placing substantial stress upon these structures (see *Figure 5*). Substantially greater loads are placed on the hard and soft tissues of these structures with the additional weight of a helmet, NVG and a counterweight (Adam, 2004). Given these statistics, neck pain is a major occupational issue for CF aircrew affecting their quality of life and requiring further investigation.

### *1.9 Use of Exercise Therapy in Pilot Populations*

A recent survey assessing neck pain in CH-146 Griffon helicopter pilots, found that less than half of these pilots sought treatment for their pain. The majority were uncomfortable with medical staff, and/or cited a fear of grounding as reasons for their



*a:* Flight engineer checking the tail rotor while wearing NVG



*b:* Flight engineer checking load



*c*: Flight engineer checking the main rotor

*Figure 5:* Routine duties performed during the course of a flight for CH-146 Griffon flight engineers, illustrating cervical postures.

hesitancy to seek medical attention (Adam, 2004). The prevalence of neck pain in this aircrew provides incentive for developing a suitable treatment option that would be used by the aircrew. Conservative interventions such as physiotherapy, manual therapy, massage, exercise therapy, and ergonomic interventions have been the focus of research involving female office workers. Despite the use of these therapies, systematic reviews and meta-analyses of the literature with regards to the effectiveness of these types of treatments indicate that most studies contain design flaws and the beneficial effects of these interventions have yet to be rigorously supported (Kay, Gross, Santiguida, Hoving, & Bonfort, 2007; Sarig-Bahat, 2003; Smidt, de Vet, Bouter & Dekker, 2005).

Exercise therapy is cited as the most commonly used treatment modalities for non-specific neck disorders. Exercise therapy is defined as "therapy [that] involves the prescription of muscular contraction and bodily movement ultimately to improve the overall function of the individual and help meet the demands of daily living" (Smidt et al., 2005, p. 71). For the occupational environments associated with flight, research examining the role of exercise therapy in the alleviation of neck pain has focused mainly on fighter pilots (Alricsson et al., 2004; Burnett et al., 2004; Hämäläinen et al., 1998; Hämäläinen, 1999; Jones et al., 2000; Sovelius et al., 2006). For rotary winged pilots, research examining neck pain is in its infancy despite its high prevalence. Based on previous fitness scores, flight experience, and the number of individuals reporting neck pain, Harrison et al. (*in press-b*) could not isolate differences between helicopter pilots and flight engineers despite the discrepancies in their occupational duties. Given the common link between the development of flight related neck pain in both CH-146 pilots and flight engineers, a cervical training program that addresses this common issue has the potential to beneficial for both.

### *1.10 Chronic Neck Pain and Presence of Strength and Weaknesses*

Numerous studies have illustrated reduced cervical strength and endurance capacity in the cervical extensor and flexor muscles in individuals with neck pain (Barton & Hayes, 1996; Placzek, et al., 1999; Treleaven, Jull, & Atkinson, 1994; Watson & Trott, 1993). Falla et al. (2003) assessed the fatigue of the superficial flexors, SCM and the anterior scalene (AS) muscles during sustained cervical flexion contractions at 25% and 50% of the MVC in patients with chronic neck pain. Greater myoelectric manifestations of muscle fatigue in the SCM and AS muscles in individuals with neck pain relative to the controls was supported by the greater slope in the EMG mean frequency (Falla et al., 2003). Gogia and Sabbahi (1994) similarly established greater cervical flexor muscle fatigability during low load sustained contractions (25 % MVC).

To examine muscle activation patterns, Falla, Jull, Dall'Alba, Rainoldi and Merletti (2003-a) developed an EMG technique capable of measuring the activation levels of the deep cervical flexors. This technique was used to determine whether activation levels varied during the performance of the cranio-cervical flexion (C-CF) test for those reporting neck pain compared to controls (Falla et al., 2004-b). Neck pain patients exhibited disturbances in the neck flexor synergy "where impairment in the deep muscles, important for segmental control and support, appeared to be compensated for by increased activity in the superficial muscles (SCM and AS)" (Fall, 2004, p. 128).

Given that increased EMG activity has been identified in the superficial cervical musculature of individuals with neck pain (Jull, 2000; Jull, Kristjansson, & Dull'Alba, 2004), the neuromuscular efficiency (NME) of the SCM and AS muscles must also be investigated. The NME was defined as the quotient of force and the integrated EMG during a cervical flexion contraction at 25% and 50% of MVC in individuals with and without neck pain (van der Hoeven, van Weerden, & Zwarts, 1993). These authors found less NME for the SCM and AS when contracting at 25% MVC in individuals with neck

pain. The reduced NME illustrates that individuals with neck pain required higher levels of muscular electrical activity to produce an equivalent force relative to the controls, or conversely with a comparable level of electrical muscular activity, neck pain patients generate a lower force output (Falla et al., 2004).

### *1.11 Neck Strength in Fighter Pilots and Helicopter Pilots with Neck Pain*

To understanding neck pain in military pilots, it is essential to investigate and quantify strength, myoelectric, and physiological characteristics of the muscles that support and stabilize the cervical spine. However, there is little literature that examines the strength characteristics of the cervical musculature in pilots who are suffering from neck pain or asymptomatic. Äng et al., (2005), assessed neck strength using myoelectric properties of the muscles as an index of fatigue (EMG) in 16 fighter and 15 helicopter pilots with neck pain. Neck extensor and flexor strength were measured using MVC and submaximal isometric contractions to fatigue. A decline in slope of EMG median frequency power spectra was used as an index of fatigue in the cervical muscles. Fighter pilots with pain had lower extensor MVC strength than controls but no difference was found in helicopter pilots. No differences were found for the flexor/extensor ratio or flexor strength in either group. However, helicopter pilots with neck pain presented with altered myoelectirc manifestations. They had unexpectedly flatter EMG-slopes for the flexors than the controls. Given these results, muscle fatigue and weakness is implicated as a risk factor in the development of neck pain (Äng et al., 2005). Harisson et al. (*in press-a*) used the % change in the median frequency of the EMG activity of the right and left splenius capitis, SCM, and upper trapezius as an indicator of fatigue. Rotary wing aircrew performed a 70% submaximal isometric contraction until fatigue. Fatigue occurred in the smallest agonist muscles in the four directions tested flexion, extension, left and right lateral flexion.

### *1.12 Research Relating Neck Pain and Exercise Training in the General Population*

Conley et al. (1997b) assigned 22 college students to either a control or two resistance groups (RESX and RES), both performed squats, deadlifts, push press, bent row and mid thigh pull. The RESX program also included head extension exercises with a load of 3 x 10 repetitions maximum. The cross sectional area of nine individual muscles/muscle groups was taken using MRI prior to and on completion of the 12 week program. The results showed an increase in cross sectional area of the cervical extensors for the RESX group only. The provision of neck specific exercises in a conventional resistance training program is necessary to increase the cross sectional area of the neck musculature or improve head extension strength. The isometric contractions of the muscles of the cervical spine required for stabilization during a conventional resistance training program was insufficient to generate neck muscle hypertrophy. These results support the conclusion that the gravitational load placed on the cervical musculature is only modest in an upright posture, thus necessitating the need for neck specific exercises to stimulate increases in cervical musculature cross sectional area (Conley et al., 1997a).

In a 12 month study assessing the effects of cervical resistance training on 180 female office workers, maximal isometric neck strength improved in flexion by 110%, rotation 76%, and extension by 69%. Relative to the endurance training and control groups, the resistance training group was the only treatment group with statistically significant increases in lateral flexion, flexion, and extension (Ylinen et al., 2003). Ylinen et al. (2006) showed that the greatest gains in neck strength coincided with decreases in neck pain and disability occurred during the first 2 months of the intervention. However, improvements continued until the end of the 12 month training period. There was a negative linear association between neck pain and disability indices with improved isometric neck strength (Ylinen et al., 2006). The large improvements in strength found in the female office workers are not likely reflective of the strength increases expected in the

military aircrew. During each flight, CH-146 helicopter pilots and flight engineers submit their cervical musculature to mechanical stressors in the form of the flight helmet, or flight helmet and NVG and the 1-2 +Gz forces. These mechanical stressors should result in some form of hypertrophy of the cervical musculature. It would therefore be expected that this military aircrew would have higher baseline cervical strength and endurance and should therefore see smaller gains in strength compared to studies looking a female office workers.

In contrast to the results of Ylinen et al. (2003; 2006) work on female office workers, Nikander et al. (2006) found that both strength and endurance training decreased perceived neck pain and disability equally regardless of intervention assignment. They found a 20 mm decrease in neck pain on the visual analog scale for those who trained for more than 8.75 metabolic equivalent hours ($MET \cdot h^{-1}$) per week, or 35 $MET \cdot h^{-1}$ of training per month. The greatest dosage of the specific training had the largest effect on neck pain symptoms. However, all patients who complied with more than 35 $MET \cdot h^{-1}$ per month belonged to the resistance training group; "[i]t may be more probable to complete 40 min of specific higher-load strength exercises than 60 min of endurance training with 2-kg dumbbells" (Nikander et al., 2006, p. 2072). Randløv et al. (1998) evaluated the neck pain associated with the activities of daily living. They found reductions in pain scores only for subjects in the higher intensity program and not in the lower intensity program after a 12 month follow-up period.

### *1.13 Neck Pain and Exercise Training Regimes in the Air Force*

Research looking at the self-reported risk factors related to the prevalence and development of neck pain in helicopter pilots illustrates that those who engage in regular aerobic exercise (Wickes & Greeves, 2005) and muscle strength training (Äng & Harms-Ringdahl, 2006) were at a decreased risk. When examining the effect of posture, head

mounted equipment, and vibration in the neck muscle of helicopter pilots, exercise therapy could potentially reduce the impact of these stressors (Harrison et al., 2007; Thuresson, 2005). The proposed benefit of exercise therapy would be the effect that increased muscular strength and endurance could have in reducing neck pain related to flight.

Little research exists regarding the use of effective exercise therapy treatment programs for helicopter pilots; therefore, previous research conducted using fighter pilots should be taken into consideration for program development. At the same time, the unique flight environment in which the rotary winged aircrew operate and how the occupational conditions differ from fighter pilots must be considered. Rotary aircrew are exposed to vibration, low level +Gz, more total flight hours, and long periods of suboptimal postures with the neck held in a forward flexed position (Äng et al., 2005; Forde, 2008; Wickes & Greeves, 2005). EMG analysis during an aerial combat sorties showed that the mean muscle strain during flight was in the range of 5-20% of the MVC with the highest values recorded for the cervical flexors (Oksa, Hamalainen, Rissanen, Myllyniemi, & Kuronen, 1996). Trampolines have been used in previous research in laboratory environments as a means to induce impulsive gravitational forces in the range of 0 to +4Gz (Sovelius et al., 2006, 2008). Sovelius et al., (2008) examined the load placed upon the right and left SCM, cervical and thoracic erector spinae and the trapezius using this approach for subjects wearing a helmet and a helmet and NVG. They examined the impact of the helmet and NVG on muscle strain as a percentage of MVC during basic bouncing, back bouncing and hand and knee bouncing. The results showed that the muscles in question did not in any condition exceed a maximal contraction force of 26.2% of the MVC. Although both of these studies were conducted on fighter pilots, the results clearly illustrate that during an aerial sortie, the cervical musculature is operating at a submaximal threshold. I expect the neuromuscular activity of cervical musculature in

helicopter aircrew would be even lower than the results of the previous two studies as helicopters operate within a gravitational range of 1 - +2Gz. Given the low level of contraction seen in the cervical muscle during flight, the importance of muscular endurance over ability to develop high levels of force (i.e., strength) is demonstrated.

In a study of Swedish fighter pilots, Alricsson et al. (2004), found no reduction in neck pain with increased strength and endurance of the neck musculature after completing a strength training program. However, the reinforced group and the non-reinforced group differed in baseline strength measurements, groups were not randomized (rather they were geographically determined), and there was no true non-treatment control group. The existence of confounding variables, such as differential history and maturation effects, provide alternative explanations for the observed changes in the neck muscle strength and endurance seen in the reinforced and non-reinforced groups.

These findings are in contrast to the research on fighter pilots comparing the effects of a weighted helmet training program versus a dynamic program. The study showed that pilots in the dynamic training program had fewer workdays lost and restriction in +Gz flights than the weighted helmet group. Isometric neck muscle strength increased similarly irrespective of training group status, and no differences were identified in the passive cervical range of motion. However, the study lacked both a control and had a small sample size (Hämäläinen & Vanharanta, 1998). Sovelius et al. (2006) found that both trampoline and strength training had positive effects on the cervical spine through decreases in muscle strain experienced in-flight and during cervical loading tests. After a three month follow-up, the improvements in muscle strain remained for both training interventions

Additional research using non-air force military individuals showed improvements in cervical strength as a result of resistance training program. Taylor et al. (2006) employed a Multi-Cervical Unit (MCU), a pin loaded machine that is able to apply

constant resistance through the full range of motion, 3 days a week for 12 weeks. Improvements in isometric strength and dynamic strength occurred within 4 weeks and continued until the final assessment when compared to the control group. Despite the success of the MCU, Taylor et al. acknowledged potential drawbacks including the cost of the MCU and that use was limited to one individual at a time (Taylor et al. 2006). Burnett, Naumann and Burton (2005) compared the effectiveness of the MCU relative to Thera-Band (Pro-Med Products, Inc.) tubing as training methods to improve cervical muscle strength determined using MVC. The MCU produced the greatest improvements in isometric strength: 64% in flexion, 63% in extension, 53% in left lateral flexion and 24% in right lateral flexion. The strength improvements in the Thera-Band tubing group were 42% in flexion, 30% in extension, 26% in left lateral flexion and 24% in right lateral flexion. When compared to the control, the MCU group achieved improvements in all directions tested while the Thera-Band tubing group was limited to flexion direction. However, the authors noted that a limitation of the study was the baseline and post-test assessments for all subjects were conducted using the MCU, which favoured those individuals training on the MCU. Despite the lack of statistically significant results, the authors concluded that the Thera-Band tubing was a valid training mode given its low cost and portability. This conclusion was supported by Netto, Burnett and Coleman (2007), who compared elastic tubing and a resistance machine at different intensities to muscle activation during aerial combat manoeuvres. They speculated that neck resistance training using elastic tubing "may be useful for pilots who fly low +Gz missions or tend to keep their head in a more neutral posture ... [such as] transport, bomber and rotary wing pilots" (Netto et al., 2007, p. 482).

### *1.14 Changes in Cervical Muscles in Response to Resistance Training*

The provision of an exercise program consisting of strength and endurance components has the capacity to increase the isometric strength and endurance of the flexors and extensor muscles of the cervical spine (Alricsson et al., 2004; Hämäläinen et al., 1998; Sovelius et al., 2006). The ability of a muscle to generate force is related to the cross sectional area (Ikai, & Fukunaga, 1970; Maughan, Watson, & Weir, 1983); which includes the cervical musculature (Mayoux-Benhamou, Wybier, & Revel, 1989). It is hypothesized that an increase in the cross sectional area of the neck musculature has the potential to provide increased stability to the cervical spine, functioning to limit or prevent musculoskeletal impairment (Conley et al., 1997a). Given that 80% of cervical stabilization is provided by the cervical muscles, improving the force production from these muscles should theoretically lead to enhanced stabilization of the cervical spine.

Resistance training is a mechanical stress that stimulates various adaptations of the neuromuscular system, the most notable being increases in muscle size and strength. These adaptations are related to the characteristics of the specific exercises employed in the training program. "This is referred to as the 'specificity of training' and suggests that the greatest gains in strength are observed in activities similar to the ones used in training" (Conley et al., 1997a, p. 444).

As mentioned previously in the context of the *Cinderella Hypothesis*, small submaximal contractions engage only a fraction of the motor units with Type I fibres being the most predominant. These fibres are continuously activated; if this continues for extended periods, it can lead to fibre damage. This hypothesis is relevant to the conditions experienced during flight in the CH-146 helicopter including low level contractions and repetitive and sustained movements for extended periods of time. Given the *specificity of training* response, an appropriate program should focus on strengthening the Type I fibres

that are continuously stressed or increasing the number of Type I fibres available within the muscle to facilitate contraction.

# 2. Introduction

Flight related neck pain is a problem for air defence forces worldwide causing personal suffering, reducing operational capabilities, and incurring high financial cost due to loss of manpower and litigation (Alricsson et al., Werner, 2004; Äng et al., 2005; Hämäläinen, 1999). The current literature documents occupational injuries among rotary winged pilots, but cervical pain goes virtually unreported, mainly due to the fear that pilot's will be grounded. Research since the 1960's to the present has targeted lower back pain and its relation to the occupational hazards of flying a helicopter (de Oliveira & Nadal, 2004; de Oliveira & Nadal, 2005; Lopez-Lopez et al., 2001; Pelham et al., 2005). Only recently has the issue of neck pain in helicopter pilots and flight engineers become an aeromedical concern, receiving increasing attention as an issue that has the potential for major health implications (Äng et al., 2005; Äng, & Harms-Ringdahl, 2006; Harrison et al., 2007; Thuresson et al., 2003; Thuresson et al., 2005; Wickes & Greeves, 2005). The purpose of my project was to assess the role of occupational factors in the development and persistence of chronic neck pain in CF CH-146 Griffon aircrew and whether exercise training programs can reduce or mitigate its occurrence.

## *2.1 Training Programs*

For the purpose of this research project, I evaluated the impact of the following exercise treatment programs: (a) general neck strength and muscle coordination training program (CTP); (b) neck muscle endurance training program (ETP); and (c) non-treatment control group (NTC). The purpose of the NTC is to control for the effects of history and maturation, which would otherwise limit the internal validity of the study (Campbell & Stanley, 1963). I undertook a 12 week pilot project where NVG-Laboratory Assessment Protocol (LAP) tests were performed at baseline, and 12 weeks later to assess isometric strength and endurance changes in the cervical musculature. This time frame

was chosen to provide sufficient time for neural and hypertrophy adaptations to occur as a result of the training stimulus (Conley et al., 1997b; Powers & Howley, 2004). It also incorporates the time period necessary for increases in tissue capillary density. Previous research identified increases in the number of capillaries per fibre during a 12-week intensive training period (Hather, Tesch, Buchanan, & Dudley, 1991; McCall, Byrnes, Dickinson, Pattany, & Fleck, 1996). This capillary increase should enhance blood flow to the respective muscle fibres, enhancing oxygen availability which has been hypothesised to reduce the occurrence of neck pain (Visser & Dieen, 2006).

*2.11 Neck muscular endurance training program.* Muscle fatigue is a major risk factor in the development and occurrence of muscle injuries. For aircrew, it has been suggested that fatigued muscles increase the risk of neck injury and have the potential to reduce mission effectiveness. Muscle activity of the cervical spine related to flying a helicopter is generally lower than that generated during a high +Gz flight (Äng et al., 2005). Loading factors such as posture, whole body vibration and increased head mass associated with helicopter flight are factors that also contribute to muscle fatigue (de Oliveira & Nadal, 2004; de Oliveira & Nadal, 2005; Harrison et al., 2007; Thuresson et al., 2003; Thuresson et al., 2005).

Endurance training focuses on increasing the ability of muscles to continue prolonged submaximal work. To maintain these repeated prolonged submaximal contractions over time, there must be a transition to more fatigue resistant muscle fibres, which rely predominantly on aerobic metabolism (Powers & Howley, 2004). To combat the generation of fatigue, I incorporated a dynamic endurance training intervention into the research design. The theory behind this is based on the predicted effect that increasing the number of fatigue resistant fibres in the cervical musculature will have on the development of neck pain.

Previous research examining the effect of dynamic endurance training programs on fighter pilots found fewer sick leaves and +Gz restrictions compared to resistance training programs (Hämäläinen et al., 1998). Ylinen et al. (2003) identified endurance training improvements in maximal isometric neck strength during a 12 month training period to be 28% in flexion, 29% in rotation and 16% in extension. For rotary winged aircrew, the ability to sustain a low level contraction is of greater importance than generating high levels of force (Harrison et al., *in press-a*).

*2.12 General neck strength and muscle coordination training program.* This program was proposed in a review on the complexity of muscle impairment in chronic neck pain conducted by Falla (2004). In contrast to the ETP, this exercise regime uses low load exercises to train and re-establish coordination between the deep and superficial layers of the neck musculature. The first aspect of this program focuses on muscle control, where low load exercises are used to enhance the coordination between the layers of cervical musculature (Falla, 2004). This motor control aspect "is based on biomechanical evidence of the functional interplay of the deep and superficial neck muscles and on the physiological and clinical evidence of impairments in these muscles in neck pain patients" (Falla, 2004, p. 131). The first stage of training focuses on specific exercises designed to isolate the deep segmental cervical stabilizers that support and maintain the neutral cervical lordosis curvature. The second stage integrates limb motion while the deep segmental stabilizers are challenged to maintain a neutral cervical spine. The third and final stage focuses on the interplay of the deep stabilizers and the superficial prime movers. The deep stabilizers are required to support the neutral lordotic curve of the cervical spine while the superficial musculature is employed for segmental control during neck motion (Kennedy, 1998). The development of this program follows from the suggestions of Äng et al. (2005) who reported altered myoelectric manifestations in the form of flatter EMG-slopes for helicopter pilots reporting neck pain relative to pain

free controls. It is common for helicopter pilots to maintain suboptimal static postures for extended periods of time. Äng et al. (2005) indicated clinical interest in using "low-load therapeutic exercise, emphasizing neck muscle control rather than strength" (p. 380) to combat neck pain in this aircrew.

In addition to neck training programs, all participants in the two treatment groups were asked to incorporate the same core strengthening component into their respective training programs. Justification for inclusion of a core strengthening component stems from a 12 month cross sectional survey of 7669 adults, where a history of low back pain was identified as a statistically significant predictor for the onset of neck pain (Croft et al., 2001). With a high prevalence of low back pain already established within the helicopter pilot community (Lopez-Lopez et al., 2001; de Oliveira & Nadal, 2004; de Oliveira & Nadal, 2005; Pelham et al., 2005), I incorporated a low back strengthening program to determine its efficacy in reducing neck pain. Previous research suggests that improvements in core stability produces significant reductions in low back pain (Pelham et al., 2005).

### *2.2 Assessment of Strength and Endurance and Neck Pain*

Improvements in strength and endurance of the cervical muscle were quantified using the force output during the NVG-LAP test. A lack of cervical strength and endurance has been cited as a risk factor in the etiology of neck pain (Falla, 2004). Changes in isometric strength and endurance were then related to the self-reported experiences of neck pain to determine the efficacy of the training programs.

Previous research addressed the success of various cervical training programs in improving cervical strength and endurance, thus the purpose of my study was not only to design a training program to improve cervical musculature function but also to reduce the occurrence of neck pain. A measure widely used in both clinical and research settings is

the visual analog scale (VAS) (Price, McGrath, Rafii, & Buckingham, 1983). This scale requires subjects to rate the intensity or severity of their pain by placing a mark on a straight line anchored with verbal labels. VAS is a valid means for assessing the change in pain (Dixon & Bird, 1981; Goolkasian, 2003; Price et al., 1983). VAS has also been employed in training studies to assess reductions in neck pain due to different cervical training programs (Ahlgren et al., 2001; Nikander et al., 2006; Ylinen et al., 2003; Ylinen et al., 2006). Nikander et al. (2006) found that both a cervical strength and endurance training protocols were capable of a 40 mm decrease on the VAS (± 20 mm). This is supported by Ylinen et al. (2003) who assessed strength and endurance training in women suffering from neck pain. There was a reduction in both training programs relative to the control group; subjects in the endurance and strength programs achieved a mean reduction of 35 mm and 40 mm respectively.

## Specific Hypotheses

$H_A$: Compared to the control, the general neck strength and muscle coordination training program would show improvements neck muscle performance as assessed with the NVG-LAP test.

$H_A$: Compared to the control, the neck muscle resistance training program would show improvements in neck muscle performance as assessed with the NVG-LAP test.

$H_A$: Compared to the control, the general neck strength and muscle coordination training program would reduce reported neck pain as a result of NVG use.

$H_A$: Compared to the control, the neck muscle resistance training program would reduce reported neck pain as a result of NVG use.

## Research Hypothesis

Based on the evidence provided by previous research, I hypothesized that the greatest reduction in neck pain based on the visual analog scale would occur in the general neck strength and muscle coordination training program.

# 3. Methods

## *3.1 Subjects*

Participants were recruited on a volunteer basis from the CH-146 Griffon aircrew in the 403 Squadron at the Canadian Forces Base (CFB) in Gagetown, NB. Due to a limited sample size, it was decided that the subjects from the 403 Squadron in Gagetown would act as the treatment population. Further volunteers were recruited from the 430 Tactical Helicopter Squadron at CFB Valcartier, QC for the NTC group comprised of CH-146 military aircrew. The NTC was encouraged to maintain as similar fitness level as the treatment population. The 430 Squadron was selected as a control due to its close proximity to the 403 Squadron and to standardize the flying environment/conditions experienced by both populations. Ethical approval was obtained from the University of Regina's Ethics Review Board. Prior to baseline data collection, each subject was fully briefed with written information and verbally about the goals of the project by a member of the research team and CF personnel before being asked to sign an informed consent form. The voluntary nature of the project and the confidentiality of outcomes and medical information were emphasized to all participants. Participants were excluded if they exhibited any of the: a) persistent signs and symptoms of a previous musculoskeletal injury to the cervico-thoracic spine; b) cervicobrachialgia or shoulder pain during neck movement; c) radiating pain when overloading the cervical spine; d) currently grounded (Falla et al., 2006; Nikander et al., 2006). All participants were required to fill out a Physical Activity Readiness Questionnaire which screened for any health problems identified as positive risk factors for which participation in a training study might have been contraindicated (Thomas, Reading & Shephard, 1992).

At the outset of the study participants ($n$ = 42: 39 males (m) and 3 females (f); 29 pilots and 13 flight engineers) were assigned to one of three categories based on their total NVG hour flying time. A threshold value of 150 NVG hours flight time has been

proposed for the onset of subjective neck pain in CF helicopter aircrew (Adam, 2004). Based on this, total NVG hours were used as a blocking variable to equally distribute individuals who were at a higher risk of suffering from neck pain among the treatment groups. Three NVG hour categories were selected: <100hrs, 100-200hrs, and >200hrs. Subjects from each category in the 403 Squadron were then randomly assigned to one of the two treatment groups. The initial subject cohort for each of the groups was the following: ETP $n$ = 15 (m = 14, f = 1), CTP $n$ = 13 (m = 11, f = 2) and NTC $n$ = 14 (m =14).

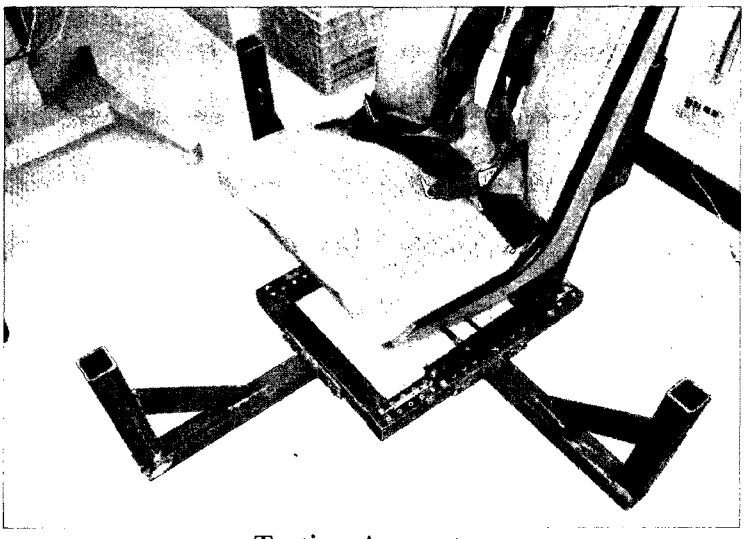
Eleven subjects; 4 from the ETP, 3 from the CTP and 6 from the NTC did not complete the post-test assessment. Reasons for withdrawal were: time commitments, ongoing neck pain, posting to a different base, away on course and personal reasons. At the completion of the 12-week study, the attrition rates for the respective programs were 27% for the ETP, 23% for the CTP, and 43% for the NTC. The final cohort consisted of 11 participants (10m, 1f) in the ETP, 10 participants (8m, 2f) in the CTP and 8 participants (8m) in the NTC.

### *3.2 Laboratory Assessment Protocol (LAP)-Procedures*

The laboratory assessment protocol employed a double-blinded format, whereby all testing was conducted by an independent researcher who was blinded to participant group status. During testing, participants were asked to refrain from providing any information indicating their training group status. Prior to both baseline and post-test assessment, subjects were asked to warm-up their neck and upper back muscles for 5 min using shoulder shrugs and neck rotations. Once they had completed the warm-up, they were seated in a standard CH-146 cockpit seat designed for the LAP Test. The subjects were secured in the seat using a 4-point harness to minimize trunk movement during the isometric testing. The cockpit seat was attached to a mount that had four pole slots

welded to the bottom of its frame. A 2.5 cm square steel pole fit into each of the 4 slots and was positioned into the appropriate slot depending on the direction being tested. A 5 cm webbing strap was secured with Velcro$^{TM}$ around the subjects head to enable MVC testing. The strap was aligned just above the eyebrows for the flexion direction, directly above the occipital protuberances for the extension direction, and under the top of the ears for the left and right lateral flexion direction. In all cases it was then attached to a SSM-AJ-100 force transducer (Interface, Scottsdale, Arizona) attached at eye level to a 2.5 cm square steel pole. The pole was secured in place 40 cm directly behind the subject (see *Figure 6*) (Harrison et al., *in press-a*).

*3.21 Muscle maximal voluntary contraction (MVC) testing.* Using the protocol and software designed at US Army Aeromedical Research Laboratory (USAARL, Ft Rucker, AL), an isometric MVC was performed during the LAP test for each of the following directions: flexion, extension, and left and right lateral flexion. The MVC testing order for each participant was determined randomly. For each direction tested, the contraction was held for a period of 5 seconds with 3 miutes rest intervals between trials. The maximum force achieved during the 5 seconds was recorded as MVC force for each of the four tested directions. Subjects were instructed to *gradually ramp up* to maximal force production (approximately 3 seconds) and avoid rapid jerking movements that could result in neck injury or artifacts measurements. Subjects were provided with a familiarization practice trial using the extension contraction as this was deemed to be the movement pattern that was the least susceptible to fatigue. Only one MVC trial was used because previous research using this protocol demonstrated that the maximal force was normally produced in the first trial (Harrison, 2008). Following each MVC trial, subjects were asked on a modified (0-10) Borg scale (1998) to rate their level of perceived exertion. The resulting force output for each on the baseline MVC trials was used to calculate the subsequent 70% target force used in the submaximal muscular



Testing Apparatus



a. Extension



c. Right Lateral Flexion



b. Flexion



d. Left Lateral Flexion

*Figure 6*: The standard CH-146 Cock pit seat used during Laboratory Assessment Protocol Procedures in the Four Cervical Testing Directions during Pre- and Post-Test Assessments

endurance trial for both baseline and the post-test assessment. The use of the baseline 70% MVC value in the post-test allowed for comparisons between the pre- and post-test scores to determine whether or not improvements in endurance capacity had occurred as a result of the respective training programs.

*3.22 Submaximal muscular endurance trials.* Subjects were then asked to perform a submaximal endurance trial in each test direction: flexion, extension, left and right lateral flexion. The same, randomized testing order used in the MVC trails was also used for the 70% submaximal endurance trials. During the test, subjects were asked to maintain an isometric force equal to 70±10% of their baseline MVC force for as long as they were able or for a period of no more than 3 minutes. Trials ended when a subject's force production fell outside this range for more than 2 seconds or at the 3 min mark. Visual feedback was available from a monitor positioned at eye level. After each endurance trial, subjects were again asked to rate their level of exertion on a modified Borg Scale (0-10). See Appendix A for the schedule of events for the NVG-LAP test (Harrison et al., *in press-a*).

### *3.3 NVG-LAP Test Measurement Instrumentation – Apparatus*

The LAP test apparatus was comprised of a load cell (Precision Transducers, Auckland, New Zealand), an amplifier (MF60, Micron Meters, Simi Valley, CA), and custom software written by Greg Dickinson (University of New Brunswick, Fredericton, NB).

***Force output.*** Custom software written by Greg Dickinson (University of New Brunswick, Fredericton, NB) using LabVIEW 7.1 (National Instruments) was used to calculate the area under the force curve, initial and final force variance maximum and minimum, a 5 second mean of initial and final force variance, and time to fatigue for the submaximal muscular endurance trial. The area under the force curve was used as index

of cervical muscle fatigue during each of the pre- and post-test submaximal muscular endurance trials.

### *3.4 Training Program Interventions*

Prior to the start of the training program, a familiarization session was conducted where each participant was asked to complete a Disability of Arm and Shoulder (DASH) Questionnaire and a Neck Disability Index Questionnaire. These are the standard questionnaires used by CF physiotherapists. Participants were also asked to complete a questionnaire put together by the research team that inquired as to their basic anthropometric measurements such as height and weight, aircrew position, counterweight usage and dominant hand. Personal information regarding the subject's flight experience with fixed wing and rotary wing aircrafts as a function of years and hours, total NVG experience in hours, average NVG mission length in hours, and longest NVG mission length in hours was also collected. Neck pain history was gathered through a combination of questionnaires designed by the research team to identify neck pain sufferers as well as standardized questionnaires currently in use by the CF medical personnel when dealing with neck pain among aircrew. These questionnaires detailed subjective pain severity as well as information regarding activities of daily living for which the aircrew members felt their performance was impaired as a result of neck pain (Harrison et al., *in press-b*).

Perceived neck pain was recorded using a visual analog scale (VAS), which is based on a 0-100 mm scale with end points anchored on the left side with 'no pain at all' (0 mm) and on the right with 'worst possible pain' (100 mm) (Ahlgren et al., 2001; Nikander et al., 2006; Ylinen et al., 2006). The use of these verbal cues is a reliable and valid measure of sensory intensity relating to neck pain (Goolkasian, 2003). Three VAS scales representing 'pain at present', 'pain in general', and 'pain at worst' were used prior to the pre- and post-test assessment (Ahlgren et al., 2001).

After the baseline assessment, subjects in each exercise intervention attended a familiarization session where the exercises that comprised their respective programs were demonstrated in detail. Alricsson et al. (2004) found that training supervised by a physiotherapist or trained individual (a pilot who was instructed by the physiotherapist) coincided with increases in strength and endurance of the cervical musculature compared to un-supervised training sessions. Based on the positive association between training supervision and strength improvement, supervision was provided during all training sessions. Supervision ensured that participants clearly understood the program, performed proper technique, and understood when to increase the resistance load. All participants were asked to undertake their respective programs 3x per week in addition to any other exercise that they were doing in their daily routine. If they were unable to attend one of four sessions offered on the training days, they were asked to complete the session on their own. On the familiarization day, each participant was given a booklet outlining their respective programs with visual and verbal illustrations of each of the exercises. They were also asked to record their training progress in a training diary (i.e., the reps and sets of each exercise performed). Diaries were collected at the end of the training period to assess training attendance and ensure that each participant had progressively increased their training load. To ensure maximal physiological adaptations in the cervical musculature, the progressive overload protocol whereby training load was increased by increasing the number of reps and intensity of the exercises. The NTC group was asked to refrain from performing any neck strengthening or endurance training programs during the 12-week training intervention, but were encouraged to continue their regular training regime.

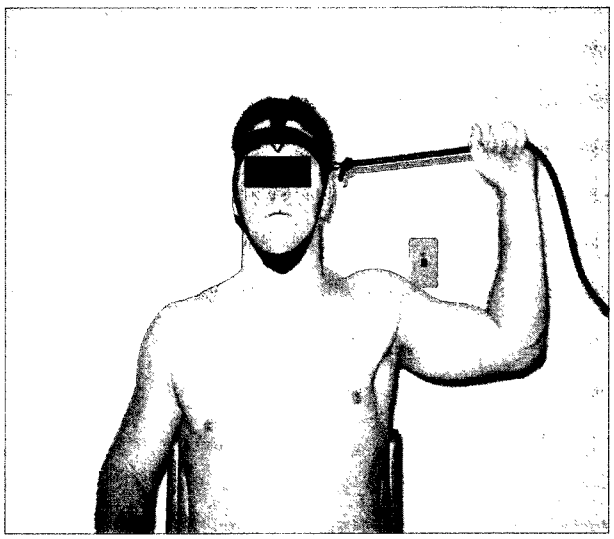
Each treatment intervention began with 5 minutes of active warm-up where participants were guided through a series of movements of the cervical spine and shoulders. Each movement was repeated 10 times in each of the 3 sets and consisted of

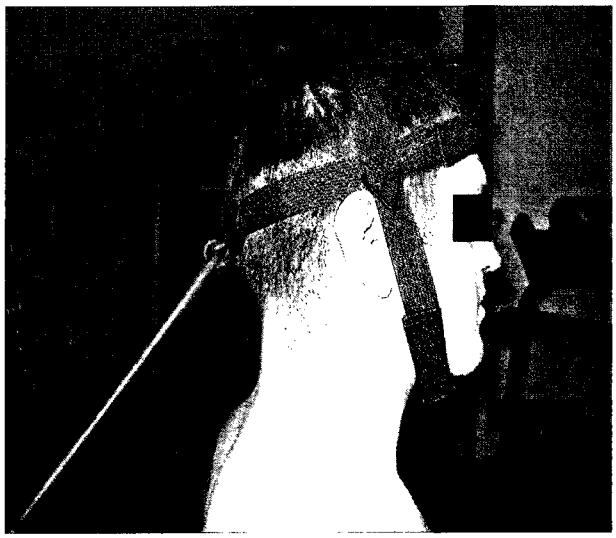
shoulder shrugs, shoulder circles, shoulder protraction and retraction and neck half circles performed in each direction. This was followed by approximately 15 minutes of specific neck exercises that varied depending on the treatment designation, concluding with approximately 15 minutes of abdominal exercises including abdominal curls with weights, the plank, core work with a swiss ball, side plank, back bridges, and *good mornings* with weights. These exercises initially began with 2 sets of 10 reps. The number of sets was increased to 3 after 5 weeks. Each session finished with approximately 5 minutes of stretching where each stretch was performed 3x and held for a total of 20 seconds. The stretching routine included: scalene muscle, upper trapezius muscle, sternocleidomastoid muscle, standing head drop stretch, and a dumb bell lateral stretch.

*3.41 Group A: Neck muscle endurance training program (ETP).* In this program, participants used elastic rubber tubing (Theraband, Hygiene Corp, Akron, OH) to resist the dynamic movements of cervical flexion, extension, right and left lateral flexion. Participants assumed a seated position wearing a 2.5 cm webbing head harness to which rubber tubing was attached via carabineers that was equivalent to their 30% MVC determined during baseline testing. The resistance was provided by rubber tubing that was cut to 70cm length. Given the floor mounts and the mean height (1.8 m) of the subjects in the seated position attached to the head harness, the approximate stretch of the rubber tubing was 25%. Given this 25% stretch, the tubing resistances were calculated to be: blue (3lbs), black (4lbs), and silver (5lbs). For the extension direction, the tubing was attached via a carabineer to a weighted mount on the floor in front of the participant. For the flexion direct, the participants turned the chair in the opposite direction and the tubing was attached to a clip on the back of the head harness and the opposite end remained attached to the same floor mount. Left and right lateral flexion required the participants to attach the tubing to the clips located on the left and right sides of the head harness. For the lateral flexion directions, the participants were asked to hold the tubing with the same

hand with their arm abducted and their elbow bent to 90° (see *Figure 7*). No directions were provided by the supervising member of the research team with regards to performance posture or ideal movement patterns to minimize any intentional effort to activate the deep neck stabilizers.

For each of the four directions, participants performed 3 sets of 10 reps maintaining a resistance level of 30% MVC (Conley et al., 1997a; Nikander et al., 2006). One minute of rest was given between each set (Conley et al., 1997a; Nikander et al., 2006; Ylinen et al., 2003). If a participant was able to perform 12 consecutive reps of an exercise, the load was increased by 5% as necessary to maintain 10RM on subsequent performed sets. A rhythmic cadence of 2 sec for concentric and eccentric contractions was maintained. At the end of range of motion, the contraction was held for 1 sec (Taylor et al., 2006). Timing for the cadence was provided by the supervisor.





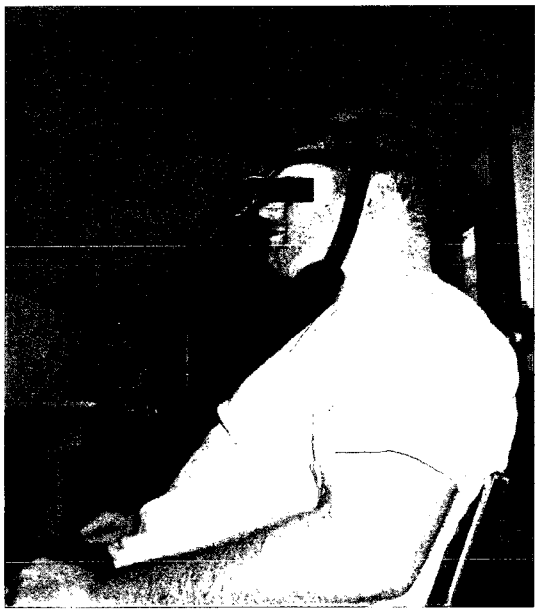


*Figure 7*: Exercises Comprising the Endurance Training Program Top Left Posture Adopted for Right and Left Lateral Flexion, Top Right Extension Contraction, Bottom Flexion Contraction.

*3.42 Group B: General neck strength and muscle coordination training program (CTP).* The purpose of this program was muscle control. Low load exercises were used to train the coordination between the layers of cervical muscle (Falla, 2004). The motor control aspect "is based on biomechanical evidence of the functional interplay of the deep and superficial neck muscles and on the physiological and clinical evidence of impairments in these muscles in neck pain patients" (Falla, 2004, p. 131). This exercise regime uses low load exercises to place specific emphasis on strengthening and re-educating the deep postural cervical musculature. Once the imbalance between deep and the superficial muscle was addressed, general neck strengthening exercises were introduced (Jull, 2000).

This exercise program was developed with the guidance of physiotherapist, Carol Kennedy BSc PT, DipManipPT, FCAMT who specializes in exercise prescription for patients with cervical spine dysfunction. Three stages of progression were used in this program. Stage 1 extended for the first two weeks of the training program and focused on isolating the deep segmental stabilizers of the cervical spine. The primary exercises of this stage used isometric contractions to maintain a neutral cervical spine (slight lordosis of the cervical spine) in supine, standing and sitting. Other exercises in this initial stage focused on proper segmental movement of the deep extensor cervical musculature in a return to neutral posture in four point kneeling and a sitting position. Stage 2 occurred during weeks 2-6 and focused primarily on maintaining a neutral cervical spine while integrating limb motion into the exercises. The exercises consisted of the windmill, internal and external shoulder rotation and return to neutral posture in four point kneeling with pure rotation. Stage 3, weeks 7-12, the programs focus changed to include only two exercises that focused specifically on the deep cervical musculature: the deep neck flexor head nod with a head lift in supine and the return to a neutral posture in four point kneeling with pure rotation. During this phase of the program, the emphasis changed to

strengthening the superficial muscles of the neck through resisted flexion, extension, and left and right lateral flexion through a controlled segmental movement pattern which incorporated the deep cervical muscle through the maintenance of proper posture and a slight chin nod. Resistance was applied using the same method used in the neck muscle endurance training program. Resistance was increased when subjects were able to perform 3 sets of 10 reps while maintaining segmental control throughout the movement.

The exercise schedule for this program was progressed on an individual basis based upon each participant's progress. All participants were given continual feedback with regards to exercise performance and posture from the supervisor. Between each set, 1 min of rest was given (Conley et al., 1997a; Nikander et al., 2006; Ylinen et al., 2003). If the participant was able to perform 12 consecutive reps for 3 sets of an exercise, the resistance was increased by approximately 5% as necessary to maintain 10RM on subsequent performed sets. A rhythmic cadence of 2 seconds for concentric and eccentric contractions was maintained for each exercise. At the end of range of motion, the contraction was held isometrically for a period of 1 second (Taylor et al., 2006). *Figure 8* illustrates examples of the exercises performed.

*3.44 Group C: Non-treatment control group (NTC).* The participants in this control group were asked to refrain from participating in any neck specific strength or endurance training. However, they were encouraged to continue with their normal daily exercise routine. They were given no information about the potential benefits of neck specific exercises or stretching.

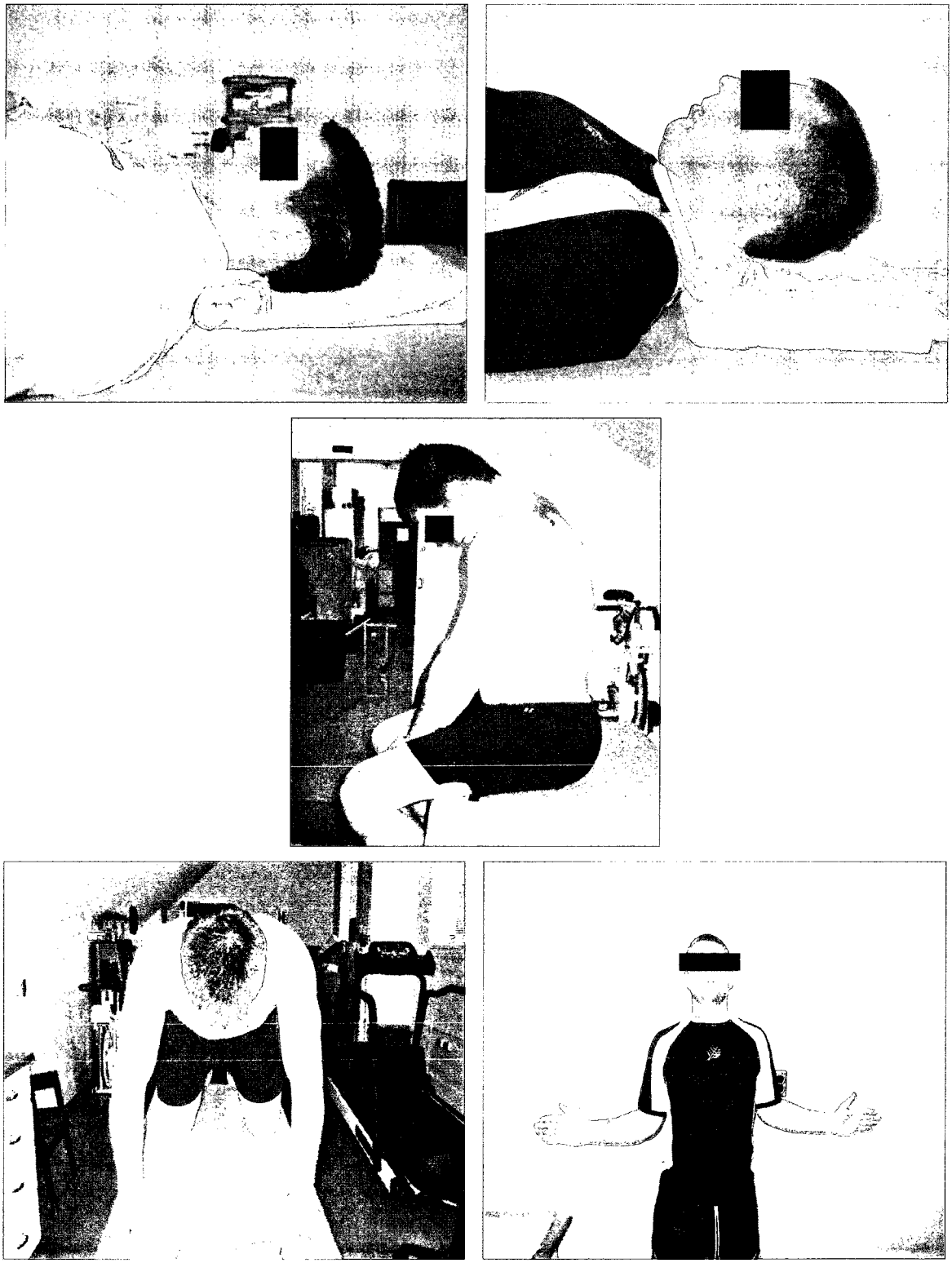


*Figure 8*: Exercises Used in the Coordination Training Program: Top Left Deep Neck Flexor Head Nod, Top Right Deep Neck Flexor Head Nod with Head Lift, Middle Seated Return to Neutral, Bottom Left Return to Neutral in Four Point Kneeling, Bottom Right External Shoulder Rotation

### *3.5 Data analysis*

Statistical Analysis was performed using SPSS 16.0 (SPSS Inc. Chicago, IL). Descriptive statistics are reported as means ± one standard deviation. All results were examined for the presence of outliers exceeding ± 2 SDs. A baseline comparison of MVCs, anthropometric measures and neck pain VAS scores was made between the two treatment groups and the control using a one-way analysis of variance (ANOVA). Pre- and post-test MVCs for flexion, extension and right and left lateral flexion were analyzed using a 3x2 mixed-model ANOVA. Times to fatigue in the 70% endurance trial and area under the 70% force curve were also analyzed using a 3x2 mixed-model ANOVA. Visual inspection of the mean pre- and post-test values for the MVC, times to fatigue and the area under the curve trials was used to identify interaction contrasts of interest which were then tested using methods described by Jaccard and Guilamo-Ramos (2002). Single degree-of-freedom contrasts were used to determine significant changes for either of the two treatment groups or the control (Jaccard & Guilamo-Ramos, 2002). Given previous research regarding the effect of neck training programs on isometric strength, endurance and self-reported neck pain, statistical significance for the interaction and single degree of freedom contrast were determined using a directional $\alpha$ level. VAS pain scores for the treatment groups were analyzed using within-subjects ANOVA. An alpha value of $\leq .05$ was used to determine statistical significance.

# 4. Results

## *4.1 Characteristics of sample*

Subject characteristics pertaining to anthropometric values, personal characteristics and flight experience for the ETP, CTP and NTC are presented as sample means and standard deviations in Table 1. At baseline, the ETP, CTP and NTC were found to be equivalent for all personal, anthropometric and flight experience variables measured (all $p > 0.05$). Based on training diaries, compliance with the exercise programs was 52.78% ± 32.39% for the ETP and 76.11% ± 20.75 for the CTP.

## *4.2 Maximal voluntary contractions*

Pre-intervention, there were no statistically significant differences between the three groups for the four isometric MVC directions tested. Both pre- and post-test results for the ETP, CTP and the NTC were visually examined for the presence of outliers that exceeded ±2 *SD*. The isometric MVC values recorded for the ETP, CTP and NTC pre- and post-intervention are presented in Table 2 and *Figure 9*. Visual inspection of the graphs displaying the means for flexion, extension and left and right lateral flexion for the ETP, CTP and the NTC were used to isolate potentially significant interaction contrasts between the three groups. Graphs were also examined for single degree of freedom changes for the ETP, CTP or NTC that occurred during the intervention.

Table 1: Baseline Anthropometric Values, Personal Characteristics and Flight Experience for the Endurance (ETP), Coordination (CTP)Training Groups and Control Group (NTC) Subjects.

| | ETP | | CTP | | NTC | |
|---|---|---|---|---|---|---|
| | *M* | *SD* | *M* | *SD* | *M* | *SD* |
| Age (yrs) | 37.18 | 4.5 | 35.40 | 8.22 | 37.12 | 6.31 |
| Height (m) | 1.80 | 0.08 | 1.74 | 0.077 | 1.79 | 0.07 |
| Weight (kgs) | 86.03 | 12.28 | 77.33 | 24.07 | 90.05 | 11.24 |
| Total Years Flying | 8.09 | 6.76 | 7.67 | 7.14 | 9.69 | 7.59 |
| Total Hours Flying | 1664.66 | 1696.82 | 1676.70 | 1741.06 | 2348.28 | 2115.19 |
| Rotary Wing Years | 5.73 | 5.35 | 5.61 | 6.63 | 8.75 | 7.64 |
| Rotary Wing Hours | 1210.00 | 1184.64 | 1281.60 | 1611.08 | 2225.09 | 2083.23 |
| NVG Hours | 167.91 | 165.86 | 148.50 | 153.84 | 225.49 | 186.69 |
| Average NVG Flight (hrs) | 1.36 | 1.12 | 1.58 | 0.83 | 1.59 | 0.76 |
| Maximum NVG Flight (hrs) | 2.55 | 2.29 | 2.45 | 1.64 | 3.25 | 2.04 |

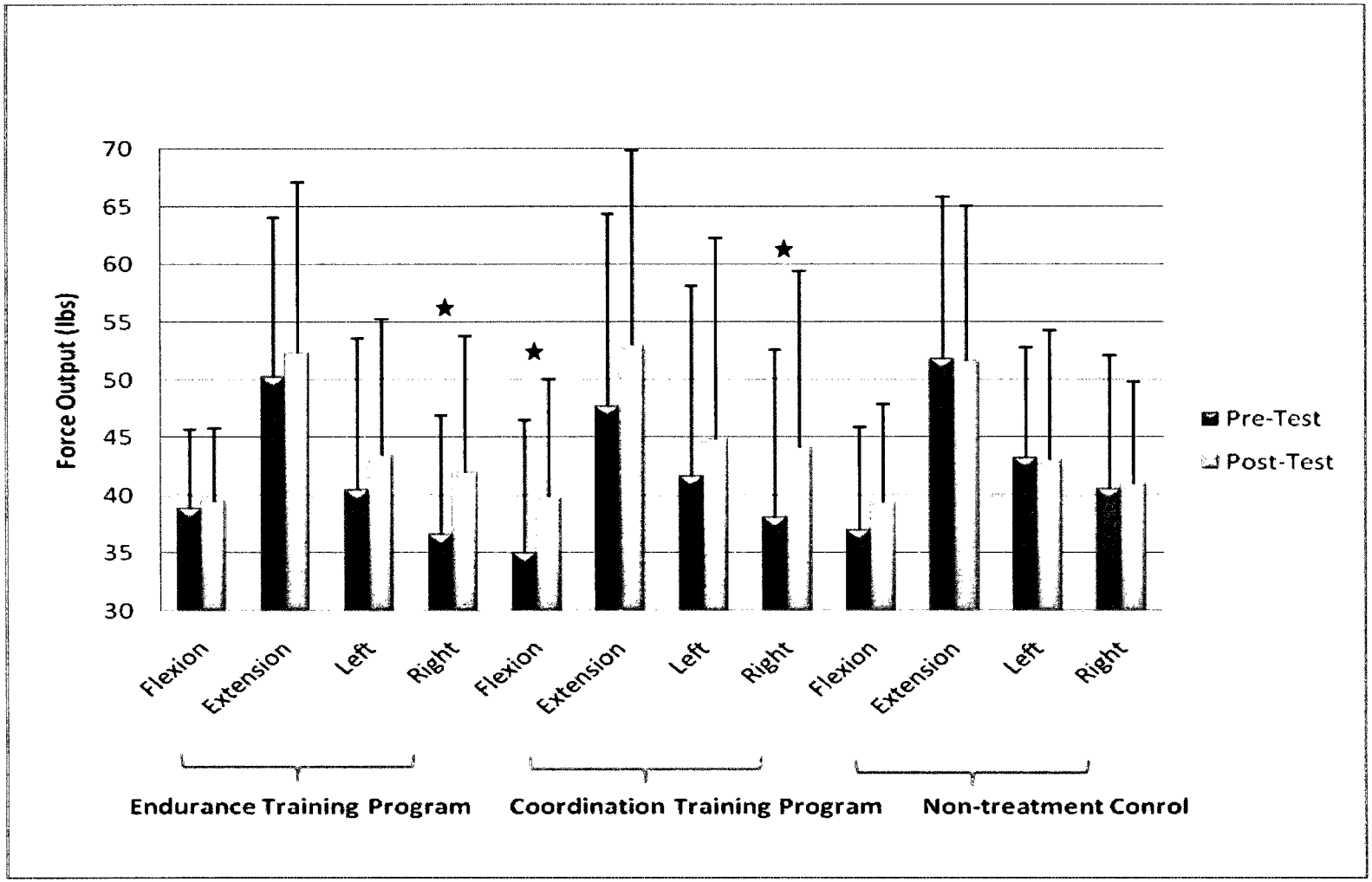



*Figure 9:* Summary Statistics for Isometric MVC (lbs) Measured Pre- and Post-Training Intervention for the ETP, CTP and NTC Groups

★Denotes statistical significance $p \leq .05$ between pre- and post-test MVC scores

Table 2: Summary Statistics for Isometric MVC (lbs) Measured Pre- and Post-Training Intervention for the ETP, CTP and NTC Groups

| | | ETP | | CTP | | NTC | |
|---|---|---|---|---|---|---|---|
| Directions Tested | | *M* | *SD* | *M* | *SD* | *M* | *SD* |
| Extension | Pre-test | 50.21 | 13.82 | 47.68 | 16.67 | 53.18 | 13.63 |
| | Post-test | 52.31 | 16.67 | 53.04 | 16.84 | 51.67 | 13.39 |
| | Difference | 2.1 | | 5.36 | | -1.51 | |
| Flexion | Pre-test | 38.84 | 6.82 | 35.03 | 11.44 | 37.95 | 8.14 |
| | Post-test | 39.44 | 6.38 | 39.85* | 10.15 | 39.34 | 8.51 |
| | Difference | 0.6 | | 4.82 | | 1.39 | |
| Left Lateral Flexion | Pre-test | 40.44 | 13.18 | 41.61 | 16.49 | 43.38 | 8.90 |
| | Post-test | 43.46 | 11.77 | 44.79 | 17.53 | 42.99 | 11.31 |
| | Difference | 3.02 | | 3.18 | | -0.39 | |
| Right Lateral Flexion | Pre-test | 36.65 | 10.23 | 38.07 | 14.54 | 40.93 | 10.76 |
| | Post-test | 41.91* | 11.90 | 44.13* | 15.32 | 40.95 | 8.88 |
| | Difference | 5.26 | | 6.06 | | 0.02 | |

* Denotes statistically significant results between pre- and post-test mean values

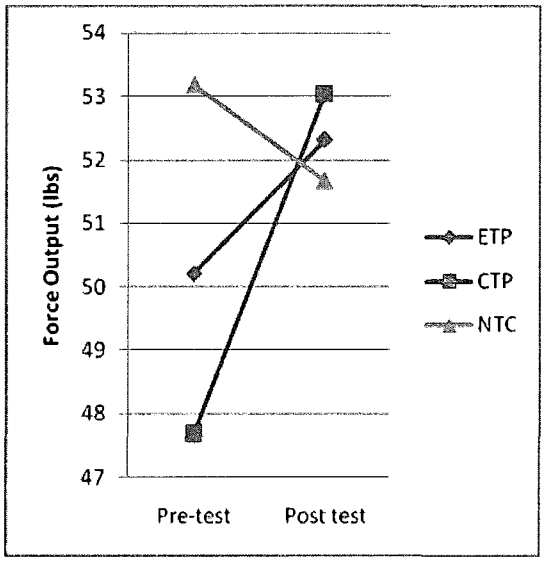
*Extension.* The MVC results for extension were statistically non-significant for the main effect, time, $F(2,26) = 2.21, p = 0.15, \eta^2 = 0.078$, indicating no improvement in MVC isometric force from pre- to post-test assessment. There were no differences between ETP, CTP, and NTC for extension as the main effect for group was also statistically non-significant, $F(2,26) = 0.044, p = 0.96, \eta^2 = 0.003$. The Time x Group was not statistically significant, $F(2,26) = 2.06, p = 0.15, \eta^2 = 0.14$ (refer to *Figure 10a*). Visual examination revealed an apparent change in the CTP from pre- to post-test assessment; but a paired t-test showed it was not statistically significant, $t(9) = 1.67, p = 0.06$. An interaction contrast was conducted to compare mean pre- and post-test scores for the CTP and the NTC. There was no statistically significant difference, $t(9) = 1.21, p = 0.13$.

*Flexion.* Analysis of the pre- and post- MCV through a 3 (Group) x 2 (Time) mixed-model ANOVA for flexion was statistically significant for the main effect, time, $F(2, 26) = 5.82, p = 0.023, \eta^2 = 0.18$. This indicated a change in isometric MVC force from baseline to post-treatment for the flexion direction regardless of group distinction. No statistically significant difference for the main effect of Group was found, $F(2,26) = 0.114, p = 0.89, \eta^2 = 0.01$. This suggests an absence of strength differences between the groups. The Time x Group interaction was also statistically non-significant, $F(2,26) = 2.02, p = 0.15, \eta^2 = 0.14$. Neither treatment group exhibited a change in flexion isometric strength relative to the NTC (refer to *Figure 10c*). Visual examination of means for the ETP, CTP and NTC suggested a large change from pre- to post-test for the CTP. A paired t-test showed this to be statistically significant, $t(9) = 2.51, p = 0.02$.

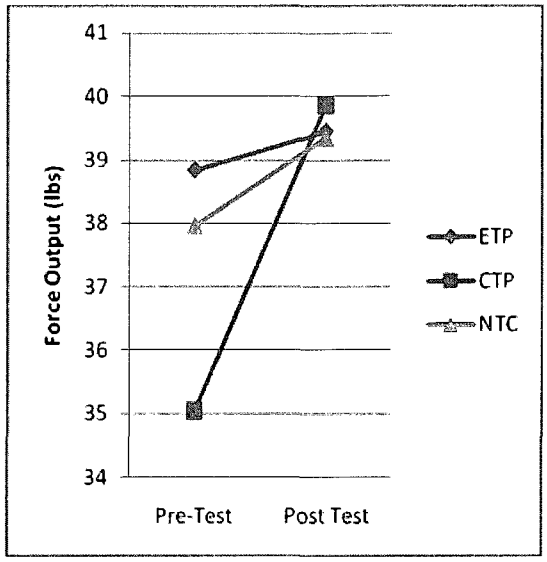
*Left lateral flexion.* Left lateral flexion isometric MVC produced similar results to those found in the extension trials. The main effect for Time was not statistically significant $F(2,26) = 1.85, p = 0.19, \eta^2 = 0.07$, indicative of no improvement from pre- to post-test. No differences were isolated for left lateral flexion between the three groups,

illustrated by a lack of statistical significance for the main effect for Group $F(2,26) = 0.03, p = 0.97, \eta^2 = 0.002$. A statistically non-significant effect was also obtained for the Time x Group interaction, $F(2,26) = 0.61, p = 0.55, \eta^2 = 0.04$, suggesting no change in the pre- and the post-test values for left lateral flexion MVC for any of the three groups tested (refer to *Figure 10b*). Two potentially interesting single degree of freedom contrasts; one for the CTP pre- and post-test and the other for the ETP were revealed. Analysis using paired t-tests produced non-statistically significant changes for both the CTP and the ETP as revealed respectively by the following results, $t(9) = 1.04, p = 0.16$ and $t(10) = 1.74, p = 0.06$; the ETP was close to a statistically significant improvement from pre- to post-test. An interaction contrast was conducted for CTP and NTC mean pre- and post-test scores revealing a non-statistically significant interaction, $t(16) = 0.70, p = 0.25$. Another interaction contrast for the mean ETP and NTC pre- and post-test scores was conducted, but failed to achieve statistical significance, $t(17) = 0.83, p = 0.21$.

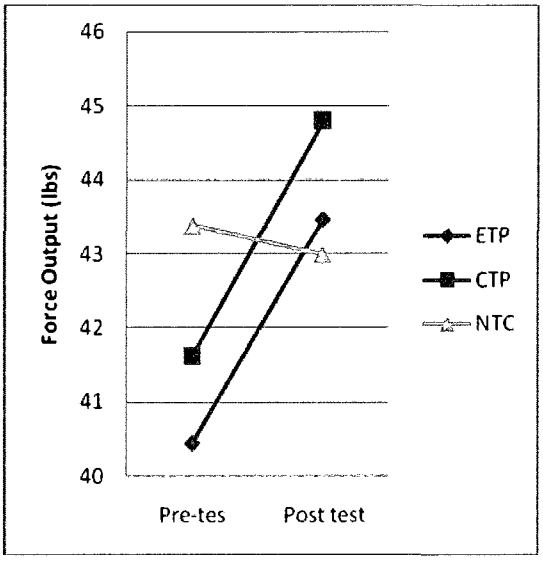
*Right lateral flexion.* Statistically significant difference for right lateral flexion was obtained for the main effect for time, signifying a change from pre- to post-test, $F(2,26) = 6.47, p = 0.02, \eta^2 = 0.20$. However, there were no differences identified between the ETP, CTP and the NTC as indicated by the non-statistically significant main effect for Group, $F(2,26) = 0.078, p = 0.93, \eta^2 = 0.01$. Likewise the Time x Group interaction for right lateral flexion was not statistically significant, $F(2,26) = 0.08, p = 0.25, \eta^2 = 0.01$. There was a change from pre- to post-test in the right lateral flexion MVC; but the two treatment conditions were not significantly different from each other or relative to the NTC as shown in *Figure 10d.* Graphs of the mean pre- and post-test values suggested two potentially relevant improvements for right lateral flexion in the ETP and the CTP. These changes were examined as single degrees of freedom contrasts using a paired t-test which indicated statistically significant improvement for the ETP, $t(10) = 2.65, p = 0.01$ and the CTP $t(9) = 1.77, p = 0.05$. The interaction contrasts for right lateral



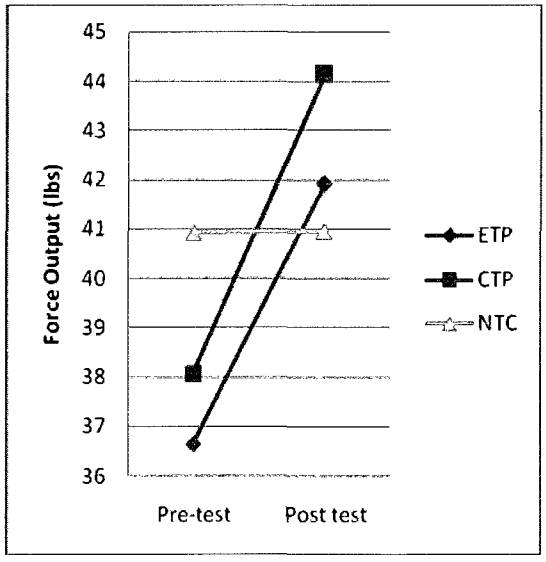

*a.* Extension




*c.* Flexion




*b.* Left Lateral Flexion



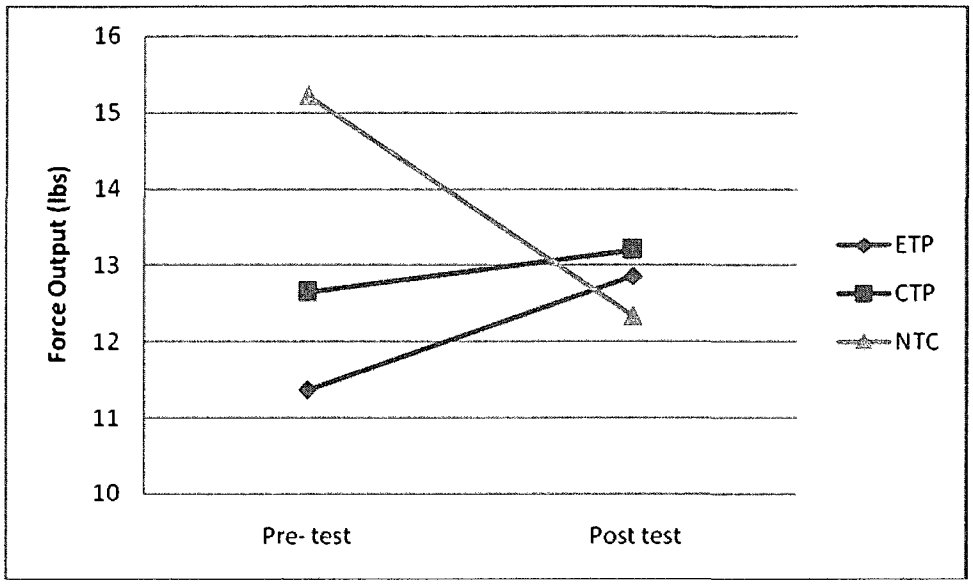

*d.* Right Lateral Flexion

*Figure 10.* Mean Force Output (lbs) for a 5 seconds Seated Maximal Voluntary Isometric Contraction in the Four Tested Directions Pre- and Post-Intervention for the ETP, CTP and NTC Groups.

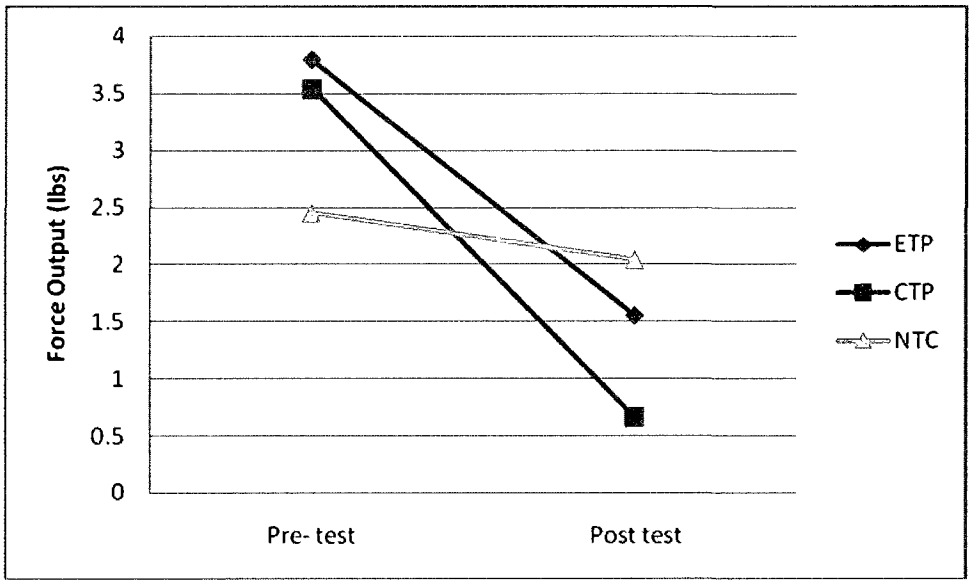
flexion were not statistically significant for CTP versus NTC but were for the ETP versus NTC, $t(16) = 1.03, p = 0.16$ and $t(17) = 1.37, p = 0.05$ respectively.

*Extension/flexion differences.* A 3 (Group) x 2(Time) ANOVA assessing the MVC pre- and post-test strength differences over the 12 week intervention in the sagittal plane showed that there was no statistically significant main effect for Time, $F(2, 26) = 0.04, p = 0.84, \eta^2 = 0.002$, or Group, $F(2, 26) = 0.07, p = 0.93, \eta^2 = 0.01$. The Time x Group interaction was also not statistically significant, $F(2, 26) = 0.86, p = 0.93, \eta^2 = 0.01$ (see *Figure 11a*). Visual inspection of the *Figure 11a* revealed a large decrease in the mean difference scores from pre- to post-test for the NTC. This decrease was examined using a paired t-test and was not statistically significant, $t(7) = 1.33, p = 0.11$. The decrease in mean scores for the NTC was contrasted against the increase for the ETP using an interaction contrast. The outcome was not statistically significance, $t(17) = 1.26, p = 0.11$.

*Left lateral flexion/right lateral flexion differences.* The strength differences in the coronal plane were not statistically significant for the main effect for Time, $F(2, 26) = 1.64, p = 0.21, \eta^2 = 0.002$, or Group, $F(2, 26) = 0.06, p = 0.94, \eta^2 = 0.004$. These analyses showed no statistical difference in the mean values from Time 1 to Time 2 regardless of the group and no distinction could be made between the two interventions and control when the mean scores were examined with respect to Group. The Time x Group interaction also failed to achieve statistical significance, $F(2, 26) = 0.24, p = 0.79, \eta^2 = 0.02$ (refer to *Figure 11b*). Visual inspection of the interaction between the CTP and NTP was examined using a interaction contrast, but was not statistically significant, $t(16) = 0.47, p = 0.32$.




*a.* Extension/Flexion




*b.* Left lateral flexion/Right lateral flexion

*Figure 11.* Mean Isometric MVC (lbs) Differences for the ETP, CTP and the NTC Groups Pre- and Post-Treatment Intervention

### *4.3 70% Endurance trial times to fatigue*

The endurance trials required individuals to sustain a resistance load equivalent to 70% of their MVC for as long as they were able to a maximum cutoff of 180 seconds. To evaluate whether or not there was an improvement in an individual's endurance capacity over the treatment period, the pre-test 70% resistance level was also used for the post-test evaluation. The four directions were inspected for outliers but none were found. During data collection, four of the data files (1 from CTP in extension, 1 from ETP and 1 from NTC for left lateral flexion and 1 from ETP for right lateral flexion) were corrupted during the data transfer and I could not obtain values for the times to fatigue, thus these files were excluded from the analysis. A one-way ANOVA was conducted to examine the pre-test times to fatigue, no statistically significant differences were found between the two treatment groups and the control. The times to volitional fatigue for the 70% isometric endurance trial are displayed in Table 3.

*Extension.* The analysis using a 3 (Group) x 2 (Time) ANOVA of the extension data failed to yield a statistically significant main effect for either Time, $F(2,25) = 0.43$, $p = 0.52$, $\eta^2 = 0.02$ or Group, $F(2,25)= 1.90$, $p= 0.17$, $\eta^2= 0.13$. Consistent with the previous results, the Time x Group interaction was also not statistically significant, $F(2,25) = 0.75$, $p = 0.48$, $\eta^2 = 0.06$ (refer to *Figure 12a*) Visual inspection revealed a potentially interesting change in the CTP that was analyzed using a paired t-test, but it was not statistically significant, $t(8) = 1.36$, $p = 0.10$. The interaction contrast between the CTP and the NTC was also not statistically significance, $t(15) = 0.94$, $p = 0.18$.

Table 3: Time (s) to Volitional Fatigue for the 70% Isometric Endurance Trial Pre- and Post-test for ETP, CTP and NTC Groups

| Direction Tested | | ETP | | CTP | | NTC | |
|---|---|---|---|---|---|---|---|
| | | *M* | *SD* | *M* | *SD* | *M* | *SD* |
| Extension | Pre-test | 68.92 | 24.94 | 77.74 | 44.52 | 60.40 | 19.45 |
| | Post-test | 69.62 | 40.18 | 95.47 | 51.18 | 56.63 | 19.67 |
| | Δ | 0.7 | | 17.73 | | -3.77 | |
| Flexion | Pre-test | 49.21 | 17.54 | 53.76 | 33.68 | 39.04 | 9.24 |
| | Post-test | 52.48 | 18.46 | 80.10 | 54.40 | 39.72 | 14.49 |
| | Δ | 3.27 | | 26.34* | | 0.68 | |
| Left Lateral Flexion | Pre-test | 92.15 | 49.85 | 78.51 | 40.39 | 85.87 | 46.49 |
| | Post-test | 105.13 | 58.64 | 102.05 | 54.33 | 58.43 | 29.23 |
| | Δ | 12.98 | | 23.54* | | -27.44* | |
| Right Lateral Flexion | Pre-test | 99.16 | 44.25 | 86.74 | 44.60 | 55.43 | 21.39 |
| | Post-test | 104.16 | 34.60 | 115.46 | 49.48 | 39.47 | 16.31 |
| | Δ | 5.00 | | 28.72* | | -15.96 | |

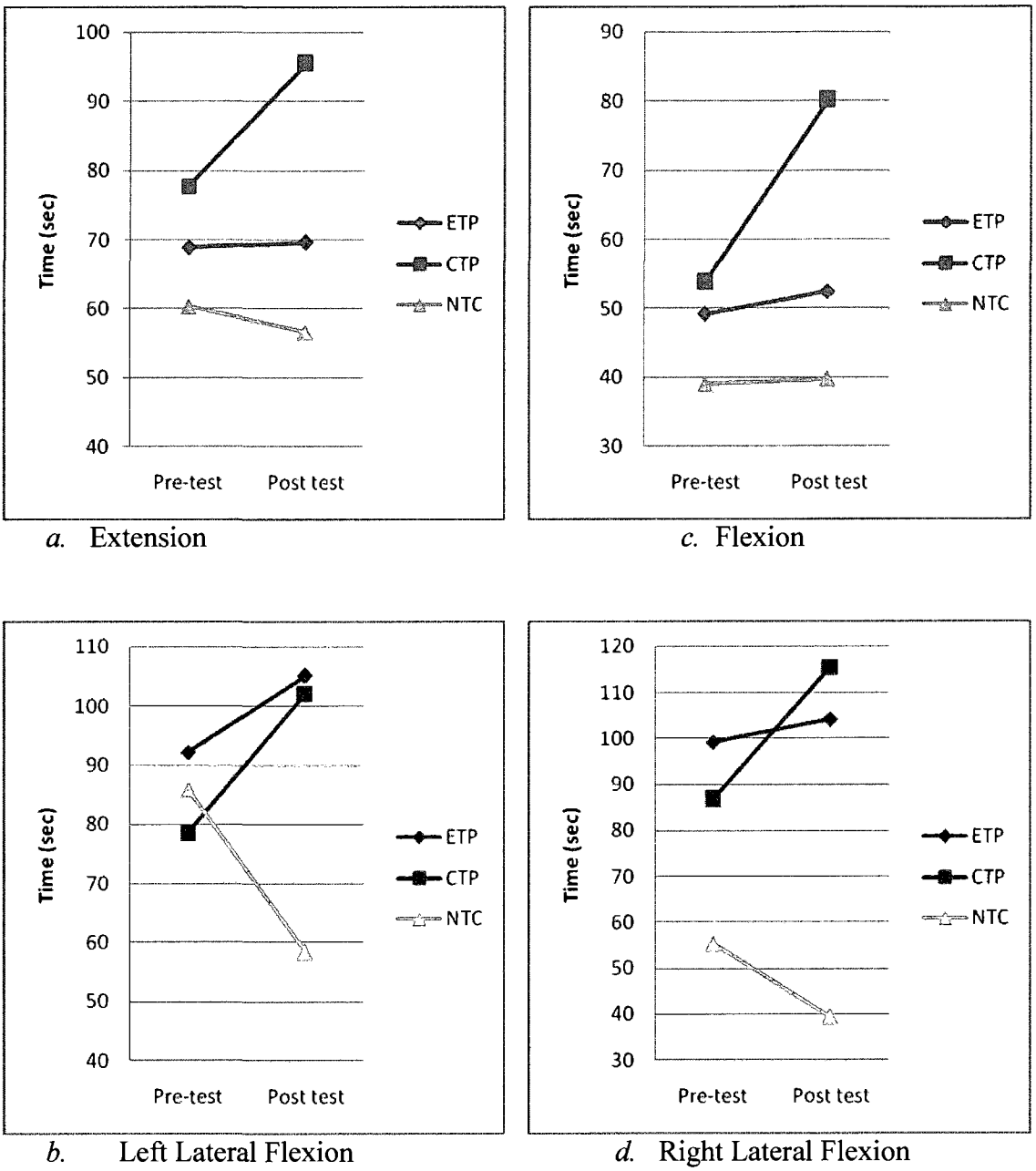
* Denotes statistical significance from pre- to post-test scores at $p \leq .05$

*Flexion.* The flexion direction produced a statistically significant main effect for Time, $F(2,26) = 7.42$, $p = 0.01$, $\eta^2 = 0.22$, and for the Time x Group interaction, $F(2, 26) = 4.93$, $p = 0.02$, $\eta^2 = 0.28$. The main effect for Group failed to achieve statistical significance, $F(2,26) = 2.42$, $p = 0.13$, $\eta^2 = 0.15$. The means for the pre- and post-test values are displayed in *Figure 12c.* The single degrees of freedom contrast for the CTP was statistically significant, $t(9) = 3.04$, $p = 0.01$. An interaction contrast was also conducted to examine the relationship between the CTP and the NTC, but failed to reach statistically significant, $t(16) = 1.60$, $p = 0.06$.

*Left lateral flexion.* Left lateral flexion showed no statistically significant improvements in endurance times for the main effect, Time, $F(2,24) = 0.20$, $p = 0.66$, $\eta^2 = 0.01$, or Group, $F(2,24) = 0.72$, $p = 0.50$, $\eta^2 = 0.06$. However, there was a statistically significant Time x Group interaction, $F(2,24) = 4.73$, $p = 0.02$, $\eta^2 = 0.28$ (refer to *Figure 12b*). Visual inspect of the resulting graph revealed an increase in left lateral flexion for the CTP and a decrease for the NTC. These changes were examined using a paired t-test. Statistically significant results were found for both the increase seen for the CTP group, $t(9) = 2.13$, $p = 0.03$, and the decrease that occurred for the NTC, $t(6) = 2.66$, $p = 0.02$. An interaction contrast comparing the CTP and the NTC yielded statistically significant results, $t(15) = 2.27$, $p = 0.02$, indicating that the time increase for left lateral flexion for the CTP differed from that seen for the NTC. The second interaction contrast for the ETP and the NTC was not statically significant level, $t(15) = 1.68$, $p = 0.06$.

*Right lateral flexion.* There was no statistically significant main effect for Time for right lateral flexion, $F(2,25) = 1.51$, $p = 0.23$, $\eta^2 = 0.06$, indicating no change in the endurance times from pre- to post-test. However, a statistically significant main effect for Group was isolated, $F(2,25) = 5.62$, $p = 0.01$, $\eta^2 = 0.31$. The interaction between Time x Group was also statistically significant, $F(2,25) = 6.71$, $p = 0.01$, $\eta^2 = 0.35$. The pre- and post-test means are displayed in *Figure 12d.* Visual inspection suggested the CTP was

potentially undergoing a significant improvement for pre- to post-test. This was analyzed and was found to be statistically significant, $t(9) = 3.51, p = 0.003$. An interaction contrast was conducted for the CTP versus the NTC and was found to be statistically significant, $t(15) = 2.70, p = 0.01$, indicating the presence of a significant difference between the CTP and NTC. Another interaction contrast was run comparing the CTP to the ETP; however, this interaction was not statistically significant, $t(18) = 0.9, p = 0.19$. However, the interaction contrast examining the mean scores pre- and post-test for the ETP and the NTC was statistically significant, $t(16) = 1.24, p = 0.01$, illustrating the presence of a difference between these two groups.




*a.* Extension

*c.* Flexion

*b.* Left Lateral Flexion

*d.* Right Lateral Flexion

*Figure 12.* Mean Times (seconds) Produced During a Fatigue Trial Conducted for the Four Tested Directions using a Resistance Load of 70% an Individuals MVC Pre- and Post-Treatment for the ETP, CTP and the NTC Groups.

### *4.4 Area Under the 70% Endurance Trial Force Curve*

A 3 (Group) x 2 (Time) mixed model ANOVA was used to assess the area under the force curve for the 70% endurance trials. For pre- and post-test results refer to Table 4 or *Figure 13*. The four directions examined were inspected for outliers exceeding the ±2 *SD* but none were found. During data collection, four of the data files (1 from CTP in extension, 1 from ETP and 1 from NTC for left lateral flexion and 1 from ETP for right lateral flexion) were corrupted during the data transfer and therefore, were excluded from the analysis.

*Extension.* Using a 3 x 2 ANOVA, no statistically significant Time main effect for extension was found, $F(2,25) = 0.03, p = 0.86, \eta^2 = 0.001$, indicating no change in the area under the curve had occurred pre- to post-training intervention for the three groups. No differences in the area under the curve were found for the two treatment conditions relative to the control as illustrated by the statistically non-significant main effect for Group, $F(2,25) = 0.93, p = 0.41, \eta^2 = 0.07$. The Time x Group interaction for the extension area under the curve values was also not statistically significant, $F(2,25) = 0.79$, $p = 0.46, \eta^2 = 0.060$ (refer to *Figure 14a*). Visual inspection of *Figure 14a* revealed a large increase from pre- to post-test for the CTP. This was examined using a paired t-test but the change was not statistically significant, $t(8) = 0.99, p = 0.18$. An interaction contrast comparing only the CTP to NTC failed to achieve statistical significance, $t(15) = 0.84, p = 0.21$.

*Flexion.* The flexion results for the area under the force curve yielded a statistically significant main effect for Time, $F(2,26) = 3.93, p = .05, \eta^2 = 0.13$. There was an overall increase in the area of the force curve from Time 1 ($M$=1043.46) to Time 2 ($M$=1207.60). There was no statistically significant main effect for Group,

Table 4: Area Under the Force Curve (lbs/sec) of the 70% Endurance Trial for ETP, CTP and NTC Groups. Values are Mean (*M*) and Standard Deviation (*SD*). Delta change (Δ) between the Pre- and Post-test Scores.

| | | ETP | | CTP | | NTC | |
|---|---|---|---|---|---|---|---|
| Directions Tested | | *M* | *SD* | *M* | *SD* | *M* | *SD* |
| Extension | Pre-test | 2132.81 | 978.97 | 2138.04 | 888.77 | 1963.32 | 675.13 |
| | Post-test | 2030.24 | 679.15 | 2552.86 | 1409.39 | 1767.85 | 588.64 |
| | Δ | -102.57 | | 414.82 | | -195.47 | |
| Flexion | Pre-test | 1156.90 | 473.79 | 1067.55 | 463.12 | 905.94 | 372.43 |
| | Post-test | 1203.79 | 431.48 | 1495.02 | 784.38 | 923.99 | 416.13 |
| | Δ | 46.89 | | 427.47* | | 18.05 | |
| Left Lateral Flexion | Pre-test | 1990.02 | 552.79 | 1848.14 | 742.38 | 2387.31 | 1515.05 |
| | Post-test | 2234.25 | 1086.12 | 2321.30 | 1353.58 | 1565.13 | 920.86 |
| | Δ | 244.23 | | 473.16 | | -822.18* | |
| Right Lateral Flexion | Pre-test | 2009.06 | 812.74 | 1971.20 | 918.12 | 1411.24 | 748.52 |
| | Post-test | 2353.31 | 492.25 | 2660.96 | 1341.31 | 935.90 | 403.27 |
| | Δ | 344.25 | | 689.76* | | -475.34 | |

* Denotes statistical significance from pre- to post-test scores at $p \leq .05$

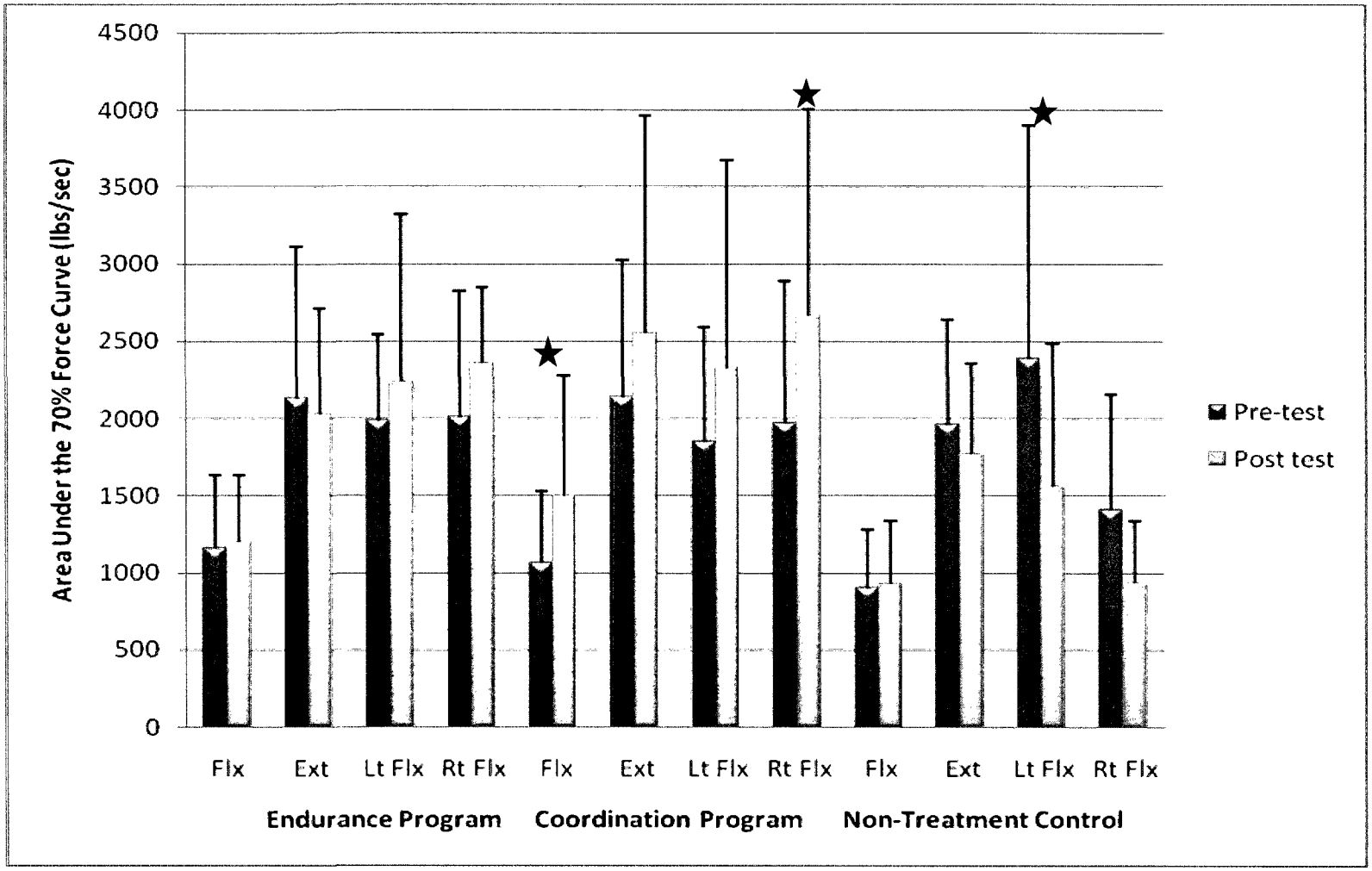



*Figure 13:* Area Under the 70% Endurance Force Curve (lbs/sec) for the Pre- and Post-Test for Flexion (Flx), Extension (Ext), Left lateral Flexion (Lt Flx) and Right lateral Flexion (Rt Flx) in ETP, CTP and NTC Groups

★ Denotes Statistical Significance between Pre- and Post-test Scores

$F(2,26) = 1.44, p = 0.26, \eta^2 = 0.10$. The Time x Group interaction also failed to achieve statistical significance, $F(2,26) = 2.60, p = 0.09, \eta^2 = 0.17$, illustrating a lack of difference in the pre- and post-test values for the ETP, CTP and the NTC. The means for the pre- and post-test values for ETP, CTP and NTC are displayed in *Figure 14c.* Despite the lack of statistical significance isolated for the main effect for Group, *Figure 14c* illustrates a large improvement in flexion strength for the CTP group. This improvement was examined using a paired t-test and the change from pre- to post-test for the CTP was found to be statistically significant, $t(9) = 2.67, p = 0.01$. However, no statistically significant interaction contrasts occurred between the CTP versus ETP and the NTC, $t(19) = 1.49, p = 0.08$ and $t(16) = 1.20, p = 0.12$, respectively. Although not statistically significant, the interaction contrast reveals a large difference in the areas under the force curve between the pre- and post- test mean scores for the CTP and the ETP.

*Left lateral flexion.* The area under the force curve results for left lateral flexion failed to produce a statistically significant main effect for both Time, $F(2,24) = 0.04, p = 0.85, \eta^2 = 0.002$, and Group, $F(2,24) = 0.04, p = 0.96, \eta^2 = 0.004$. However, a statistically significant result was found for the Time x Group interaction, $F(2,24) = 4.26, p = 0.03, \eta^2 = 0.26$. Examination of the cell means for the area under the force curve indicated an increase in the ETP from Time1 ($M$=1990.02 ± 552.79 lbs/sec) to the Time 2 values ($M$=2234.25 ± 1086.12 lbs/sec), but a larger increase was seen for the CTP Time 1 values ($M$=1848.14 ±742.38 lbs/sec) to Time 2 values ($M$=2321.20 ± 1353.58 lbs/sec). This is contrasted against the decrease seen from Time 1 ($M$=2387.31 ± 1515.05 lbs/sec) to Time 2 ($M$=1565.13 ± 920.86 lbs/sec) for the NTC (refer to *Figure 14b*). The interaction contrast for the CTP and NTC was statistically significant, $t(15) = 2.15, p = 0.02$, but no statistically significant interaction for the ETP vs. NTC was found, $t(15) = 1.62, p = 0.06$. The finding for the CTP and NTC interactions indicates that the pre- and post-test values differ for the area under the force curve. While the pre- and post-test values for the ETP

and the NTC differed in regard to the area under the curve, this difference was not statistically significant. Using a paired t-test, a single degree-of-freedom contrast was conducted for the CTP and the NTC. The improvement seen in the CTP failed to reach statistical significance, $t(9) = 1.61, p = 0.07$, but a statistically significant decrease was seen for the NTC, $t(6) = 2.54, p = 0.02$.

*Right lateral flexion.* The right lateral flexion area under the force curve results yielded a non-statistically significant main effect for Time, $F(2,25) = 1.95, p = 0.18, \eta^2 = 0.072$. However, the main effect for Group was statistically significant, $F(2,25) = 4.92, p = 0.02, \eta^2 = 0.282$, indicating the existence of differences between the ETP, CTP, and the NTC whose scores were respectively $M$=2181.19 lbs/sec, $M$=2316.08 lbs/sec, $M$=1173.57 lbs/sec. A statistically significant finding was also found for the Time x Group interaction, $F(2,25) = 5.96, p = 0.01, \eta^2 = 0.32$. The ETP increased from Time 1($M$=2009.06 ± 812.74 lbs/sec) to Time 2 ($M$=2353.31 ± 492.25 lbs/sec), and a similar but slightly larger increase was seen for the CTP from Time 1 ($M$=1971.20 ± 918.12 lbs/sec) to Time 2 ($M$=2660.96 ± 1341.31 lbs/sec). Both of these increases for the CTP and the ETP were analyzed using single degree-of-freedom contrasts (see *Figure 14d*). The paired t-test results for the CTP were statistically significant, $t(9) = 2.78, p = 0.01$, but the increase for the ETP was not statistically significant, $t(10) = 1.73, p = 0.06$. An interaction contrast was conducted to compare the pre- and post-test results for the CTP and the NTC, which was statistically significant, $t(16) = 2.44, p = 0.01$. The increase in the area under the force curve from pre- to post-test for the CTP was different when compared to the decrease from Time 1 ($M$=1411.24 ± 748.52 lbs/sec) to Time 2 ($M$=935.90 ± 403.27 lbs/sec) experienced by the NTC. The interaction contrast comparing the ETP to the NTC was also statistically significant, $t(15) = 1.89, p = 0.04$, indicating the presence of a difference between the right phase shift for the ETP and the left phase shift for the NTC.

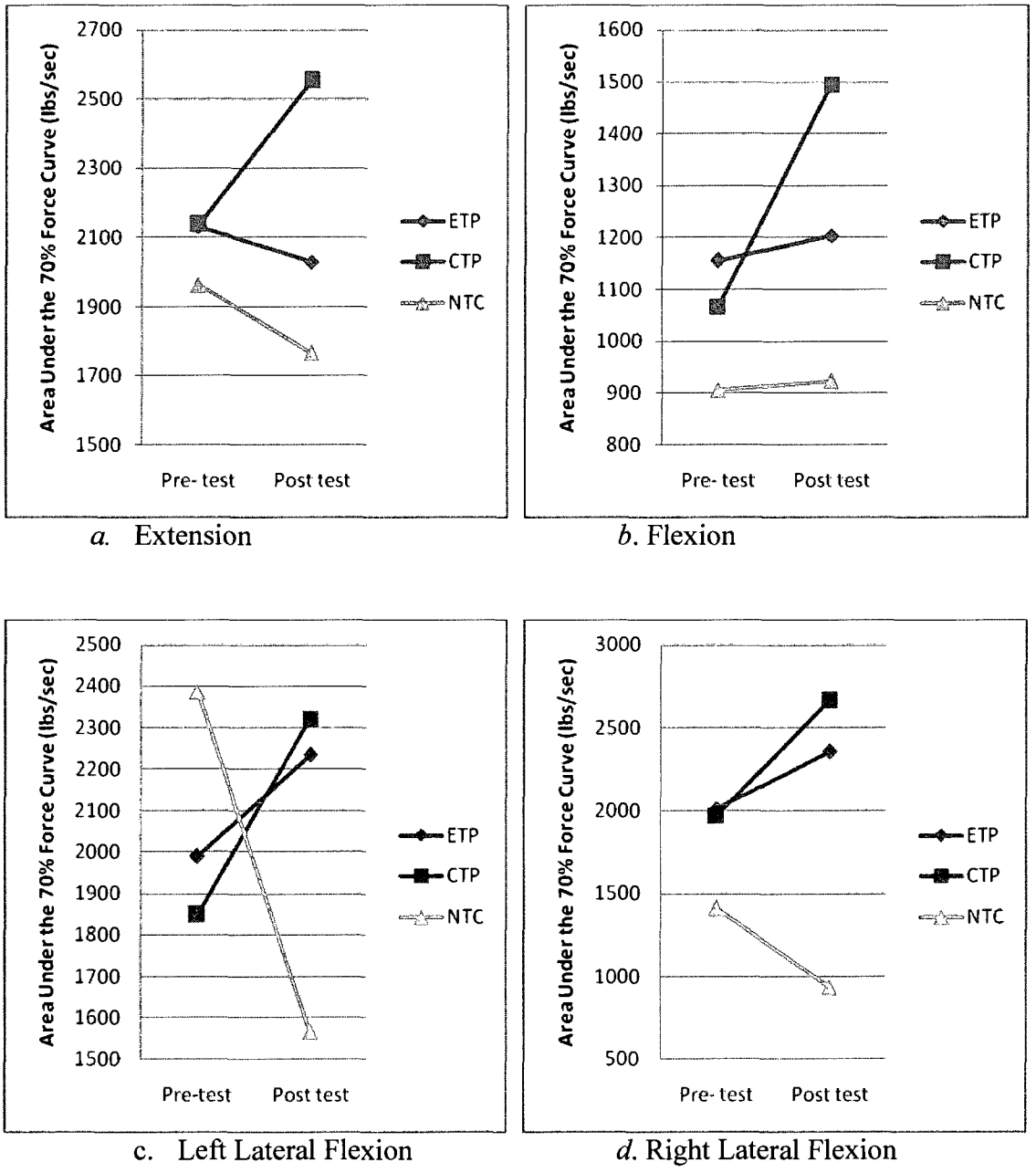



*Figure 14.* The Mean Areas Under the 70% Endurance Force Curve (lbs/sec) for the Four Tested Direction in the ETP, CTP and the NTC Groups During the Pre- and Post-test Assessments.

### *4.5 Visual analog scale (VAS) self-reported neck pain scores*

In the familiarization session, the first baseline self-reported neck pain scores were collected for 'pain at present', 'pain in general' and 'pain at worst' on a VAS scale for both treatment groups. The control group was asked to rate their baseline neck pain on VAS sheets during their baseline assessment, but only three of the 14 volunteers completed the required form. Due to the large amount of missing data, we were unable to include the NTC data in the analysis of neck pain scores. It was also asked that the entire sample complete a neck pain VAS sheet after every night and day flight. However, only a few individuals complied with this request, therefore we were not able to obtain a clear indication of self-reported neck pain over the course of the treatment condition. With the data collected, the analysis of the self reported neck pain scores was limited to the baseline and post-test assessment for the ETP and the CTP.

Based on previous research analyzing self reported neck pain on a 100m VAS pre- and post-training intervention, a 10mm decrease in the neck pain VAS score was viewed as a clinically important finding (Linton, 1986). The ETP produced a 10mm drop in self-reported neck pain scores in 1 individual in the 'pain at present' category, 2 individuals in the 'pain in general' category and 6 individuals in the 'pain at worst' category. For the CTP, a 10mm reduction was achieved by 2 individuals in the 'pain at present' category, 3 individual in the 'pain in general category' and 5 individuals in the 'pain at worst category'.

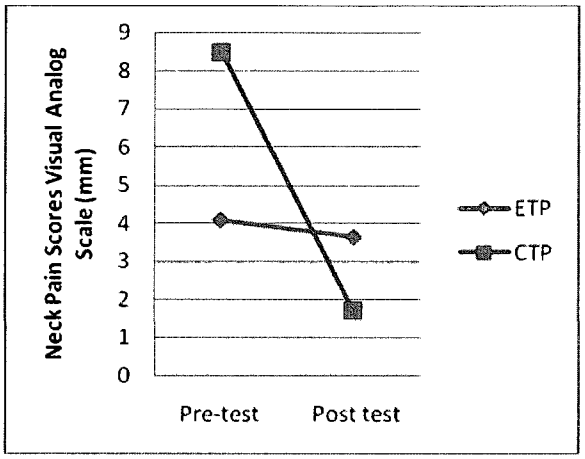
*Pain at present.* A 2 (Group) x 2 (Time) ANOVA was conducted for the self-reported neck pain scores for 'pain at present' and showed a no statistically significant main effect for Time, $F(1,19) = 2.30, p = 0.15, \eta^2 = 0.11$, or Group, $F(1,19) = 0.17, p = 0.67, \eta^2 = 0.01$. The Time x Group interaction was also statistically non-significant, $F(1,19) = 1.76, p = 0.20, \eta^2 = 0.09$, indicating no statistical difference in the neck pain

scores for 'pain at present' between the two treatment groups from the pre- to the post-test assessment. *Figure 15a* illustrates the mean scores pre- and post-intervention for the ETP and the CTP. The mean VAS scores for the ETP for Time 1 were $M$=4.09 ± 9.17mm and Time 2 were $M$=3.60 ± 5.04mm. The Time 1 and Time 2 mean VAS score for the CTP were $M$= 8.50 ± 13.75mm and $M$= 1.7 ± 3.94mm, respectively. The decrease seen in CTP was examined using a paired t-test, but was not statistically significant, $t(9) = 1.59$, $p$= 0.07. Although, the findings failed to achieve statistical significant for 'pain at present' they came close to approximating $p = .05$ and therefore should be mentioned.

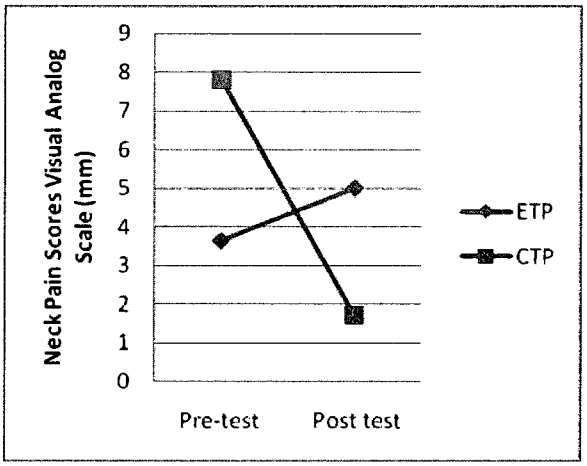
*Pain in General.* The neck pain score results for the main effect for Time and the main effect for Group from the 2 x 2 ANOVA for 'pain in general' were $F(1,19) = 5.25$, $p = 0.16$, $\eta^2 = 0.10$, and $F(1,19) = 0.03$, $p = 0.87$, $\eta^2 = 0.002$, respectively. These results indicated that with respect to 'pain in general', there was no statistical difference from pre- to post-test assessment or between the ETP and the CTP. A statistically significant Time x Group interaction was produced, $F(1,19) = 5.25$, $p = 0.03$, $\eta^2 = 0.22$ (refer to *Figure 15b*). This interaction was the result of the increase seen for Time 1 ($M$=3.64mm ±5.04mm) to Time 2 ($M$= 5.00mm ±7.42mm) for the ETP contrasted against the decrease from Time 1 ($M$= 7.8mm ±9.58mm) to Time 2 ($M$= 1.7mm ±3.94mm) revealed by the CTP. The results of a paired t-test looking at the decrease in the VAS score from pre- to post-test found a statistically significant decrease had occurred for the CTP, $t(9) = 2.47$, $p = 0.02$, (see *Figure 16*).

*Pain at worst.* The 2 x 2 ANOVA 'pain at worst' VAS scores relating to the self-reported neck pain of the treatment population produced a statistically significant result for the main effect for Time, $F(1,19) = 8.61$, $p = 0.01$, $\eta^2 = 0.31$, indicating a difference from the pre-to post-test assessment. However, there was no statistically significant main effect for Group, $F(1,19) = 0.06$, $p = 0.81$, and $\eta^2 = 0.003$, indicating that no statistical difference existed between the two treatment conditions. There was no statistically

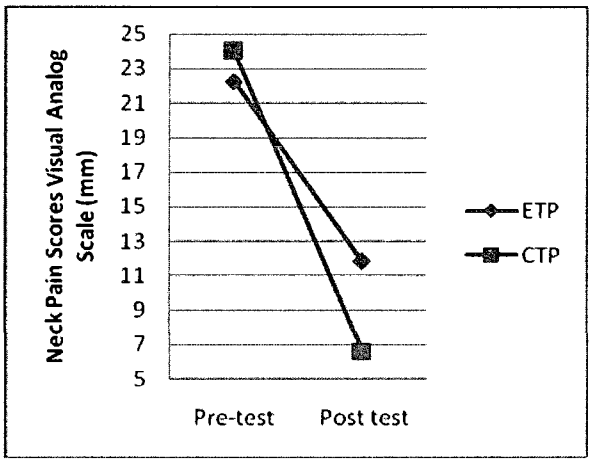
significant result for Time x Group interaction, $F(1,19) = 0.06$, $p = 0.47$, and $\eta^2 = 0.003$ (see *Figure 15c*). Visual inspect of *Figure 15c* revealed that both treatment conditions underwent large decreases in the self-report 'pain at worst': the CTP decreased from $M$=22.27 ± 22.73 mm to $M$=11.82 ±18.34mm, and the ETP decreased from $M$= 24 ± 24.13mm to $M$= 6.6 ± 8.58mm. Single degree-of-freedom contrasts were conducted for both the CTP and the ETP using paired t-tests to determine if the respective decreases were statistically significant. Both CTP and ETP underwent statistically significant decreases in 'pain at worst', $t(9) = 2.06$, $p = 0.04$, and $t(10) = 2.18$, $p = 0.03$, respectively (see *Figure 16*).




a. Pain at Present




b. Pain in General




c. Pain at Worst

*Figure 15.* Mean Pre- and Post-test Scores of Self-reported Neck Pain on a Visual Analog Scale for 'Pain at Present' 'Pain in General' and 'Pain at Worst' for the ETP, CTP, and NTC Groups.

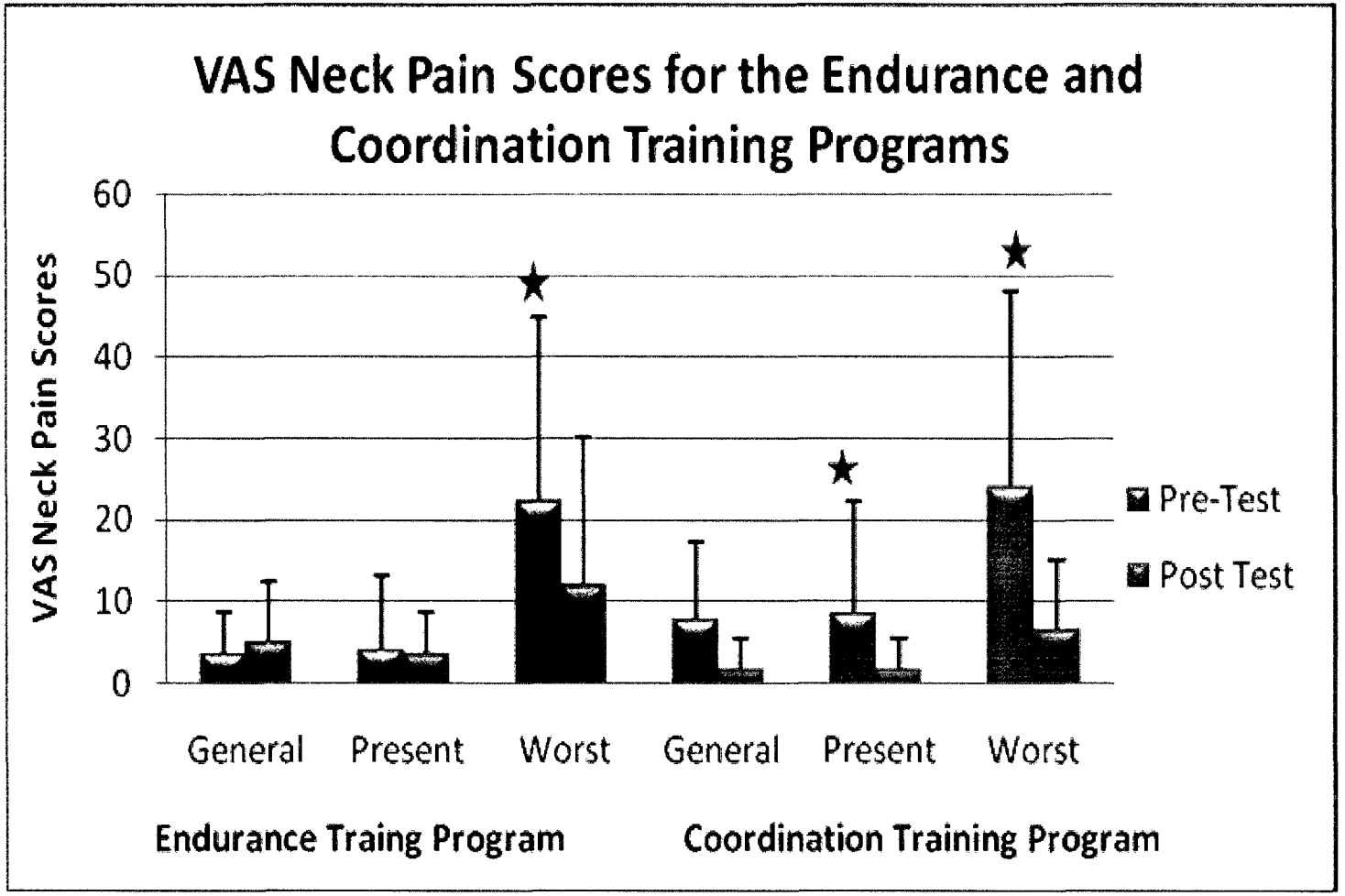



*Figure 16.* Mean Self Reported VAS Neck Pain Scores (mm) for 'Pain in General', 'Pain at Present', and 'Pain at Worst' Pre- and Post-Intervention for the ETP and CTP Treatment Groups.
★ Denotes Statistically Significance Decrease Between Pre- to Post-test Scores.

# 5. Discussion

Previous research has established that neck pain is a medical concern with occupational relevance for the Canadian Forces CH-146 helicopter aircrew (Adam, 2004; Forde, 2008; Harrison et al., 2007; Harrison, *in press-a, in press-b, in press-c*). The provision of cervical exercise training programs has been found to reduce in-flight muscle strain (Sovelius et al., 2006), work days lost (Hamalainen et al., 1998), and to improve isometric and dynamic strength (Burnett et al., 2005; Taylor et al., 2006) in the cervical musculature of military populations with an emphasis on fighter pilots. To the best of my knowledge, there has been no research examining the effects of a cervical exercise training programs for helicopter aircrew and the occurrence of neck pain. Thus, the purpose of my study was to evaluate the effects of two different cervical exercise training programs on improving isometric strength and endurance while reducing self-reported neck pain in a cohort of CH-146 aircrew. The results of this study indicated that both a low load muscular endurance training program (ETP) and a muscular coordination training program (CTP) achieved clinically and statistically significant reductions in self-reported neck pain for 'pain at worst' on a visual analog scale (VAS). However, only those subjects in the CTP achieved statistically significant reductions in 'pain in general' on the VAS. The prescribed training programs produced a positive trend towards improved cervical strength and endurance relative to the non-treatment control group (NTC). The greatest improvements in isometric strength and endurance occurred for those subjects in the CTP.

## *5.1 Maximal voluntary contractions*

Research has demonstrated marked hypertrophy of the cervical musculature following neck specific training programs when assessed using magnetic resonance imaging (Conley et al., 1997a; Conley et al., 2007b) and computerised tomography

(Portero, Bigard, Gamet, Flageat, & Guezennec, 2001). There is improved MVC strength after resistance and endurance training programs in the cervical muscle for both non-military (Nikander et al., 2006; Ylinen et al., 2003; Ylinen et al., 2006) and military subjects (Alricsson et al., 2004; Burnett et al., 2005; Sovelius et al., 2006; Taylor et al., 2006). Consistent with the previous research, the current study found improved isometric strength for both the CTP and ETP. The aircrew randomly assigned to the CTP experienced the greatest improvements in isometric force. This group exhibited improvement in cervical extension (11.24%), flexion (13.76%), left (7.64%) and right lateral flexion (15.9%). The aircrew in the ETP underwent slightly smaller improvements in isometric force: 4.18% for extension, 1.54% for flexion, 7.47% for left lateral flexion and 14.35% for right lateral flexion. The percent improvement found in my study was smaller than improvements recorded in other training studies that employed higher intensity training interventions. Burnett et al. (2005), found improvements in cervical flexion (62.9%), extension (64.4%), left (53.3%) and right lateral flexion (49.1%); while Taylor et al. (2006) found cervical strength improvements of 72.1%, 45.9%, 83.0%, 71.8%, respectively. The use of high load resistance exercises would have produced a fibre transition and hypertrophy favouring fast twitch and fast-oxidative-glycolytic fibres, allowing the muscles to contract at higher levels of force (Powers & Howley, 2004). This is consistent with data from computerised tomography study showing that a resistance load of 70% of the MVC resulted in muscle hypertrophy of the cross-sectional area of the paravertebral muscles. In contrast to a resistance load of 30% MVC that elicited no such hypertrophy in the paravertebral muscles (Danneels et al., 2001).

Another plausible explanation for the large improvements reported by Burnett et al. (2005) and Taylor et al. (2006) is the initial baseline isometric MVC levels recorded. Burnett et al. (2005) reported the smallest initial mean isometric MVC values ranging from 13.4 - 24.0 lbs for the two treatment conditions and the control. Taylor et al. (2006)

reported a mean range of 26.5 - 41.2 lbs for the treatment and the control. In the current study, mean values ranged from 35.05 - 53.18 lbs (see Table 2). These results are comparable to previous work conducted using CH-146 aircrew whose mean values ranged from 38.13 - 51.63 lbs (Harrison, *in press-b*). The lower initial isometric force values recorded at baseline would permit a greater percent improvement in cervical force over the course of the intervention (Sharkey, 1970). Therefore the elevated force values found in our study at baseline may provide a possible explanation for smaller improvements in isometric force.

The discrepancies in the baseline isometric values between the three studies may be explained by the loads imposed on the cervical musculature in the occupational environments of the different subjects. The CH-146 aircrew are expected to wear a helmet and commonly use NVG which increases the mass of the head, thereby enhancing the load supported by the cervical musculature. During flight, CH-146 aircrew must adopt suboptimal postures (Forde, 2008) and are exposed gravitational forces within the range of +1-2 Gz (Äng & Harms-Ringdahl, 2006). Exposure to these stressors leads to the expectation that some form of strength adaptation in the cervical musculature will occur. In comparison, the gravitational load imposed on the cervical spine in upright posture is insufficient to generate muscle hypertrophy as found during a typical resistance training program that excluded neck specific exercises. To improve neck muscle function, neck specific exercises are required (Conley et al., 1998a; Conley et al., 1998b). Taylor et al. (2006) studied military personnel in the Navy who maintain a similar fitness level, but do not wear a flight helmet, NVG or face exposure to gravitational forces. Burnett et al. (2005) used a sample of 36 males but did not report their occupations. I expected that the subjects Burnett et al. (2005) and Taylor et al. (2006) studied would have lower baseline MVC scores as no previous neck specific exercises experience had been reported and their occupational environments placed little mechanical loads on the spine.

Elastic tubing as a form of resistance has been used for resistance/endurance exercise training and has successfully increased neck isometric strength (Burnett et al., 2005; Ylinen et al., 2006). However, difficulties with this form of resistance arise when attempting to quantify the resistance load. Ylinen et al. (2006), they were able to quantify the load to 80% of the subject's MVC for flexion and extension using an isometric strength testing device. Variability in the resistance during the exercise was avoided by performing isometric contractions where the head and neck were "held strictly stable in relation to the trunk" (Ylinen et al., 2006, p. 8) while the trunk was bent obliquely forwards and backwards. Given the specificity response of musculature to resistance training, the maintenance of isometric contractions is a movement pattern with little occupational relevance for our subject population. Helicopter pilots and flight engineers are constantly required to perform low level dynamic contractions to scan and view flight panel instruments and the outside environment.

Burnett et al. (2005) compared two training modalities: a Multi-Cervical Unit and Thera-Band tubing (Pro-Med Products, Inc.) over a 10-week training period relative to a control group. The Thera-Band tubing group used the thickness and length of four different bands of tubing to provide resistance load during dynamic cervical flexion, extension, left and right lateral flexion. The Multi-Cervical Unit group exhibited significant improvements in isometric MVCs. Statistical significance was achieved in all four testing directions relative to the control. Those assigned to the Thera-band modality experienced a 42.0% improvement in flexion, 29.8% for extension, 26.9% for left lateral flexion and 24.3% for right lateral flexion. When the Thera-band modality was compared to the non-treatment control, flexion was the only direction that reached significance. This finding is similar to the results for the ETP in the current study, where only one direction significantly improved relative to the control. In both studies, the significant finding was isolated for the direction that demonstrated the largest improvement. For the current

study, this occurred for right lateral flexion compared to Burnett et al. (2005) where the change was found for flexion.

The results of our study are similar to those seen using Finnish air cadets looking at the effect of trampoline exercise versus strength training in reducing in-flight muscle strain, cervical load and improving isometric MVC strength (Sovelius et al., 2006). The strength training group used a resistance load of approximately 15 to 30% of MVC for flexion, extension and isometric rotation that was performed 2 to 3 times a week for 6 weeks. Sovelius et al (2006) reported improvements of 2.3% for flexion and 6% for extension. However, this study had no control group and the time frame used was half the length of our training intervention. Yet the increases were approximately equivalent to those seen in the ETP group albeit smaller than the improvements seen for the CTP. Sovelius et al. (2006) found improved cervical strength "decreased muscle strain in-flight and during the cervical load testing" (p. 23). The interventions appeared to be most effective for those muscles responsible for head and neck movements, the cervical erector spinae and the sternocleidomastoid, and less effective for the larger posture stabilizers, such as the trapezius and thoracic erector spinae (Sovelius et al., 2006). The environment in the helicopter likely elicits less in-flight cervical muscle strain compared to +Gz forces experienced in fighter jets. Additional factors such as the load of the flight helmet, NVG and vibration must be considered when examining cervical muscle strain in helicopter aircrew. As such, the ability to reduce in-flight muscle strain for CH-146 aircrew may potentially have a significant impact on flight related neck pain.

Both CTP and the ETP training programs employed low intensity exercises, leading to the expectation that the improvements in maximal cervical muscle force would be small. The literature suggests that resistance training in the cervical spine using submaximal loads can have positive effects on maximal force (Hämäläinen & Vanharanta, 1998; Sovelius et al., 2006). This is supported by the small improvements in

isometric maximum force made in both the ETP and the CTP as predicted. I hypothesized that the ETP would have exhibited the greatest improvements in maximal isometric force as a 30% resistance load was used throughout the intervention. In contrast to the CTP where the first 6 weeks of the training were focused on strengthening the deep cervical stabilizers using only the mass of the head as resistance. When compared to the NTC, this held true; the ETP achieved the only statistically significant result for mean pre- and post-test scores in right lateral flexion. However, the post-test results for the four directions tested revealed that the greatest percent improvement in force occurred for the CTP. This was unexpected as the ETP had been training at a resistance load of 30% of their MVC from the outset. It was therefore expected that the greatest improvements in maximal isometric force would be in the ETP. These results could be reflective of the fact that the attendance for the ETP was just over 50% compared to 76% for the CTP.

Previous work with CH-146 and other rotary winged aircrew identified discrepancies between the cervical musculature on the right and left sides (Dieën, 1996; Forde, 2008; Harrison et al., 2007; Harrison, *in press-a, in press-b*; Lopez-Lopez et al., 2001). The mean baseline values for left lateral flexion were higher than those seen for right although the difference was only significant in the ETP. This differs from research looking at a sample of healthy 20-84 year old males ($n$=91) where isometric muscle strength for right lateral flexion was greater than left lateral flexion (Chiu, Lam & Hedley, 2002). It is assumed that this force discrepancy would be the norm for the general population, which is predominantly right-handed. The ergonomic design of the CH-146 Griffon cockpit forces aircrew to display a trend towards greater isometric force on left side despite the predominant right handedness pattern displayed by 85% of our aircrew (Forde, 2008; Harrison, *in press-b*). The exercise training interventions (i.e., CTP and ETP) displayed a positive trend toward reducing the left-right side difference seen at baseline, but neither decrease was statistically significant. Research using near infrared

spectroscopy (NIRS) to measure tissue oxygenation shows that the posture adopted by helicopter pilots imposes a larger metabolic demand on the right side (Harrison et al., 2007). The greatest improvements in maximal force for the present study were found for right lateral flexion in both the CTP and ETP. The improved strength of the cervical muscle on the right side would enhance the functional capacity of the muscle and could plausibly play a role in reducing the onset of fatigue and pain.

### *5.2 70% Endurance trial times to fatigue*

Research has shown muscular endurance to be most relevant to occupational environment of the CH-146 aircrew (Äng et al., 2005; Forde, 2008; Harrison, *in press-a*; Netto et al., 2007). The importance of muscular endurance is based on previous research examining the relationship between force properties of the cervical muscle and the presence of neck pain in CH-146 aircrew. No associations were identified for CH-146 aircrew reporting neck pain and decreased cervical strength or endurance (Harrison et al., *in press-b*). Altered myoeletric manifestations were identified during submaximal contractions to fatigue in helicopter pilots reporting neck pain when compared to pain free counterparts (Äng et al., 2005).

The literature has shown that the use of submaximal contractions of the cervical muscle accurately depicts muscular fatigue (Alricsson et al., 2004; Conley et al., 1997a; Conley et al., 1997b; Randløv et al., 1998; Taylor et al., 2006; Ylinen et al., 2003). Submaximal isometric contractions in the extension direction have been used to evaluate fighter pilots time to fatigue. They were found to improve with participation in a neck strengthening program. Alricsson et al. (2004) used a predetermined load of 196 N to evaluate cervical fatigue, making comparison to our study difficult because the resistance load used was determined as a percentage of each subject's MVC. Isometric submaximal contractions to fatigue have been examined in a helicopter pilots and were found to be

reliable for evaluating of cervical muscle fatigue (Harrison, *in press-a, in press-b*; Thuresson et al., 2005). Harrison (*in press-a*) examined the normalized median frequency EMG signal of the cervical muscle of 40 CH-146 aircrew members as an indicator of neuromuscular fatigue. The initial baseline times to fatigue for the four directions tested in our study were similar to the times reported by Harrison (*in press-a*). The longest time to fatigue in both studies occurred during left lateral flexion and the shortest for flexion. The actual times to fatigue recorded during the baseline assessment in our study were longer in every direction compared to those reported by Harrison (*in press-a*).

Over the course of the intervention, the CTP showed the greatest improvements in times to fatigue, reporting increases in all four directions; with the exception of extension all improvements were significant. The improvements in flexion, left and right lateral flexion for the CTP also proved to be statistically significant when compared to the NTC. While the ETP showed no change in times to fatigue for extension, there was an increase for flexion, left and right lateral flexion albeit not significant when compared to baseline. However, the left and right lateral flexion time improvements were statistically significant when compared to the control. The significant changes for the ETP for left and right lateral flexion may reflect the large decreases seen in the NTC rather than actual improvements in the ETP. The large decreases in post-test times to fatigue in left and right lateral flexion for the NTC could be attributed to reports of delayed onset muscle soreness in the cervical muscle after the baseline assessment. As left and right lateral flexion contractions are performed to a smaller extent than flexion and extension contractions in normal daily activities, stiffness and pain may have been more pronounced with these movements.

Based on the increased endurance times for the CTP, it can be postulated that the provision of muscle coordination program which focussed on strengthening the deep cervical stabilizers prior to beginning a general neck endurance program enhanced

muscular endurance of the cervical musculature. During submaximal endurance trials to volitional fatigue with a resistance load equivalent to the one used in our study, bilateral EMG monitoring of the splenius capitis, sternocleidomastoid and upper trapezius was used to identify neuromuscular fatigue (Harrison et al., *in press-a*). The results indicated that the small muscles were the first to fatigue, while no decrease in the normalized median frequency was observed for the upper trapezius. Harrison et al. (*in press-a*) recommended a neck training program that targeted the endurance capacity of the smaller cervical muscles e.g., splenius capitis and the sternocleidomastoid. The CTP had small improvements in isometric force, but large increases in times to fatigue for flexion, left and right lateral flexion. These results combined with the neuromuscular data of Harrison (*in press-a*) suggests a plausible physiological explanation. The improved endurance capacity of the smaller cervical muscles like the splenius capitis and the sternocleidomastoid could have positively impacted the maintenance of the submaximal load during the endurance trial to fatigue, while minimally improving maximal force production. This is speculative, but future research should examine the physiological response of the cervical musculature to a training intervention using neuromuscular and metabolic measures.

### *5.3 Area Under the 70% Endurance Trial Force Curves*

To compliment the data on endurance times to fatigue, I calculated the area under the 70% endurance force curves which is a reflection of the force maintained throughout the fatigue trial. For neither treatment group the extension direction did not change significantly as was the case for the times to fatigue. The CTP experienced a right phase shift after the training intervention; however, this phase shift was the smallest mean increase of the four directions tested. For the extension direction, the ETP and the NTC both exhibited left phase shifts. CTP produced the only statistical significant right phase

shift for the flexion, increasing the area of the force curve by 427.47lbs/s, while the ETP and the NTC remained relatively unchanged. Although not statistically significant, there was a trend towards pre- and post-treatment difference between the CTP and ETP. For left lateral flexion, both the ETP and CTP produced right phase shifts, with the CTP phase shift being approximately double that of the ETP. Compared to the statistically significant left phase shift in the NTC, the right phase shift of the CTP was significantly different for left lateral flexion. The significant right phase shifts for right lateral flexion were the largest produced for both the ETP ($M$ = 344.25 lbs/s) and the CTP ($M$ = 689.76 lbs/s); when compared to the NTC. The results indicate that the provision of an ETP had minimal effect on improving the area of the 70% force curve for extension and flexion. Right phase shifts were produced for both left and right lateral flexion in the ETP but were not statistically significant. The CTP incurred post-test increases in the area under the 70% force curve for all four directions. Statistical significance was achieved for flexion and right lateral flexion, while left lateral flexion approximated statistical significance.

After the 12-week intervention, subjects in the CTP were successfully able to sustain 70% isometric force load for a greater length of time before experiencing fatigue. Surface EMG identifies the small agonist muscles of the cervical spine as the first muscles to fatigue during submaximal endurance protocols (Harrison, *in press-a*). Given these results, one could hypothesize that exercises focused initially on muscle coordination, enhanced the strength and endurance capacity of the deep and small cervical musculature.

### *5.4 Summary*

To the best of my knowledge, this research represents the first study to examine the influence of specific exercise training programs on neck pain and physical function in

military helicopter aircrew. Work in a laboratory has examined the neuromuscular effect of wearing a helmet, NVG and a counterweight in a variety of flexed and rotated cervical postures, and suggests that activity levels of 16% MUR occur in the most extreme cervical postures (Thuresson et al., 2005). In-flight EMG analysis at +1Gz in fighter pilots identified upper limit cervical activity levels of approximately 15% of the maximal isometric force (Netto et al., 2007). Although occupational conditions vary between the two modes of flight, helicopter aircrew should experience gravitational forces of approximately +1Gz. Given the previous results, upper limit cervical activity threshold levels may be comparable.

Research on lumbar neuromuscular activity of helicopter pilots during flight in the right and left erector spinae muscle s found that 88% of the subjects experienced contraction levels below 5% of MVC. During an average 2 hr flight, the erector spinae of the entire sample population operated at a level well below the threshold of 14% MVC (de Oliveira & Nadal, 2004). Based on this previous research, it can be concluded that during a flight, CH-146 aircrew cervical spines' are contracting at low submaximal levels. The *Cinderella Hypothesis* proposes a stereotypical motor recruitment pattern occurs during low level submaximal tasks. This recruitment pattern leads to the selective overloading of Type I muscle fibres. If this continues for extended periods, it would lead to the development of muscle damage and potentially ragged red fibres (Visser & Dieën, 2006). This is one plausible hypothesis for the development flight related neck pain in CH-146 aircrew. Hagg (2000), reviewed ten biopsy studies of which eight focused on the upper trapezius, and one examined the first dorsal interosseous muscle, and one the extensor carpi radialis brevis muscle. These studies consisted of predominantly female populations with myalgic complaints as a result of static or highly repetitive work. Those reporting myalgic complaints when compared to pain free controls demonstrated decreased capillary density per fibre cross-sectional area. There was increased association

between the number of ragged red fibres (CytOx negative fibres) and the severity of the myalgic complaint (Hagg, 2000). At baseline, our subjects reported a mean range of rotary hour flight experience of 1210-2225 hrs, with the average flight lasting 1.36-1.59 hrs, and the longest NVG flight ranging from 2.45-3.25hrs. Given the flight histories and low level repetitive dynamic movements that occur in flight, Hagg (2000) conclusions could plausibly have implications for the current population.

The requirements imposed on the cervical spine of CH-146 aircrew includes the ability to withstand vibration, sub-optimal postures, and supporting the mass of head, helmet, NVG and for some, counterweights while continually scanning the cockpit and the outside environment. Adam (2004) found the prevalence of neck pain in this aircrew to be the range of 81%. In response, I tested two training programs in a sample of CH-146 aircrew. Based on neuromuscular activation level from laboratory data, a resistance load of 30% of MVC was selected (Netto et al., 2007; Sovleius et al., 2008; Thursseson et al., 2005). The principle of training *specificity* state that the adaptations occurring in the exercised muscle will reflect the type of training (Powers & Howley 2004). Therefore, the greatest improvement in the CTP and ETP were expected in the submaximal trial to fatigue, which is supported by our results. Few changes occurred in maximal force for either training group. Rather, the greatest improvements were in the times to fatigue and the area under the 70% force curve.

Improvements observed over the course of the 12-week training program suggest that a resistance load of 30% is a sufficient stimulus to produce small improvements in maximal force and larger improvements in isometric endurance. The greatest improvements in endurance capacity were seen in the general neck strength and muscle coordination group. The initial focus of the CTP was on strengthening the deep cervical muscles and retraining the muscle coordination, progressing to improving muscular endurance. Given the time frame of the program (12-weeks), changes in muscle strength

and fatigue could mostly be attributed to physiological adaptations of the neuromuscular system. Gerdle, Henriksson Larsen, Lorentzon, and Wretling (1991) showed that reduced post-treatment fatigue resulted from modification of the recruited motor unit pool, where recruitment of Type I fibres increased relative to Type II fibres. Alterations in the motor program used by CTP subjects may have enabled them to perform the movement more efficiently, which could provide an alternate explanation. For example, CTP subjects could have employed a more effective control strategy to maintain the 70% submaximal force. The control strategy could have been altered by an improved ability to recruit motor units, an improved synchronization of the motor units firing, modification of the neural activation patterns required for the movement or improved coordination of the muscles involved in the movement (Falla et al., 2006). To a lesser extent, the improved endurance capacity may be due to physiological adaptations such as the hypertrophy of Type I fibres, which was found to occur in the vastus lateralis after 12 weeks of endurance training (Putman, Xu, Gillies, MacLean, & Bell, 2004). To date, there is no research that has biopsied the cervical muscle after a training intervention; therefore, I can only speculate about the fibre type distribution and hypertrophy.

### *5.5 Visual analog scale self-reported neck pain scores*

At baseline, 76.2% of the treatment population reported some form of neck pain, comparable to values reported by Adam (2004); 81.2% for CH-146 helicopter pilots and 84.5% flight engineers. These values are higher than the 53% reported by Harrison (*in press-b*). Previous research on flight related neck pain has been conducted using dichotomized scales to identify participants as either suffering from neck pain or pain free (Alricsson et al., 2004; Äng et al., 2005; Harrison, *in press-b, in press-c*). To our knowledge, the use of a VAS for assessing neck pain is a novel approach for studies of aircrew. This scale has been used to examine the effects of different neck training

modalities on self-reported neck pain in female office workers with chronic neck pain (Niklander et al., 2006; Randløv et al., 1998; Ylinen et al., 2006). Niklander et al. (2006), Randløv et al. (1998) and Ylinen et al. (2006) all found the greatest decreases in self-reported neck pain based on the VAS occurred in subjects who performed high load resistance training. The model used in the present study was based on Ahlgren et al. (2001), who used 'pain at present', 'pain in general' and 'pain at worst' to assess the effects of three dynamic training programs on women with work-related trapezius myalgia. They found no differences between the treatment interventions in the reducing perceived 'pain at present' and 'pain in general'. However, in comparison to the endurance and coordination training programs, the strength training program produced the greatest reductions in 'pain at worst' (Ahlgren et al., 2001).

The finding of our study differ from Ahlgren et al. (2001) who found that individuals in the coordination and endurance groups reported lower scores for all three VAS scales, but isolated no differences in the reductions between the two groups. Over the course of our training intervention, the results for 'pain at present' did not significantly change for either the CTP or the ETP. However, the decrease in self-reported neck pain for the CTP from pre- to post-test came close to achieving statistical significance with a $p$ value of 0.07. The two treatment groups differed in response to 'pain in general', the ETP mean scores actually increased relative to the statistically significant decrease produced by CTP. Both training programs elicited statistically significant decreases in 'pain at worst' over the 12-weeks. Thus based on VAS scores CTP subjects reported the greatest reduction in overall neck pain for all three categories (i.e., 'pain at present', 'pain in general' and 'pain at worst').

Alricsson et al. (2004) tested the effects of a supervised cervical strength training program on the occurrence of neck pain in fighter pilots. In contrast to our results, they found that despite improved maximal force in the neck extensors and flexors, and

improved endurance in the neck extensors, there was no change in the frequency of self-reported neck pain. However, neck pain was determined using a frequency scale with possible responses ranged from 0-3 based on the number of neck pain occurrence during the past 3 months (Alricsson et al., 2004).

The reduction in 'pain at worst' for both the ETP and CTP and 'pain in general' for the CTP were statistically significant; however, the impact of these in a clinical environment is unknown. Linton (1986) suggested that a reduction of 10mm on the VAS is the minimal value required to indicate clinical importance. This contradicts Forouzanfar, Weber, Kemler, and van Kleef (2003) who found a relative pain reduction of 50% or more and an absolute pain reduction of at least 30 mm on the VAS was needed to accurately predict pain reduction after a treatment. Given that the mean reported baseline value for 'pain at present' and 'pain in general' in our population did not exceed 8.5 mm, a 30 mm decrease was not possible. Therefore a 10 mm decrease was taken to indicate whether a reduction was clinically important. 20% of the individuals in the CTP achieved a 10 mm decrease for 'pain at present', 30% for 'pain in general' and 50% for 'pain at worst'. A 10 mm reduction for 'pain at present', 'pain in general' and 'pain at worst' for the ETP occurred in 9%, 18%, and 54% of the group subjects, respectively. In comparison to the 10 mm decrease, a relative pain reduction of 50% was also used to determine whether the treatment was successful. Based on this criteria, ETP achieved a 50% reduction in VAS scores in 18% of the subjects for both 'pain at present' and 'pain in general' and in 45% of the subjects for 'pain at worst'. This decrease was achieved by 30% of the CTP cohort for 'pain at present', 40% for 'pain in general', and 60% for 'pain at worst'. The decrease in self-reported neck pain scores seen in conjunction with the improved maximal cervical force and muscular endurance is consistent with "chronic non-specific neck pain is to a considerable extent neuromuscular in origin" (Ylinen et al., 2006, p. 10). However, Falla et al. (2006), found reductions in self-reported neck pain in a

treatment intervention with no coinciding improvement in cervical flexor strength or endurance suggesting the existence of other physiological or psychological mechanisms which contribute to reductions in neck pain (Falla et al., 2006). Alricsson et al. (2004) improved neck strength and endurance in fighter pilots with a supervised training program, but found no reductions in reported neck pain. Other studies have reported reduced neck pain symptoms in non-treatment control subjects and have linked this improvement to spontaneous recovery (Ahlgren et al., 2001). These findings illustrate the complexities surrounding the experience of neck pain and the role psychological and physiological factors may play in its reduction.

### *5.6 Limitations*

There are several limitations in the current study. First, only a small number of 403 Squadron aircrew volunteered. Given the small subject size, I could only randomize 403 Squadron participants to the two treatment conditions, while volunteers for the non-treatment control were recruited from CH-146 aircrew at the 430 Squadron in Valcartier, QC. Despite this the experimental groups and the NTC did not differ in baseline anthropometric values, personal characteristics, flight experience, isometric MVC strength or submaximal endurance values.

Second, there were a large number of drop-outs over the course of the training intervention. The majority of these drop-outs occurred as a result of individuals being transferred or gone on course, making them unavailable to complete the post-test assessment. Unfortunately, this also forced interruptions in the completion of some individuals exercise training as seen in the training diaries, which highlights the importance of portability for an exercise program completed by this aircrew.

Third, during the baseline assessment, some individuals experienced pain and fatigue associated with the laboratory assessment protocol (LAP). This may have

influenced the post-test assessment even though there was an interval of 12 weeks. However, because the testing protocol was consistent among the groups, the effects should have been equivalent. During the NVG LAP-test, volunteers were asked to produce MVC for the four directions tested: flexion, extension, left and right lateral flexion. For any untrained population, when producing a maximal voluntary muscular contraction there maybe questions as to whether or not the contraction was truly maximal. This maybe particularly relevant for the cervical musculature as production of maximal contraction would rarely, if ever be performed by these subjects.

Fourth, a higher percentage of training sessions were completed by the CTP subjects. This may explain for the greater improvements seen when compared to the ETP. Analysis of the training diaries indicated a large drop in attendance in the final third of the training period. The drop in attendance coincided with the onset of a pilot and flight engineer training course at the 403 Squadron. Some of the subjects were instructors in these courses and did not complete their exercise training sessions during this time.

Fifth, the exact force produced with the Thera-band™ elastic tubing was unknown. Although a 30% resistance load was selected the resistance of the tubing varied with the length and thickness, thus the 30% resistance load was an approximation rather than a precise estimate. Despite this the elastic tubing is portable, enabling individuals to perform the exercises on their own time. The use of elastic tubing enabled the principal researcher to supervise and monitor the exercise training programs of multiple individuals.

One final limitation of my study was the possible under-reporting of neck pain severity (VAS score). Some subjects may have been comfortable identifying themselves as neck pain suffers, while others may have been hesitant to report the actual severity of their neck pain. The lack of control group VAS scores is a major limitation in the study. The absence of such scores prevented us from controlling for spontaneous recovery that

occurs for some pain suffers. However, there is little likelihood of spontaneous recovery occurring in this population given that occupational exposure during each flight would likely re-aggravate the symptoms.

### *5.7 Future Directions*

This study demonstrated that both an ETP and a CTP improved maximal isometric force, enhanced muscular endurance and reduced self-reported neck pain in CH-146 aircrew. The greatest improvement in force and reductions in neck pain scores occurred for the CTP group. Future research should therefore expand upon the finding of this investigative study and examine the effects in a larger population over an extended time frame. It would also be useful to examine the effects of the two treatment programs on in-flight muscle strain.

The first six weeks of the CTP focused on strengthening the deep cervical musculature and muscle coordination, before progressing to general endurance exercises for the neck. Given the change in focus of the CTP, the degree of activation of the deep cervical flexors needs to be determined pre- and post-intervention compared to the level in the ETP. Additional analysis of the prime movers of flexion, extension, left and right lateral flexion is needed to assess the two treatment interventions with neuromuscular and metabolic measurements.

## 6. Conclusion

Flight related neck pain has been identified as a concern with high prevalence in CH-146 aircrew (Adam, 2004; Forde, 2008; Harrison et al., 2007; Harrison, 2008). The provision of an ETP and CTP 3x a week for 12 weeks resulted in improvements in maximal isometric force and muscular endurance in the cervical spine. The greatest improvements occurred for those subjects in the CTP. These improvements in cervical muscular function reduced self-reported neck pain based on a VAS for 'pain in general' and 'pain at worst'. Reductions in self-reported neck pain for the ETP were limited to 'pain at worst'.

## 7. References

Adam, J. Results of NVG-induced neck strain questionnaire study in CH-146 Griffon aircrew. Defence R&D Canada Toronto ON. DRDC Toronto TR 2004-153, 2004.

Alderton, W. K., Cooper, C. E., & Knowles, R. G. (2001). Nitric oxide synthases: structure, function and inhibition. *Biochemical Journal, 357,* 593-615.

Ahlgren, C., Kerstin, W., Kadi, F., Djupsjobacka, M., Thornell, L., & Sundelin, G. (2001). Effects on physical performance and pain from three dynamic training programs for women with work-related trapezius myalgia. *Journal of Rehabilitation Medicine, 33*, 162-169.

Alricsson, M., Harms-Ringdahl, Larsson, B., Linder, J., & Werner, S. (2004). Neck muscle strength and endurance in fighter pilots: Effects of a supervised training program. *Aviation, Space, and Environmental Medicine, 75,* 23-28.

Anderson, G. B., & Ortengren, R. (1984). Assessment of back load in assembly line work using electormyography. *Ergonomics, 27,* 1157-1168.

Äng, B., & Harms-Ringdahl, K. (2006). Neck pain and related disability in helicopter pilots: a survey of prevalence and risk factors. *Aviation, Space, and Environmental Medicine, 77,* 713-719.

Äng, B., Linder, J., & Harms-Ringdahl, K. (2005). Neck strength and myoelectric fatigue in fighter and helicopter pilots with a history of neck pain. *Aviation, Space, and Environmental Medicine, 76,* 375-380.

Barbe, M. F., Barr, A. E., Gorzelany, I., Amin, M., Gaughan, J. P., & Safadi, F. F. (2003). Chronic repetitive reaching and grasping results in decreased motor performance and wide-spread tissue response in a rat model of MSD. *Journal of Orthopaedic Research, 21* 167-176.

Beckman, J. S. & Koppenol, W. H. (1996). Nitric oxide, superoxide, and peroxynitrite: The good, the bad, the ugly. *American Journal of Physiology, 271,* 1424-1437.

Bhambhani, Y. N. (2004). Muscle oxygenation trends during dynamic exercise measured by near infrared spectroscopy. *Canadian Journal of Applied Physiology, 29,* 504-523.

Bogduk, N. (2003). The anatomy and pathophysiology of neck pain. *Physical Medicine and Rehabilitation Clinics of North America, 14*, 455-472.

Bonger, P. M., Hulshof, C. T., Dijkstra, L., Boshuizen, H. C., Groenhout, H. J., & Valken, E. (1990). Back pain and exposure to whole body vibration in helicopter pilots. *Ergonomics, 33,* 1007-1026.

Bronfort, G., Evans, R., Nelson, B., Aker, P. D., Goldsmith, C. H., & Vernon, H. (2001). A randomized clinical trial of exercise and spinal manipulation for patients with chronic neck pain. *Spine, 26,* 788-799.

Brown, G. C., & Borutaite, V. (2002). Nitric oxide inhibition of mitochondrial respiration and its role in cell death. *Free Radical Biology and Medicine, 33,* 1440-1450.

Burnett A. F., Naumann, F. L., & Burton, E. J. (2004). Flight-training effect on the cervical muscle isometric strength of trainee pilots. *Aviation, Space, and Environmental Medicine, 75,* 611-615.

Butler, B. P. (2000). Long duration exposure criteria for head supported mass. Fort Rucker, AL; U.S. Army Aeromedical Research Laboratory. Report No. USAARL-1997-34.

Campbell, D. T. & Stanley, J. C. (1963). *Experimental and quasi-experimental designs for research.* Chicago: Rand McNally.

Carayon, P., Smith, M. J., & Haims, M. C. (1999) Work organization, job stress, and work-related musculoskeletal disorders. *Human Factors, 41,* 644-663.

Chen, Y., Wickramasinghe, V., & Zimcik, D. (2007). Adaptive mount approaches for helicopter seat vibration control. *Proceedings of ICAST 2007.*

Chiu, T., Lam, T., & Hedley, A. (2002). Maximal isometric muscle strength of the cervical spine in healthy volunteers. *Clinical Rehabilitation, 16,* 772-779.

Conley, M. S., Meyer, R. A., Bloomberg, J. J., Feeback, D. L., & Dudley, G. A. (1995). Non-invasive analysis of human neck muscle function. *Spine, 23,* 2505-2512.

Conley, M. S., Stone, M. H., Nimmons, M., & Dudley, G. A. (1997a). Specificity of resistance training responses in neck muscle size and strength. *European Journal of Applied Physiology, 75,* 443-448.

Conley, M. S., Stone, M. H., Nimmons, M., & Dudley, G. A. (1997b). Resistance training and human cervical muscle recruitment plasticity. *Journal of Applied Physiology, 83,* 2105-2111.

Costes, F., F. Prieur, L. Feasson, A. Geyssant, J. C., Barthelemy, & Denis, C. (2001). Influence of training on NIRS muscle oxygen saturation during submaximal exercise. *Medical Science in Sport and Exercise, 33,* 1484–1489.

Côté, P., Cassidy, J. D., & Carroll, L. (1998). The Saskatchewan health and back pain survey: The prevalence of neck pain and related disability in Saskatchewan adults. *Spine, 23,* 1689-1698.

Craig, J., Task, L., & Filipovich, D. (1997). Development and evaluation of the panoramic night vision goggle. *Air Force Research Laboratory Wright-Patterson AFB,* OH., ASC97-0507.

Cram, J.R., Kasman, G. S., & Holtz, J. (1998). *Introduction to surface electromyography.* Gaithersburg, MD: Aspen Publishers.

Croft, P., Lewis, M., Papegeorgiou, A. C., Thomas, E., Jayson, M. V., Macfarlane, G. J., & Silman, A. J. (2001). Risk factors for neck pain: a longitudinal study in the general population. *Pain, 93,* 314-325.

Danneels, L. A., Cools, A. M., Vanderstraeten, G. G., Cambier, C., Witvrouw, E. E., Bourgois, J., & de Cuyper, H. J. (2001). The effects of three different training modalities on the cross-sectional area of the paravertebral muscles. *Scandinavian Journal of Medicine and Science in Sports, 11,* 335-341.

de Oliveira, C. G., Simpson, D. M., & Nadal, J. (2001). Lumbar back muscle activity of helicopter pilots and whole-body vibration. *Journal of Biomechanics, 34,* 1309-1315.

de Oliveira, C. G., & Nadal, J. (2004). Back muscle EMG of helicopter pilots in flight: Effects of fatigue, vibration, and posture. *Aviation, Space, and Environmental Medicine, 75,* 317-322.

de Oliveira, C. G., & Nadal, J. (2005). Transmissibility of helicopter vibration in the spines of pilots in flight. *Aviation, Space, and Environmental Medicine, 76,* 576-580.

Dieën, J. H. (1996). Asymmetry of erector spinal muscle activity in twist postures and consistency of muscle activation patterns across subjects. *Spine, 21,* 2651-2661.

Dixon, J. S., & Bird, H. A. (1981) Reproducibility along a 10-cm vertical visual analogue scale. *Annuals of the Rheumatic Diseases, 40,* 87-89.

Docherty, D., & Sporer, B. (2000). A proposed model for examining the interference phenomenon between concurrent aerobic and strength training. *Sports Medicine, 30,* 385-394.

Dupuis, H. & Zerlett, G. (1987). Whole body vibration and disorders of the spine. *International Archives of Occupational and Environmental Health, 59,* 323-336.

Falla, D., Dall'Alba, P., Rainoldi, A., Merletti, R., & Jull, G. (2002). Location of innervations zones of sternocleidomastoid and scalene muscles—a basis for clinical and research electromyography applications. *Clinical Neurophysiology, 113,* 57-63.

Falla, D., Jull, G., Dall'Alba, P., Rainoldi, A., & Merletti, R. (2003-a). An electormyographic analysis of the deep cervical flexor muscle s during cranio-cervical flexion. *Physical Therapy, 83,* 899-906.

Falla, D., Rainoldi, A., Merletti, R., & Jull, G. (2003-b). Myoelectric manifestations of sternocleidomastoid and anterior scalene muscle fatigue in chronic neck pain patients. *Clinical Neurophysiology, 114,* 488-495.

Falla, D. (2004). Unravelling the complexity of muscle impairment in chronic neck pain. *Manual Therapy, 9,* 125-133.

Falla, D., Jull, G., & Hodges, P. (2004). Patients with neck pain demonstrate reduced electromyographic activity of the deep cervical flexor muscle s during performance of the craniocervical flexion test. *Spine, 29,* 2108-2114.

Falla, D., Jull, G., Hodges, P., & Vicenzino, B. (2006). An endurance-strength training regime is effective in reducing myoelectric manifestations of cervical flexor muscle fatigue in females with chronic neck pain. *Clinical Neurophysiology, 17,* 828-837.

Ferrari, M., Mottola, L., & Quaresima, V. (2004). Principles, techniques, and limitations of near infrared sprectroscopy. *Canadian Journal of Applied Physiology, 29,* 463-487.

Forde, K. A. (2008). Neck loads and postures experienced by Canadian forces helicopter pilots during simulated day and night flights. Un-published master's thesis, University of New Brunswick, Fredericton, New Brunswick, Canada.

Forsman, M., Kadefors, R., Zhang, Q., Birch, L., & Palmerud, G. (1999). Motor-unit recruitment in the trapezius muscle during arm movements and in VDU precision work. *International Journal of Industrial Ergonomics, 24,* 619-630.

Forouzanfar, T. Weber, W., Kemler, M., and van Kleef, M. (2003). What is a meaningful pain reduction in patients with complex regional pain syndrome type 1? *The Clinical Journal of Pain, 19,* 281-285.

Galen, G. P., Muller, M. L., Meulenbroek, R. G., & Gemmert, A. W. (2002). Forearm EMG response activity during motor performance in individuals prone to increased stress reactivity. *American Journal of Industrial Medicine, 41,* 406-419.

Gerdle, B., Henriksson Larsen, K., Lorentzon, R., & Wretling, M. L. (1991). Dependence of the mean power frequency of the electromyography on muscle force and fibre type. *Acta Physiologica Scandinacica, 142,* 457-465.

Gissel, H. (2000). $Ca^{2+}$ accumulation and cell damage in skeletal muscle during low frequency stimulation. *European Journal of Applied Physiology, 83,* 175-180.

Gogia, R. P., & Sabbahi, M. A. (1994). Electromyographic analysis of neck muscle fatigue in patients with osteoarthritis of the cervical spine. *Spine, 19,* 502-506.

Goolkasian, P. (2003). Neck pain and disability scale: A critical evaluation. *Expert Review of Pharmacoeconomics and Outcomes Research, 3,* 379-382.

Green, N. C., & Brown, L. (2004). Head positioning and neck muscle activation during air combat. *Aviation Space and Environmental Medicine, 63,* 709-716.

Hagg, G. M. (2000). Human muscle fibre abnormalities related to occupational load. *European Journal of Applied Physiology, 83,* 159-165.

Hämäläinen, O., Vanharanta, H., & Bloigu, R. (1994). Determinants of +Gz-related neck pain: A preliminary survey. *Aviation, Space, and Environmental Medicine, 64,* 651-652.

Hämäläinen, O., Heinijoki, H., & Vanharanta, H. (1998). Neck training and +Gz-related neck pain: A preliminary study. *Military Medicine, 163,* 707-708.

Hämäläinen, O. (1999), Thoracolumbar pain among fighter pilots. *Military Medicine, 164,* 595-596.

Hansson, T., Magnusson, M., & Brange of motionan, H. (1991). Back muscle fatigue and seated whole-body vibrations: An experimental study in man. *Clinical Biomechanics, 6,* 173-178.

Harms-Ringdahl, K., Ekholm, J., Schuldt, K., Nemeth, G., & Arborelius, U. P. (1986). Load moments and myoelectric activity when the cervical spine is held in full flexion and extension. *Ergonomics, 29,* 1539-1552.

Harms-Ringdahl, K., Linder, J., Spångberg, C., & Burton, R. R. (1999). Biomechanical considerations in the development of cervical spine pathologies. *NATO Research and Technology Organization RTO-TR, RTO-TR-4,* 49-65.

Harrison, M. F., Neary, J. P., Albert, W. J., Veillette, D. W., McKenzie, N. P., & Croll, J. C. (2007). Trapezius muscle metabolism measured with NIRS in helicopter pilots flying a simulator. *Aviation, Space, and Environmental Medicine, 78,* 110-116.

Harrison, M. F., Neary, J. P., Albert, W. J., Kuruganti, U., Croll, J. C., Chancey, V. C., & Bumgradner, B. A. (in press-a). Measuring neuromuscular fatigue in cervical spinal muscle of military helicopter aircrew. *Military Medicine.*

Harrison, M. F., Neary, J. P., Albert, W. J., McKenzie, N. P., Chancey, V. C., Bumgradner, B. A., & Croll, J. C. (in press-b). Neck pain and physiological capacities in helicopter aircrew. *Aviation, Space, and Environmental Medicine.*

Harrison, M. F., Neary, J. P., Albert, W. J., McKenzie, N. P., Chancey, V. C., Bumgradner, B. A., & Croll, J. C. (in press-c). A predictive logistic regression equation for neck pain in helicopter aircrew. *Aviation, Space, and Environmental Medicine.*

Hather, B. M., Tesch, P. A., Buchanan, P., & Dudley G. A. (1991). Influence of eccentric actions on skeletal muscle adaptations to resistance training. *Acta Physiologica Scandinavica, 143,* 177-185

Henneman, E., Somjen, G., & Carpenter, D. O. (1982). Excitability inhibitability of motorneurons of different sizes. *Journal of Neurophysiology, 28,* 599-620.

Hermans, V., & Spaepen, A. J. (1997). Influence of electrode position on changes in electromyographic parameters of the upper trapezius during submaximal sustained contractions. *European Journal of Applied Physiology, 75,* 319-325.

Holmstrange of motion, E. B., Lindell, J., & Mortiz, U. (1992). Low back and neck/shoulder pain in construction workers: Occupational workload and psychosocial risk factors: Part 2 relationship to neck and shoulder pain. *Spine, 17,* 672-677.

Holte, K. A., & Westgaard, R. H. (2002). Daytime trapezius muscle activity and shoulder-neck pain of service workers with work stress and low biomechanical exposure. *American Journal of Industrial Medicine, 41,* 393-405.

Hansen, J. (2002). Sympathetic neural control of skeletal muscle blood flow and oxygenation. *Danish Medicine Bulletin, 49,* 109-129.

Ikai, M. & Fukunaga, T. (1970). A study on training effects on strength per unit cross sectional area of muscle by means of ultrasonic measurement. *European Journal of Applied Physiology, 28,* 173-180.

Immke, D. C. & McCleskey, E. W. (2001). Lactate enhances the acid-sensing $Na^+$ channel on ischemia-sensing neurons. *Nature Neuroscience, 4,* 869-870.

Issberner, U., Reeh, P. W., & Steen, K. H. (1996). Pain due to tissue acidosis: A mechanism for inflammatory and ischemic myalgia. *Neuroscience Letters, 208,* 191-194.

Jaccard, J., & Guilamo-Ramos, V. (2002). Analysis of variance frameworks in clinical child and adolescent psychology: Issues and recommendations. *Journal of Clinical Child and Adolescent Psychology, 31*, 130-146.

Jävholm, U., Palmerud, G., Karlsson, D., Herberts, P., & Kadefors, R. (1991). Intramuscular pressure and electromyography in four shoulder muscles. *Journal of Orthopaedic Research, 9,* 609-619.

Jensen, B. R., Jørgensen, K., Huijing, P. A., & Sjøgaard, G. (1995). Soft tissue architecture and intramuscular pressure in the shoulder region. *European Journal of Morphology, 33,* 205-220.

Jones, J., Hart, S. F., Baskin, D. S., Effenhauser, R., Johnson, S. L., Novas, M. A., Jennings, R., & Davis, J. (2000). Human and behavioral factors contributing to spine-based neurological cockpit injuries in pilots of high-performance aircraft: recommendations for management and prevention. *Military Medicine, 165,* 6-12.

Jonsson, B. (1982). Measurement and evaluation of local muscular strain in the shoulder during constrained work. *Journal of Human Ergology, 11,* 73-88.

Juel, C. (2001). Current aspects of lactate exchange: Lactate/H+ transport in human skeletal muscle. *European Journal of Applied Physiology, 86,* 12-16.

Jull, G. (2000). Deep cervical flexor muscle dysfunction in whiplash. *Journal of MusculoskeletalPain, 8,* 143-154.

Jull, G., Barrett, C., Magee, R., & Ho, P. (1999). Further clinical clarification of the muscle dysfunction in cervical headache. *Cephalalgia, 19,* 179-185.

Jull, G., Kistjansson, E., & Dall'Alba, P. (2004). Impairment in the cervical flexors: A comparison of whiplash & insidious onset neck pain patients. *Manual Therapy, 9,* 89-94.

Kadefors, R., Forsman, M., Zoega, B., & Herberts, P. (1999). Recruitment of low threshold motor-units in the trapezius muscle in different static arm positions. *Ergonomics, 42,* 359-375.

Katazaki, S. & Griffin, M J. (1998). Resonance behavior of the seated human body and effects of posture. *Journal of Biomechanics, 31,* 143-149.

Kay, T. M., Gross, A., Goldsmith, C., Santaguida, P. L., Hoving, J., & Bronfort, G. (2007). Exercise for mechanical neck disorder (review). *The Cochrane Library, 2007,* 1-78.

Keller, K., Corbett, J., & Nichols, D. (1998). Repetitive strain injury in computer keyboard users: Pathomechanics and treatment principles in individual and group interventions. *Journal of Hand Therapy, 11,* 9-26.

Kennedy, C. (1998). Exercise interventions for the cervical spine. *Orthopaedic Division Review, Nov/Dec,* 13-29.

Kennedy, C. (2007). *Exercise prescription for the patient with cervical spine dysfunction.* Unpublished manuscript.

Knardahl, S. (2002). Psychophysiological mechanisms of pain in computer work: The blood vessel-nociceptor interaction hypothesis. *Work Stress, 16,* 179-189.

Legér, L. A., Mercier, D., Gadoury, C., & Lambert, J. (1988). The multistage shuttle run test for aerobic fitness. *Journal of Sport Science, 6,* 93-101.

Lexell, J., Jarvis, J., Downham, D., & Salmons, S. (1993). Stimulation induced damage in rabbit fast-twitch skeletal muscle s: A quantitative morphological study of the influence of pattern and frequency. *Cell Tissue Research, 273,* 357-362.

Lindstrange of motion, L., Magnusson, R., & Peterson, I. (1970). Muscle fatigue and action potential conduction velocity changes studied with frequency analysis of EMG signals. *Electromyography in Clinical Neuro-Physiology, 4,* 341-353.

Lindstrange of motion, L., Kadefors, R., & Peterson, I. (1977). An electromyographic index for localized muscle fatigue. *Journal of Applied Physiology, 43,* 750-754.

Linton, S. (1986). A critical review of methods for analysis of behaviour in chronic pain. *Scandinavian Journal of Behavioural Therapy, 5,* 31-48.

Lopez-Lopez, J. A., Vellejo, P., Rios-Tejada, F., Jimenez, R., Sierra, I., & Garcia-Mora, L. (2001). Determination of lumbar muscular activity in helicopter pilots: A new approach. *Aviation, Space, and Environmental Medicine, 72,* 38-43.

Lupajärvi, T., Kuorinka, I., Virolainen, M., & Holmberg, M. (1979). Prevalence of tenosynovitis and other injuries of the upper extremities in repetitive work. *Scandinavian Journal of Work Environment, 6,* 48-55.

Maikala, R. V. & Bhambhani, Y. N. (2004). Simultaneous measurements of cerebral and muscle oxygenation and blood volume responses in healthy men during exposure to seated whole body vibration (WBV). In P. K. Singal & N. Takeda (Eds.), *Adaptation biology and medicine,* p. 317-328. India, ND: Narosa Publishing House.

Marieb, E. N., & Hoehn, K. (2007). *Human anatomy & physiology* (7th ed.). Montreal, QC: Pearson Benjamin Cummings.

Mathiassen, S. E. & Winkel, J. (1990). Electromyographic activity in the shoulder-neck region according to arm position and glenohumeral torque. *European Journal of Applied Physiology, 61,* 370-379.

Maughan, R. J., Watson, J. S., & Weir, J. (1983). Strength and cross-sectional area of human skeletal muscle . *Journal of Physiology, 338,* 37-49.

Mayoux-Benhamou, M. A., Wybier, M., & Revel, M. (1989). Strength and cross-sectional area of the dorsal neck muscle s. *Ergonomics, 32,* 513-518.

Mayer, T. G., Kondraske, G., Mooner, V., Carmichael, T., & Butsch, R. (1989). Lumbar myoelectric spectral analysis of the back extensors to fatigue. *Spine, 14,* 986-991.

McCall, G. E., Byrnes, W. C., Dickinson, A., Pattany, P.M., & Fleck, S. J. (1996). Muscle fibre hypertrophy, hyperplasia, and capillary density in college men after resistance training. *Journal of Applied Physiology, 81,* 2004-2012.

McKeon, M. D., Albert, W. J., & Neary, J. P. (2006). Assessment of neuromuscular and haemodynamic activity in individuals with and without chronic low back pain. *Dynamic Medicine, 5,* 1-8.

Moncada, S., & Erusalimsky, J. D. (2002). Does nitric oxide modulate mitochondrial energy generation and apoptosis. *Nature Reviews and Molecular Cell Biology, 33,* 214-220.

Naumann, F. L., Grant, M. C., & Dhaliwal, S. S. (2004). Changes in cervical spine bone mineral density in response to flight training. *Aviation, Space and Environmental Medicine, 75,* 255-259.

Netto, K. J., Burnett, A. F., & Coleman, J. L. (2007). Neck exercises compared to muscle activation during aerial combat maneuvers. *Aviation, Space, and Environmental Medicine, 78,* 478-483.

Nikander, R., Malkia, E., Parkkari, J., Heinonen, A., Strack, H., & Ylinen, J. (2006). Dose –response relationship of specific training to reduce chronic neck pain and disability. *Medicine and Science in Sport and Exercise, 06,* 2068-2074.

Oksa, J., Hamalainen, O., Rissanen, S., Myllyniemi, J., & Kuronen, P. (1996). Muscle strain during aerial combat maneuvering exercise. *Aviation, Space and Environmental Science, 67,* 1138-1143.

Oldervoll, L. M., Rø, M., Zwart, J-A., & Svebak, S. (2001). Comparison of two physical exercise programs for the early intervention of pain in the neck, shoulders and lower back in female hospital staff. *Journal of Rehabilitation of Medicine, 33,* 156-161.

Panjabi, M. M., Cholewicki, J., Nibu, K., Grauer, J., Babat, L. B., & Dvorak, J. (1998). Critical load of the human cervical spine: An *in vitro* experimental study. *Clinical Biomechanics, 13,* 11-17.

Pelham, T. W., White, H., Holt, L. E., & Lee, S. W. (2005). The etiology of low back pain in military helicopter aviators: Prevention and treatment. *Work, 24,* 101-110.

Pope, M. H., Wilder, D. G., Donnermijer, D. D. (1986). Muscles fatigue in static and vibrational seating environments. In *Backache and back discomfort* (pp. 1-25.9). Neuilly-Sur-Seine, France: AGARD; AGAD-CP-378.

Pope, M. H., Magnusson, M., Lindell, V., Svensson, M., & Andersson, G. B. (1990). The measurement of oxygen uptake under whole-body vibration. *Iowa Orthopedic Journal, 10,* 85-88.

Portero, P., Bigard, A., Gamet, D., Flageat, J., & Guezennec, C. (2001). Effects of resistance training on neck muscle performance and electromyogram power sprectrum changes. *Eurpoean Journal of Applied Physiology, 84,* 540-546.

Powers, S. K., & Howley, E. T. (2004). *Exercise physiology theory and application to fitness and performance (5th ed.).* New York, NY: McGraw-Hill.

Price, D. D., McGrath, P A., Rafii, A., & Buckingham, B. (1983). The validation of visual analogue scales as ratio scale measures for chronic and experimental pain. *Pain, 17,* 45-55.

Pritchard, M. H., Pugh, N., Wright, I., & Brownlee, M. (1999). A vascular basis for repetitive strain injury. *Rheumatology, 38,* 636-639.

Putman, C. T., Xu, X., Gillies, E., MacLean, I. M., & Bell G. J. (2004). Effects of strength, endurance, and combined training on myosin heavy chain content and fibre type distribution in humans. *Journal of Applied Physiology, 92, 376-384.*

Radi, R., Cassina, A., Hodara, R., Quijano, C., & Castro, L. (2002). Peroxynitrite reaction and formation in mitochondria. *Free Radical Biology & Medicine, 33,* 1451-1461.

Randløv, A. Mikkel, Ø., Manniche, C., Kryger, P., Jordan, Al., Heegaard, S., & Holm, B. (1998). Intensive dynamic training for females with chronic neck/shoulder pain: A randomized controlled trial. *Clinical Rehabilitation, 12,* 200-210.

Ray, C. A., Hume, K. M., & Shortt, T. L. (1997). Skin sympathetic outflow during head-down neck flexion in humans. *American Journal of Physiology, 273,* 1142-1146.

Sarig-Bahat, H. (2003). Evidence for exercise therapy in mechanical neck disorders. *Manual Therapy, 8,* 10-20.

Seng, K., Lam, P., & Lee, V. (2003). Acceleration effects on neck muscle strength: Pilots vs. non-pilots. *Aviation, Space, and Environmental Medicine, 74,* 164-168.

Sekersted, O. M., Hargens, A. R., Kardel, K. R., Blom, P., Jensen, O., & Hermansen, L. (1984). Intramuscular fluid pressure during isometric contractions of human skeletal muscle. *Journal of Applied Physiology, 56,* 287-295.

Sharkey, B. (1970). Intensity and duration of training and the development of cardiorespiratory endurance. *Medicine and Science in Sports, 2,* 197-202.

Short, T. L., & Ray, C. A. (1997). Sympathetic and vascular responses to head-down neck flexion in humans. *American Journal of Physiology, 272,* 1780-1784.

Sjøgaard, G., Kiens, B., Jorgensen, K., & Saltin, B. (1986). Intramuscular pressure, EMG and blood flow during low-level prolonged static contraction in man. *Acta Physiologica Scandinavica, 128,* 475-484.

Smidt, N., de Vet, H.C., Bouter, L.M., & Dekker, J. (2005). Effectiveness of exercise therapy: A best-evidence summary of systematic reviews. *Australian Journal of Physiotherapy, 51,* 71-85.

Sovelius, R., Oksa, J., Rintala, H., Huhtala, H., Ylinen, J., & Sitonen, S. (2006). Trampoline exercise vs. strength training to reduce neck strain in fighter pilots. *Aviation, Space and Environmental Science, 77,* 20-25.

Sovelius, R., Oksa, J., Rintala, H., Huhtala, H., & Sitonen, S. (2008). Neck muscle strain when wearing helmet and NVG during acceleration on a trampoline. *Aviation, Space, and Environmental Medicine, 79,* 112-116.

Spriet, L. L., Howlett, R. A., & Heigenhauser, G. J. (2000). An enzymatic approach to lactate production in human skeletal muscle during exercise. *Medical Science and Sport Exercise, 32,* 756-763.

Sutherland, S. P., Cook, S. P., & McCleskey, E. W. (2000). Chemical mediators of pain due to tissue damage and ischemia. *Progress in Brain Research, 129,* 21-37.

Taylor, M. K., Hodgdon, J. A., Griswold, L., Miller, A., Roberts, D. E., & Escamilla, R. F. (2006). Cervical resistance training: Effects on isometric and dynamic strength. *Aviation, Space, and Environmental Medicine, 77,* 1131-1135.

Thomas, S., Reading I., and Shephard, R.J. (1992). Revision of the Physical Activity Readiness Questionnaire (PAR-Q). *Canadian Journal of Sport Science, 17,* 338-345.

Thorn, S., Forsman, M., Zhang, Q., & Taoda, K. (2002). Low-threshold motor unit activity during 1-h static contraction in the trapezius muscles. *International Journal of Industrial Ergonomics, 30,* 225-236.

Thuresson M. (2005). On neck load among helicopter pilots: Effects of head-worn equipment, whole-body vibration and neck position. Karolinska Institutet Stockholm, Sweden. Dissertation.

Thuresson, M., Äng, B., Linder, J., & Harms-Ringdahl, K. (2003). Neck muscle activity in helicopter pilots: Effect of position and helmet-mounted equipment. *Aviation, Space, and Environmental Medicine, 74,* 527-532.

Thuresson, M., Äng, B., Linder, J., & Harms-Ringdahl, K. (2005). Mechanical load and EMG activity in the neck induced by different head-worn equipment and neck postures. *International Journal of Industrial Ergonomics, 35,* 13-18.

van der Hoeven, J. H., van Weerden, T. W., & Zwarts, M. J. (1993). Long-lasting supernormal conduction velocity after sustained maximal isometric contraction in human muscle. *Muscle Nerve, 16,* 312-320.

Viikari-Juntura, E. (1983). Neck and upper limb disorders among slaughterhouse workers: An epidemiologic and clinical study. *Scandinavian Journal of Work Environment, 9,* 283-290.

Visser, B., & Dieën, J.H. (2006). Pathophysiology of upper extremity muscle disorders. *Journal of Electromyography and Kinesiology, 16,* 1-16.

Vøllestad, N. K., & Roe, C. (2003). Metabolic and mechanical changes during low-intensity work and their relation to work-related pain. In H. Johansson, U. Windhorst, M. Djupsjobacka, & M. Passatore (Eds.), *Chronic work-related myalgia: Neuromuscular mechanisms behind work-related chronic muscle syndromes*. Gavle University Press, Gavle.

Watson, D. H. & Trott, P. H. (1993). Cervical headache: An investigation of natural head posture and upper cervical flexor muscle performance. *Cephalalgia, 13,* 272-284.

Weirstra, B.T. (2001). Ergonomic assessment of flight engineers at 403 SQN. Camp Gagetown NB; Physiotherapy Report 6600-1.

Westgaard, R. H. & DeLuca, C. J. (1999). Motor unit substitution in long-duration contractions of the human trapezius muscle. *Journal of Neurophysiology, 82,* 501-504.

Wickes, S., & Greeves, J. (2005). Epidemiology of flight-related neck pain in Royal Air Force (RAF) aircrew. *Aviation, Space, and Environmental Medicine, 76,* 298-311.

Wikstrange of motion, B., Kjellberg, A., & Landstrange of motion, U. (1994). Health effects of long-term occupational exposure to whole-body vibration: A review. *International Journal of Industrial Ergonomics, 14,* 273-292.

Wilder, D., Woodworth, B. B., Frymoyer, J. W., & Pope, M. H. (1982). Vibration and the human spine. *Spine, 7,* 243-254.

Winters, J. M., & Peles, J. D. (1990). Neck muscle activity and 3-D head kinematics during quasi-static and dynamic tracking movements. In J. M. Winters, & S. L. Woo (Eds.), *Multiple muscle systems: Biomechanics and movement organization* (pp. 461-480). New York, NY: Springer.

Ylinen, J. J., Takala, E., Nykanen, M., Häkkinen, A., Mälkiä, E. A., Pohjolainen, T., Karppi, S., Kautiainen, H., & Airaksinen, O. (2003). Active neck muscle training in the treatment of chronic neck pain in women a randomized controlled trial. *Journal of American Medical Association, 289,* 2509-2516.

Ylinen, J. J., Häkkinen, A. H., Takala, E., Nykänen, M. J., Kautiainen, H. J., Mälkiä, E. A., Pohjolainen, T. H., Karppi, S., & Airaksinen, O. V. (2006). Effects of neck muscle training in women with chronic neck pain: One-year follow-up study. *Journal of Strength and Conditioning Research, 20,* 6-13.

Yoshitake, Y., Ue, H., Miyazaki, M., & Moritani, T. (2001). Assessment of lower-back muscle fatigue using electromyography, mechanomyography, and near-infrared spectroscopy. *European Journal of Applied Physiology, 84, 174-179.*

Zennaro, D. Laubli, T., Krebs, D. Klipstein, A., & Krueger, H. (2003). Continuous, intermitted and sporadic motor unit activity in the trapezius during prolonged computer work. *Journal of Electromyography and Kinesiology, 13,* 113-124.

Zuylen, E. J., Gielen, C. C., Denier van der Gon, J. J. (1988). Coordination of inhomogeneous activation of human arm muscle during isometric torques. *Journal of Neurophsyiology, 60,* 1523-1548.

# Appendix A

## Schedule of Events for the NVG-LAP Test

| Elapsed Time | Task Time | Task |
|---|---|---|
| 15 min | 15 min | Welcome to lab, informed consent |
| 30 min | 15 min | Instrumentation and anthropometric measurements |
| 35 min | 5 min | Warm up stretches and exercises |
| 55 min | 20 min | MVCs |
| 70 min | 15 min | Submaximal calibration |
| 85 min | 30 min | Submaximal endurance protocol |
| 115 min | 20 min | De-instrumentation and debriefing |

# APPENDIX B

## Ethics Approval Form

UNIVERSITY OF REGINA

OFFICE OF RESEARCH SERVICES

MEMORANDUM

DATE: August 14, 2006

TO: Dr. Patrick Neary
KHS

FROM: K. Arbuthnott
Chair, Research Ethics Board

**Re: Night Vision Goggles-Induced Neck Strain and Muscle Fatigue Characteristics of Griffon Helicopter Personnel (07R0607)**

---

Please be advised that the University of Regina Research Ethics Board has reviewed your proposal and found it to be:

☑ 1. ACCEPTABLE AS SUBMITTED. Only applicants with this designation have ethical approval to proceed with their research as described in their applications. The *Tri-Council Policy Statement on Ethical Conduct for Research Involving Humans* requires the researcher to send the Chair of the REB annual reports and notice of project conclusion for research lasting more than one year (Section 1F). **ETHICAL CLEARANCE MUST BE RENEWED BY SUBMITTING A BRIEF STATUS REPORT EVERY TWELVE MONTHS.** Clearance will be revoked unless a satisfactory status report is received. Please send revised consent form for your file.

☐ 2. ACCEPTABLE SUBJECT TO CHANGES AND PRECAUTIONS (SEE ATTACHED). Changes must be submitted to the REB and subsequently approved prior to beginning research. Please address the concerns raised by the reviewer(s) by means of a supplementary memo to the Chair of the REB. Do not submit a new application. Please provide the supplementary memorandum**, or contact the REB concerning the progress of the project, before **October 14, 2006** in order to keep your file active. Once changes are deemed acceptable, approval will be granted.

☐ 3. UNACCEPTABLE AS SUBMITTED. Please contact the Chair of the REB for advice on how the project proposal might be revised.

Dr. Katherine Arbuthnott

KA/ae/ethics2.dot

** supplementary memorandum should be forwarded to the Chair of the Research Ethics Board at the Office of Research Services (AH 505) or by e-mail to research.ethics@uregina.ca

APPENDIX C

Consent Form

**FACULTY OF KINESIOLOGY & HEALTH STUDIES**

**UNIVERSITY OF REGINA**

**INFORMATION SHEET FOR NIGHT VISION GOGGLES STUDY**

**RESEARCH PROJECT TITLE:**

**NIGHT VISION GOGGLES – INDUCED NECK STRAIN AND MUSCLE FATIGUE:**

**The Use of Training Programs to Combat Neck Strain**

**PRINCIPAL RESEARCHERS:**

Dr. J. Patrick Neary (PI, University of Regina)

Dr. Wayne J. Albert (University of New Brunswick)

---

**PURPOSE:**

It has been determined and well documented by Canadian Forces (CF) expertise that there exists a significant problem in the neck musculature and spinal column of flight personnel related to wearing NVG equipment during flight in both rotary and fixed wing aircraft. This is not only a safety concern (both short-term and long-term health implications) for the Instructor Pilots and Flight Engineers involved in their occupation during these operations, but this also has significant financial and fiscal implications for the Canadian Forces. In particular, the Questionnaire Survey performed by DRDC (Toronto, ON) clearly established that 90% of the pilots who responded to this questionnaire, who had greater than 150 flying hours with NVG's, reported neck pain. Collectively, the reports and questionnaires to date demonstrated that "neck injuries range from relatively minor to serious impairments", and in some cases, the grounding of pilots for years. The majority of research to date has used fighter pilots that typically experience gravitational forces greater than 4 Gz. However, limited quantifiable and objective scientific data is available on pilots that are exposed to gravitational forces (Gz) less than 2 Gz, such as that typically experienced by the CH146 Griffon Pilots and Flight Engineers. Therefore, the current research

proposes to address this issue, i.e., collection of physiological and biomechanical data, and then provide a diagnosis and prognosis for the inherent muscular injury problem(s) associated with wearing NVG's in CF personnel. Specifically, non-invasive near infrared spectroscopy (NIRS, to monitor muscle oxygenation and blood volume changes) and electromyography (EMG, to monitor neuromuscular changes) will be used to examine the fatigue characteristics of the neck musculature that is associated with neck strain as a result of wearing NVG equipment. Both of these tools have been used in previous research in the Faculty of Kinesiology do provide minimal risk to the subjects participating in the data collection. The data collected will then allow us to provide appropriate rehabilitation programs and early intervention guidelines for prevention for these pilots and flight engineers.

**METHODS**

This testing session will require two visits, separated by 12 weeks, to the laboratory space located in the Simulator Building at 403 Squadron at Gagetown, NB. Testing will take approximately 1.5 hours/session and will require some physical exertion. You will be asked to perform standardized testing exercises to allow the researchers to determine your maximal strength in the muscle groups specific to maintaining a heads-up posture. These muscles are located in the neck and back (trapezius, sternocleidomastoid, erector spinae) and are the muscles that many pilots and flight engineers have identified as being stiff or sore after using NVG. During your time in the research laboratory, sensors for the specialized equipment (Near Infrared Spectroscopy – NIRS; Electromyography – EMG) will be placed on your skin at these muscles and secured with adhesive tape. A standard cockpit seat has been set up for you to use during these testing sessions. While seated, a Velcro strap will be placed around your head and attached to force transducer, a specialized research tool that can be used to measure the strength and endurance of a muscle or muscle group during activity. You will be asked to perform a series of "pulls" with your neck forwards, backwards, left and right while wearing this equipment and seated in the cockpit seat. These pulls will provide insight into the strength and endurance of your neck muscles. Appropriate rest will be provided between pulls and your condition will be monitored at all times by the researcher for risk of injury. All of this equipment can be adjusted for proper fit and should not cause you to be uncomfortable while wearing it.

You will also be asked to complete some questionnaire documents that will help us sort your data. These questionnaires include disability questionnaires that CF physiotherapists are currently using as well as a questionnaire designed by the research team specific to this project. This document will provide background

information about your past experiences with neck pain, helicopter flight, NVG flight, and physical activity/lifestyle choices. This information will help the research team design fitness programs to assist in rehabilitating or preventing neck strain in CH-146 aircrew in later stages of the research.

In the 12-weeks between testing sessions, you will be asked to be a member of one of three groups – the muscle strength and endurance training group, the muscle coordination group, or the control group. The two training groups will undergo 3 training sessions per week, for 12 weeks, under the supervision and guidance of a trained instructor. The goal of these groups will be to improve muscular function and decrease pain reports in flight specific situations for CH-146 aircrew. The control group will be asked to continue with flight operations and fitness regimens as they normally would.

**QUESTIONNAIRES:**

You will be asked to complete the following questionnaires:

- Physical Activity Readiness Questionnaire (PAR-Q)
- Disabilities of the Arm, Shoulder, and Hand (DASH)
- QinetiQ Neck Strain Questionnaire
- Ratings of Perceived Exertion (RPE)
- NVG Information Sheet

**POTENTIAL RISKS & PARTICIPATION:**

Your participation is completely voluntary and you may withdraw yourself as a participant at any time during the testing. Potential risks are minimal. You may experience some soreness or fatigue during the later stages of the testing and following the completion of the testing (i.e. later in the day, the next morning, up to and including 72 hours post-testing). However, this discomfort will not differ from what you may experience during the course of a normal training or duty cycle as a flight crewmember of the CH 146 Griffon and should not interfere with your operational responsibilities.

**CONFIDENTIALITY:**

All information and results collected from you will be used exclusively for scholarly or diagnostic purposes with the intent of preventing and treating the neck pain currently being reported and observed in CH 146 flight crewmembers that are exposed to night vision goggles.

**EXPLANATION OF RESULTS:**

You will be given a thorough explanation of your own test results at your request, both verbally and in written form if desired.

| | |
|---|---|
| _____ | _____ |
| Investigator's Signature | Date |
| _____ | _____ |
| Participant's Signature | Date |
| _____ | _____ |
| Witness Signature | Date |

APPENDIX D

NVG-LAP Test Questionnaire

***University of Regina – Faculty of Kinesiology & Health Studies***

***Night Vision Goggle Study Data Sheet***

Please complete the information below honestly and accurately. Your responses will be held by the research team in the strictest of confidence and your individual responses will not be provided to CF medical personnel or your superior officers for any reason.

Name & Rank: _____ Age: _____

Height: _____ Weight: _____

Position: ☐ Pilot ☐ Flight Engineer ☐ Other

Dominant Hand: ☐ Right ☐ Left

Counterweight: ☐ I use it ☐ I do not use it

***Flight experience***

Total years of flight experience: _____

Total hours of flight experience: _____

Years of rotary wing flight experience: _____

Hours of rotary wing flight time: _____

Hours of Night Vision Goggle (NVG) experience: _____

Average length of a typical NVG flight (hours): _____

Maximum length of personal NVG flight (hours): _____

***Neck Pain History***

Do you experience neck pain? (please check all that apply)

☐ Yes, constant & severe ☐ Yes, constant but moderate

☐ Occasionally ☐ During the winter months

☐ During the summer months ☐ During extended exposure to NVG

☐ During repeated exposure to NVG ☐ During actual flight time

☐ During simulator exercises ☐ Never

☐ Do you smoke tobacco? ☐ Yes ☐ No

***Physical Activity History***

How often, per week, do you engage in aerobic exercise for a period of 20 minutes or more?

☐ 0 ☐ 1-2 ☐3-4 ☐5+

How often, per week, do you engage in strength training exercise that includes at least 3 different exercises?

☐ 0 ☐ 1-2 ☐3-4 ☐5+

My physical fitness regimen includes the following: (please check all that apply)

☐ Aerobic training ☐ Strength training

☐ Stretching and flexibility ☐ Yoga

☐ Organized team sports ☐ Other

***Neck Pain Treatment History***

I have sought treatment from the following professionals for neck pain: (please check all that apply)

☐ Flight Surgeon ☐ Physiotherapist

☐ Massage therapist ☐ Acupuncture specialist

☐ Personal trainer ☐ Other

If "Other", please specify: _____

_____

_____

_____

Thank you for your time and input!